THE INVESTIGATION OF MUSCULAR FACTORS IN NIGHT VISION GOGGLE INDUCED NECK STRAIN IN CANADIAN FORCES HELICOPTER AIRCREW

A Thesis

Submitted to the Faculty of Graduate Studies and Research

In Partial Fulfillment of the Requirements

For the Degree of

Doctor of Philosophy

(Special Case)

in Kinesiology & Health Studies

University of Regina

By

Michael Frederick Harrison

Regina, Saskatchewan

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SUPERVISORY AND EXAMINING COMMITTEE

Michael Frederick Harrison, candidate for the Special Case Doctor of Philosophy Degree in Kinesiology and Health Studies, has presented a thesis titled, ***The Investigation of Muscular Factors in Night Vision Goggle Induced Neck Strain in Canadian Forces Helicopter Aircrew,*** in an oral examination held on December 16, 2008. The following committee members have found the thesis acceptable in form and content, and that the candidate demonstrated satisfactory knowledge of the subject material.

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ABSTRACT

The issue of neck strain in helicopter aircrew among the Canadian Forces is a recent research focus. To date, the majority of the investigations have focused on questionnaire data related to perceived causes of the pain. This thesis represents the first comprehensive investigation of neck strain in this population to focus on muscular responses and capacities. Using 5 key papers and 3 appendices, this thesis: 1) demonstrates the impact the additional mass of the night vision goggle (NVG) equipment has on metabolic activity as compared to normal flight missions; 2) investigates the relationships between the superficial cervical muscles as they fatigue during submaximal isometric endurance trials; 3) clearly presents the capacities and characteristics of the current aircrew population in the Canadian Forces; 4) presents a means of differentiating and predicting neck pain in symptomatic aircrew as compared to their asymptomatic colleagues. The methodology used in this thesis included the use of near infrared spectroscopy (NIRS) and electromyography (EMG) evaluations of the superficial cervical muscles, including the splenius capitis, sternocleidomastoid, and upper trapezius muscles. Questionnaire data collection included the gathering of information such as aircrew position, flight experience, age, scores from the most recent annual fitness testing, and reported neck pain history and current status. Statistical analyses ranged from ANOVA and within subjects repeated-measures ANOVA to more in-depth analyses using logistic regression and principal components analysis. The results detail significant findings (i.e. differences in EMG patterns between symptomatic and asymptomatic aircrew; NIRS variables between Day and Night (NVG) flights) and present discussion of

the practical meanings and applications of these findings with respect to the medical community and the flight-line operators of the Canadian Forces.

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I wish to acknowledge the guidance of my supervisory committee: Dr. J. Patrick Neary, Dr. Wayne J. Albert, Dr. Yagesh Bhambhani, Dr. Darren Candow, and Dr. Richard MacLennan I also wish to acknowledge those who were not officially involved but provided valuable resources in data collection, data analyses, paper preparation & revision: Dr. James C. Croll, Dr. Victoria L. Chester, Dr. Usha Kuruganti, Dr. V. Carol Chancey., Dr. Paul St. Onge, Dr. Nabeh M. Alem, Mr. Gregory T. Dickinson, Mr. Bradley A. Bumgardner, Mr. Neil P. McKenzie, Maj. Daniel W. Veillette, Maj. Mario Coutu, Capt. Daniel O'Neill, Capt. Cheryl Elvidge, MCpl Joseph Hussey, all aircrew at 403 (H)OTS and 430 ESC and all members of the 1 Wing community. I am grateful to everyone at the University of New Brunswick (Faculty of Kinesiology) for providing desk space during my time there for data collection and I am grateful to the University of Regina (Kinesiology & Health Studies) for allowing me the freedom to pursue a project that was of such interest to me. I wish to acknowledge the Canadian Forces further as the source of funding that permitted this project to exist (Canadian Forces Quality of Life Grant #1725981 & Military Health Program #W3931-050513-001-SV). I am also grateful to the Aerospace Medical Association's Life Sciences & Biomedical Engineering Branch for the generous gift and recognition through the Ross McFarland Student Research Award (2006). Most importantly, this project would not have been possible without the support of my parents. Thank you for everything.

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I would like to extend my sincerest appreciation to Dr. John S. Crowley for serving the role of external examiner. It was an honour to discuss physiology with you on 16 December 2008. My apologies about the exceptional case of Canadian weather you experienced during your time away from the pleasurable climate of Alabama in December.

DEDICATION

There is no more humbling experience than acknowledging everyone who helped in the creation of this document. With that in mind, I dedicate this thesis to my grandfather, William Frederick Harrison, who was the kindest and humblest of them all. I can still hear him telling me that education is the easiest item to carry with you and no one can ever take it away.

"You only get out of this life what you put into it."

(WFH)

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CHAPTER 1
THESIS INTRODUCTION

The cervical spine and associated injuries are the focus of research related to military aircrew. The exact causes and effects of these injuries, while not perfectly understood, have become the specific focus of scientific investigation of late. The mass supported by the head and neck by military aircrew has increased to match the increases in flight responsibilities and numerous sources of information and feedback (Hart, 1988). While this mass represents equipment, such as night vision goggles (NVG) or heads-up display (HUD) units, designed to improve an operator's ability to monitor their aircraft and their environment, this mass has also been associated with reports of neck pain. These reports represent injuries that cover a broad spectrum from relatively minor, resulting in discomfort that is addressed with rest or non-prescription anti-inflammatory medications, to more severe, sometimes resulting in long-term discomfort and disability and the associated permanent removal from flight duties (Butler, 1992; Weirstra, 2001; Adam, 2004; Ang, 2006).

THE CERVICAL SPINE: STRUCTURE & INJURIES

At any given time, it has been reported that 10-20% of Western adults are currently experiencing neck pain (Holmstrom, Lindell, & Moritz, 1992). It has also been reported that the prevalence of neck pain in the general population is 67% (Cote, Cassidy, & Carroll, 1998). Within military aircrew populations, this rate of injury varies from approximately 28% to 90%, depending upon numerous factors including the military of

study, the method of injury classification and stratification of experience (Thomae et al., 1998; Wickes & Greeves, 2005; Adam, 2004). Despite the higher than average fitness levels of military aircrew, they are not immune to musculoskeletal injury and are in fact possibly more prone to injury.

The spinal column consists of 26 bones that articulate to form a curved and flexible structure (Marieb, 1998; Coakwell et al., 2004). This structure serves multiple purposes including protection for the spinal cord, support for the axial skeleton, the transfer of the load of the trunk to the lower limbs, articulation sites for the ribs, attachments for muscles of the rib cage, the back and the shoulders, and, perhaps most importantly, support for the skull. This supporting of the head allows the body's command and control center, the brain, to observe its environment. It is this function that resulted in the phenomena that provided the motivation for this thesis.

The cervical spine is composed of seven bones that are the smallest and most delicate of the spinal column and provide the attachment sites for the muscles of the lower jaw and the neck. The structure of the bones of the cervical spine and the locations of the muscles associated with the cervical spine allow for the great deal of head movement of which the human body is capable – flexion, extension, rotation, and combinations of flexion or extension with rotation. However, it is this delicacy in the design that makes the cervical spine a fragile region when large forces are applied instantaneously (i.e. as a result of sudden impact in a motor vehicle accident or contact sports) or over an extended period of time (i.e. cumulative loading).

It has been estimated that the maximum tolerance for single exposure to compressive forces without risk of injury to be 2,414 N for individuals 20 to 40 years of

age and 1,738 N for individuals over the age of 40 (Hidalgo et al., 1992). Furthermore, the authors estimated that prolonged loads should not exceed 1% of the MVC value. The difficulty with interpreting these values as they relate to helicopter aircrew is the complexity of the job and the mobility it demands. Helicopter aircrew do not maintain a static position while in flight and, as demonstrated mathematically by Hidalgo et al. (1992), small changes in cervical posture can result in large increases in the forces placed upon the ligaments, bone, intervertebral discs, and muscles. Forde (2008) has shown these increases to be even more pronounced when the neck must support a head loaded with a fully-equipped flight helmet.

Defining neck pain with the available literature is difficult. When the pain is muscular in nature, the term neck myalgia is often used. Further to this, neck myalgia is a component of the category of injuries referred to as upper extremity muscle disorders (UEMD) (Lupajärvi, Kuorinka, Virolainen, & Holmberg, 1979). Visser and Dieën (2006) defined UEMD as "disorders of muscle tissue proper, excluding tendon disorders and disorders of the tendinous insertions". In this case, the injuries are characterized by subjective symptoms such as sensation of constant muscle fatigue, muscle stiffness, and radiating pain (Visser & Dieën, 2006). Further difficulty in defining neck pain relates to the mechanism of injury. As will be presented in the discussion related to differences between motor vehicle accident victims with whiplash associated disorder and office workers with poor workspace ergonomics or, more appropriate to this thesis, between fast-jet aircrew and helicopter aircrew, the forces and duration of application of these forces can vary greatly.

As a result of these differences, classifying all neck pain as the same presents great difficulty. As a result, most research into neck pain indicates the pain and dysfunction to be multifactoral. These numerous factors include external psychosocial factors, physical loading factors, and the psychological and biological characteristics of the particular individual (Bonfort et al., 2001; Ylinen et al., 2003; Ylinen et al., 2006). Hypothesized causative factors such as muscle degeneration and/or impaired neuromuscular function resulting from chronic overuse (Conley et al., 1997) have been proposed in the literature.

Panjabi (2006) presents a clear and logical multifactorial hypothesis for neck and back pain. The proposed mechanism of pain is derived from cumulative microtrauma to the ligamentous structures of the cervical spine as a result of an extended period of submaximal loading. The microtrauma results in minor injuries and ruptures of the ligaments of the cervical spine that results in impaired muscle function, including "muscle coordination and individual muscle force characteristics, i.e. onset, magnitude, and shut-off" (Panjabi, 2006). Another proposed mechanism for injury in the literature has been termed the "Cinderella hypothesis" (Sjoogaard et al., 2000; Knardahl, 2002; Thorn, 2005). In this case, sustained submaximal muscular contractions result in occlusion of capillaries and arterioles within the muscle. As a result, oxygen delivery is impeded and aerobic respiration at the cellular level is compromised in the muscle. Such an occurrence is indicated and measurable through a shift in the redox state of cytochrome-c oxidase, the final enzyme in the electron transport chain. This hypothesis has been shown to be particularly applicable to situations in which a prolonged head forward posture and sustained mental alertness occur (Thorn, 2005).

For military aircrew such as pilots, flight engineers, weapons operators, or others responsible for tactical responsibilities in various military forces, currently reported cases of flight-related neck pain or strain are multi-faceted in their causal factors specifically related to their job performance (Hamalainen & Vanharanta, 1992; Hamalainen, 1993; Hamalainen et al, 1994; Oksa et al, 1996; Newman, 1997; Hamalainen et al., 1998; Ang et al., 2005; Thuresson et al., 2005). Previous studies have identified increased G-force (Gz) exposure, accumulated flying hours, head position, vibration, body posture, airframe and cockpit ergonomics, and head supported devices as t he most common factors (Adam, 2004; Wickes et al., 2005; Pelham et al., 2005). The available literature to date has focused on primarily fast-jet aircrew with a more recent shift towards equality of investigation to include helicopter aircrew. This is of particular importance given the substantial differences between working environments and the impact these differences are likely to have upon the mechanism of injury and the subsequent methods of mitigation. Fast-jet aircrew operations may expose aircrew to forces between +4.0 Gz and +7.0 Gz while helicopter aircrew rarely exceeds +2.0 Gz (Hamalainen & Vanaranta, 1992; Oksa et al., 1996; Weirstra, 2001). The helmets of fast-jet aircrew range in mass from 1.31kg – 2.15kg (Hamalainen, 1993; Hamalainen et al, 1994a) while a Canadian Forces helicopter aircrew member's flight helmet, when fully-loaded with NVG and counterweight equipment, may have a mass of 3.68 kg (Weirstra, 2001).

With respect to pilots specifically, another significant difference exists to differentiate fast-jet from helicopter aircrew. The in-flight spinal posture of helicopter pilots has been described as being hunched and twisted due to the location of particular flight controls (Lopez-Lopez et al., 2001; de Oliviera and Nadal, 2004; Pelham et al.,

2005). Pelham et al. (2005) attribute this to the location of the collective control, responsible for the pitch of the rotor blades, which is located to the left and below the pilot's seat. This location factor has been identified as an ergonomic concern in the literature (Lopez-Lopez et al., 2001; de Oliviera and Nadal, 2004; Pelham et al., 2005). This thesis aims to lend further evidence to this supposition.

These factors and issues suggest an investigation into neck strain among the Canadian Forces must be as specific and comprehensive as possible to helicopter aircrew for it to be successful. To date, the investigations have largely relied on a single method of evaluation. These methods include the use of questionnaires to identify rates of injury and perceived causes (Thomae et al., 1998; Adam, 2004; Wickes & Greeves, 2005), the use of electromyography to identify neuromuscular effects of NVG use (Thuresson et al., 2005; Butler, 1992), or other means of investigation of structural tolerance and operational ergonomics (Barazanji & Alem, 2004, Pelham et al., 2005) but no study has yet endeavoured to combine measures. Given the likelihood that the source of injury is multifactoral, research to identify causes and methods of mitigation should mirror that fact.

THESIS GOALS & STRUCTURE

This thesis represents the first comprehensive analysis of the issues related to NVG-induced neck strain in the Canadian Forces. While work has been done in other militaries by Butler (1992) and by Thuresson (2005), the findings of Adam (2004) and of Fraser et al. (2006) suggest that the issue is more severe within the Canadian Forces

aviation community. Thus, the need for this body of work is justified. The goals of this thesis are three-fold:

1. To provide evidence that the mass of standard-issue NVG equipment poses a significantly increased and measurable physiological (metabolic) stress as compared to helmets and operational equipment associated with day operations.

2. To classify the characteristics of individuals in both aircrew positions, pilots and flight engineers, with respect to anthropometrics, flight experience, physiological capacity and tolerances of the musculature of the cervical spine, and self-reported neck pain. The intent is to use this data to investigate means by which neck pain could be predicted and mitigated or prevented.

3. To identify any key differences in the above listed characteristics and physiological capacities between the two primary aircrew positions within the Canadian Forces helicopter community.

This thesis is presented in a journal article format. A series of 4 peer-reviewed publications have been prepared to address these primary goals and an additional 5 publications are presented in the appendices as sources of further detail on related topics. From these papers, the following general research hypotheses will be addressed:

H_1: The additional mass of the night vision goggles on the flight helmet will result in an increased in metabolic stress of the superficial neck muscles that is measurable with near infrared spectroscopy.

H_{2a}: Neuromuscular fatigue during isometric contractions to volitional fatigue would be measureable using electromyography to evaluate the superficial cervical muscles.

H_{2b}: Increased metabolic stress during isometric contractions to volitional fatigue would be measureable using near infrared spectroscopy to evaluate the upper trapezius muscles.

H_3: Using advanced statistical methods such as principal components analysis and logistic regression, the injured aircrew would display significant differences in their physiological and flight experience variables as compared to the non-injured aircrew.

More specific hypotheses can be found within each chapter or an appendix of this thesis.

The bibliographic details for the publications related to the primary aims are as follows:

1. **Harrison, M.F.**; J.P. Neary, W.J. Albert., D.W. Veillette, N.P. McKenzie, J.C. Croll. (2007). Trapezius muscle metabolism measured with NIRS in helicopter pilots flying a simulator. Aviat Space Environ Med. 78:110-6.

2. **Harrison, M.F.** J.P. Neary, W.J. Albert, U. Kuruganti, J.C. Croll, V.C Chancey, B.A. Bumgardner. (2008) Measuring neuromuscular fatigue in cervical spinal musculature of military helicopter aircrew. Mil Med. *In Press.*

3. **Harrison, M.F.**; J.P. Neary, W.J. Albert, N. P. McKenzie, V.C Chancey, B.A. Bumgardner, J.C. Croll. (2008) Neck pain and physiological capacities in helicopter aircrew. Submitted to Aviat Space Environ Med.

4. **Harrison, M.F.**; J.P. Neary, W.J. Albert, N. P. McKenzie, V.C Chancey, B.A. Bumgardner, J.C. Croll. (2008) A predictive logistic regression equation for neck pain in helicopter aircrew. Submitted to Aviat Space Environ Med.

The bibliographic details for the supporting publications included in the appendices are as follows:

A. **Harrison, M.F.**; J.P. Neary, W.J. Albert., D.W. Veillette, N.P. McKenzie, J.C. Croll. (2007). Cockpit seat side and trapezius muscle metabolism with night vision goggles. Aviat Space Environ Med. 78:995-8.

B. **Harrison, M.F.**; J.P. Neary, W.J. Albert., D.W. Veillette, N.P. McKenzie, J.C. Croll. (2007). Physiological effects of night vision goggle counterweights on neck musculature of military helicopter pilots. Mil Med. 172:864-70.

C. **Harrison, M.F.**; J.P. Neary, W.J. Albert, N. P. McKenzie, D.W. Veillette, J.C. Croll. (2008). Trapezius cytochrome oxidase changes with night vision goggle usage. Int J Ind Erg. *In Press.*

D. **Harrison, M.F.**; J.P. Neary, W.J. Albert, V.L. Chester, V.C Chancey, B.A. Bumgardner, J.C. Croll. (2008) Differentiation of physiological measures of neck myalgia by principal component analysis. Submitted to J Electromygr Kinesiol.

Paper 1, in conjunction with Appendices A and B, addresses the first goal. This initial work in the flight simulator with near-infrared spectroscopy provides quantitative indications of the impact of NVG on the musculature of the neck during simulated flight operations. **Paper 2** investigates the neuromuscular capacities of the superficial cervical muscles with an emphasis on patterns of fatigue during submaximal isometric contractions to volitional fatigue. **Papers 3** and **4** and **Appendix D** address the second and third goals. **Paper 3** presents detailed summaries of the "who" the aircrew population, including the physiology of the aircrew population with a specific focus on the function of the musculature of the cervical spine. **Paper 4** and **Appendix D** apply multivariate statistical techniques to this data. Using logistic regression, a predictive equation with respect to self-reported neck pain is developed and cross-validated with remarkable accuracy. Using principal component analysis, a means by which aircrew members who are symptomatic for neck pain can be differentiated from their asymptomatic peers using the electromyographical signals from the superficial musculature cervical spine during a submaximal muscle contraction is identified.

Additional to fulfilling these goals, other analyses of our results have provided further insight into the issue of neck strain including possible mechanisms of dysfunction and mitigation of injury. **Appendix A** evaluates the lack of relief of the physiological stress associated with a 90-minute flight with NVG through alternating between the right and left cockpit seat on subsequent missions. **Appendix B** discusses the efficacy of employing a counterweight to balance the mass of the NVG. **Appendix C** further evaluates neuromuscular and metabolic function as they relate to fatigue and to aerobic function of the muscles involved.

CHAPTER 2

TRAPEZIUS MUSCLE OXIDATIVE METABOLISM MEASURED WITH NIRS IN HELICOPTER PILOTS FLYING A SIMULATOR

[1]Michael F Harrison, [1]J Patrick Neary, [2]Wayne J Albert, [3]Daniel W Veillette, [4]Neil P McKenzie, [2]James C Croll

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INSERT: This paper, published in Aviation Space and Environmental Medicine (2007; 78:110-6) represents the start of this research. This work was performed in the flight simulator at 403 (Helicopter) Operational Training Squadron at Canadian Forces Base Gagetown, NB. This work established the utility of near infrared spectroscopy technology in quantifying the increase in metabolic stress as a result of the use of night vision goggles during flight operations.

INTRODUCTION

Military missions often require aircrews to operate in poor visibility and high stress situations. The most common cause of poor visibility in flight operations is darkness, and as a result, night vision goggles (NVG) have become standard equipment

for aircrew. NVGs are mounted on the helmets and thereby increase the headborne weight for the duration of the operation. Because of this, the American, British, Canadian, Swedish and Finnish militaries have recently begun concerted investigations into the issue of neck strain in their aircrews as a result of airborne operations and duties (Adam, 2004; Barazanji & Alem, 2000; Thuresson, 2005; Wickes et al., 2005). To date, a great deal of this research has focused on fast jet pilots (Hamalainen et al., 1994; Hamalainen et al., 1998; Oksa et al., 1996). This focus is due to the numerous reports of discomfort as a result of the combination of head postures and the large gravitational (+Gz) forces (>7.0Gz) experienced by fighter pilots during certain maneuvers. The stress experienced by the musculature of the cervical vertebrae during flight has been monitored using surface electromyography (EMG) that is occasionally combined with basic biomechanical information related to the neck and head position (Green & Brown, 2004; Hamalainen et al., 1994).

Helicopter pilots have received some attention of late due to the factors that distinguish their occupational environment from that of their fighter pilot colleagues (Hart, 1988; Thuresson, 2005; Thuresson et al., 2005). Helicopter pilots experience smaller +Gz forces (< 2Gz) during flight but are subjected to greater aircraft vibration exposure (Hart, 1988). In fact, these vibrations have been shown to be among the worst experienced in the workplace by any occupation and literature reviews have linked this exposure to the premature onset of some serious spinal abnormalities in helicopter pilots (Butler, 1992). To compound these issues, the weight of the helmet of a military helicopter pilot can be more than double the weight of a fighter pilot's helmet. The final operational weight is determined by the instruments, such as NVGs or Nuclear-

Biological-Chemical (NBC) equipment, which are affixed to assist the helicopter pilot in completing their assigned mission. The tactical helicopter pilots of the Canadian Forces (CF) currently employ a helmet that weighs 3.7kg when fully loaded with NVG-equipment and indications are that the combination of this weight and the working environment are causing significant health issues ranging from pain and stiffness to loss of sensation in the extremities (Adam, 2004).

Canadian Forces documents state that "neck pain incidents and injuries within the tactical helicopter community have been increasing over the past decade of flying with [NVG]", equipment that is "indispensable during night operations" (Adam, 2004). Recent CF documents indicate that all CF tactical helicopter units using the CH-146 Griffon had >50% of their pilots reporting neck pain during or following flight. Furthermore, five of the seven helicopter units had >70% of their pilots reporting neck pain (Adam, 2004). However, CF pilots report that NVGs allow for increased efficiency and mission outcome success in night operations, an operating environment that is both enabling and protecting. With NVGs, military helicopter pilots are able to visually operate when their enemies may not be able to visually detect their location. As a result, NVG use is likely to increase in the coming decade(s).

To date, the limited research efforts focusing on spinal muscle pain using helicopter crewmembers as research subjects have used EMG to evaluate neuromuscular activity and fatigue of the muscles of the upper thoracic and cervical spine (Barazanjo & Alem, 2000; Bulter, 1992; Thuresson, 2005). EMG has been shown to be reliable in monitoring the neck neuromuscular activity and associated tolerances in these

crewmembers, but does not provide any indication of metabolic factors such as oxygen supply and oxygen use or haemodynamics during these helicopter operations.

Near infrared spectroscopy (NIRS) has proven useful and reliable in measuring these metabolic factors in vivo (Delpy & Cope, 1997) with whiplash sufferers (Leavins et al., 2005a; Leavins et al., 2005b), during muscular activity (Albert et al., 2004), during whole-body vibration similar to those experienced by helicopter crewmembers (Maikala & Bhambhani, 2004), and even during the piloting of high-performance fighter jets (Kobayashi & Miyamoto, 2000). However, to date, NIRS has not been used to investigate NVG-induced neck strain among helicopter crewmembers. Therefore, the purpose of the investigation was to determine the utility of using NIRS in monitoring the metabolic responses in the trapezius muscles during a simulated mission flight with and without NVG-equipment affixed to the helmet. It was hypothesized that NIRS would be capable of identifying metabolic differences in the trapezius muscles of CF pilots between NVG and non-NVG flights. This information may be helpful in assisting helicopter pilots to determine appropriate rehabilitation strategies, appropriate operational schedules and human limitations.

METHODS

Subjects. Ethical approval was obtained from university Research Ethics Board and subjects were provided verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Thirty-three CF helicopter pilots representing all Canadian tactical helicopter squadrons agreed to participate in the

study and signed informed consent forms. The pilots also agreed to provide information about their level of experience (hrs/yrs flying), and their history of exposure to NVG flights (hrs).

Protocol. The pilots were monitored during two simulated flights – one with NVG equipment mounted on the helmet and one without NVG equipment. Monitored flights occurred at approximately the same time of day on two successive days, with the day flight performed on Day 1, and the Night (NVG) flight performed on Day 2. All missions were performed in the full-motion CH-146 Griffon Flight Simulator (CAE, Canada) at Base Gagetown, New Brunswick. Mission design was determined by CF recurrency training guidelines and was not interrupted or altered in any way by this research design. However, it was fortuitous that this sequence (day before night mission) was selected to avoid any possible residual effect that the NVG equipment might cause if the night mission was performed first. Therefore, the missions were similar in task and duration (Mean (SD): Day 96.6(16.5) min; NVG 97.7(16.1) min) between conditions and subjects. Pilots wore their own helmets in their normal configuration and sat in the same seat position for both monitored flights.

Near Infrared Spectroscopy Monitoring. The trapezius monitoring was performed bilaterally using the NIRO-300 Spatially Resolved Spectroscopic (SRS) oximeter (Hamamatsu Photonics KK, Japan). NIRS is an optical tool used to measure quantitative changes in muscle oxygenation, deoxygenation, and total hemoglobin during periods of physical activity and rest. The technical description and operation of the NIRO-300 has

been described in detail elsewhere (Suzuki et al., 1999). Briefly, the NIRO-300 uses different laser diodes with 4 wavelengths (776, 826, 845, 905nm) as the light source. Utilization of four lasers increases the sensitivity of instrument measurement as more chromophores can be detected in the localized region of light interrogation. The muscles are penetrated with near infrared light (700-1000nm) and the attenuation of this light is measured. The absorbency is calculated using a modified Beer-Lambert law (Delpy & Cope, 1997).

The NIRS probes were placed bilaterally at the C7 level of the upper trapezius, with the skin marked to ensure the same location of placement on the subsequent night mission. Placement location was selected as a result of pilot work in the simulator, discussions with CF medical personnel, and previous work in our laboratory with whiplash sufferers (Leavins et al., 2005a; Leavins et al., 2005b).

A standardized one-minute of baseline data was collected at the start of each mission with the pilots wearing their helmet and seated in the cockpit of the simulator. During simulated day missions, the helmet did not include any helmet mounted devices (i.e., NVGs) or additional weight. During all simulated night missions, the pilots wore their helmets with the appropriate helmet mounted NVG devices. For those pilots whose personal preference dictated their usage, counterweight equipment was present (n=25). Counterweight equipment is an approximately 600g lead mass attached with Velcro™ to the back of the pilot's helmet to act as a balance for the mass of the NVG equipment. CF command does not dictate that this equipment is mandatory for its aircrew and its usage has been left to the discretion of each individual. In the more experienced and highly trained pilots (n = 9; Mean (SD) =1877.5(1160.4) flight hrs, 7.8 (3.9) yrs helicopter

experience), a heads-up display unit was also affixed to the helmet. Due to the two distinct phases of all the training missions that, to a degree, replicate what occurs in actual flight operations, markers were inserted into the data to address a potentially telling feature of piloting the CH-146. Event markers were imbedded in the data to separate the period in which the pilots were primarily in a head-down posture to properly prepare the aircraft's avionics for navigation, radio frequencies, and other crucial information. These markers are displayed on the graphs (Figure 2.1-2.4) with a dashed vertical line at the 20% Mission Time marker.

The NIRS optodes (source and receiver) were set at a separation distance of 5.0cm for all subjects, allowing for a penetration depth of approximately 2.5cm (Ferrari et al., 1997). Published values for differential path length factor (DPF) are not available for the trapezius muscle. Thus, a DPF was not used in this study, but instead the concentration changes from baseline are reported as a delta value in $\mu M \cdot cm^{-1}$ for total haemoglobin (tHb), oxyhaemoglobin (HbO_2), deoxyhaemoglobin (HHb), as well, the tissue oxygenation index (TOI, %) was monitored to reflect changes in the percent of Hb saturation. tHB was calculated as:

$$\mathbf{tHb = [HbO_2] + [HHb]} \qquad \mathbf{(2.1)}$$

while TOI was calculated as:

$$\mathbf{TOI = [HbO_2] \bullet ([HbO_2] + [HHb])^{-1}} \qquad \mathbf{(2.2)}$$

The sampling speed of the NIRO-300 was set at 6 Hz to account for rapid changes in the dependent variables during the separate missions.

Statistical Analyses. Prior to the statistical analyses, the raw data was visually inspected for error messages within the output. Two subjects data sets were removed due to collection errors, providing a total of 31 data sets that were used in the final analyses. The mission times were also standardized to percentages to account for minor (non-significant) differences in mission duration. Event markers were imbedded in the data files to identify significant happenings such as change in NVG position (i.e. from the non-operational "upright" position to the operational "down" position) or the removal of helmet/NVG equipment in some rare situations.

Data was compiled as running averages of the values during two minute periods at every tenth percent (i.e. 0, 10, 20... 100% mission time) and analyzed using SPSS V16.0 (SPSS, USA). Visual inspection for outlying measurements was performed to identify values that were greater than two standard deviations from the mean. A within subjects repeated-measures analysis of variance (ANOVA) was conducted to determine left and right side differences, between conditions, and over the course of each mission. Significance was set at $p<0.05$. Adjustments to the statistical model, as necessary, were performed automatically with the Huynh-Feldt epsilon adjustment in SPSS.

RESULTS

Thirty-three male subjects (Age 37.0(7.4) yrs; Height 1.78(0.39) m; Weight 80.6(10.5) kg) volunteered to be monitored during their training sessions in the flight simulator. The subjects had an average of 7.1 (6.4) yrs experience as military helicopter pilots with an average of 1559.3(1400.5) flight hrs logged. Of these flight hours, an

average of 133.3(135.7) hrs was logged with NVG. Average reported NVG flight duration was 2.0(0.4) hrs with maximum length reported NVG flights being 3.5(1.4) hrs. This research successfully monitored haemodynamic changes in the neck musculature during simulated NVG and non-NVG missions using NIRS technology. However, the data sets from two subjects were removed after visual inspection of the data due to concerns of erroneous measurement as a result of incomplete NIRS probe adherence to the skin over the trapezius muscle that resulted in unreliable data (> ±3SD's).

When controlled for the individual variability of the subjects, significant differences *(Table 2.1)* were observed for three (tHb, HbO_2, and HHb) of the four haemodynamic variables measured over the course of the missions under the two operational conditions, including tHb (Figure 2.1), HbO_2 (Figure 2.2) and HHb (Figure 2.3). No significant differences were found for TOI (Figure 2.4) when compared by operational condition.

Variable	Condition (df = 1)	Side (df = 1)	Time (df = 10)	Condition x Side (df = 1)	Condition x Time (df = 10)	Condition x Side x Time (df = 10)
TOI						
p-value	0.31	0.14	0.62	0.37	0.63	0.98
(F-score)	(0.02)	(123.1)	(4.87)	(0.42)	(0.72)	(0.31)
[η^2]	[0.00]	[0.10]	[0.04]	[0.00]	[0.01]	[0.00]
Observed Power	1.00	1.00	1.00	0.10	0.39	0.17
HbO_2						
p-value	0.00	0.00	0.00	0.01	0.76	0.98
(F-score)	(18.0)	(54.8)	(9.6)	(9.48)	(0.52)	(0.31)
[η^2]	[0.02]	[0.05]	[0.08]	[0.01]	[0.01]	[0.00]
Observed Power	0.99	1.00	1.00	0.87	0.31	0.17
HHb						
p-value	0.19	0.00	0.23	0.00	0.41	0.96
(F-score)	(0.06)	(37.7)	(0.93)	(27.0)	(0.79)	(0.38)
[η^2]	[0.00]	[0.03]	[0.01]	[0.02]	[0.01]	[0.00]
Observed Power	0.06	1.00	0.48	1.00	0.42	0.20
tHb						
p-value	0.00	0.00	0.00	0.00	0.62	0.98
(F-score)	(12.7)	(62.4)	(5.34)	(15.2)	(0.57)	(0.34)
[η^2]	[0.01]	[0.05]	[0.05]	[0.01]	[0.01]	[0.00]
Observed Power	0.95	1.00	1.00	0.97	0.30	0.18

Table 2.1: Summary of statistical results, level of significance based on the within subjects repeated-measures ANOVA.

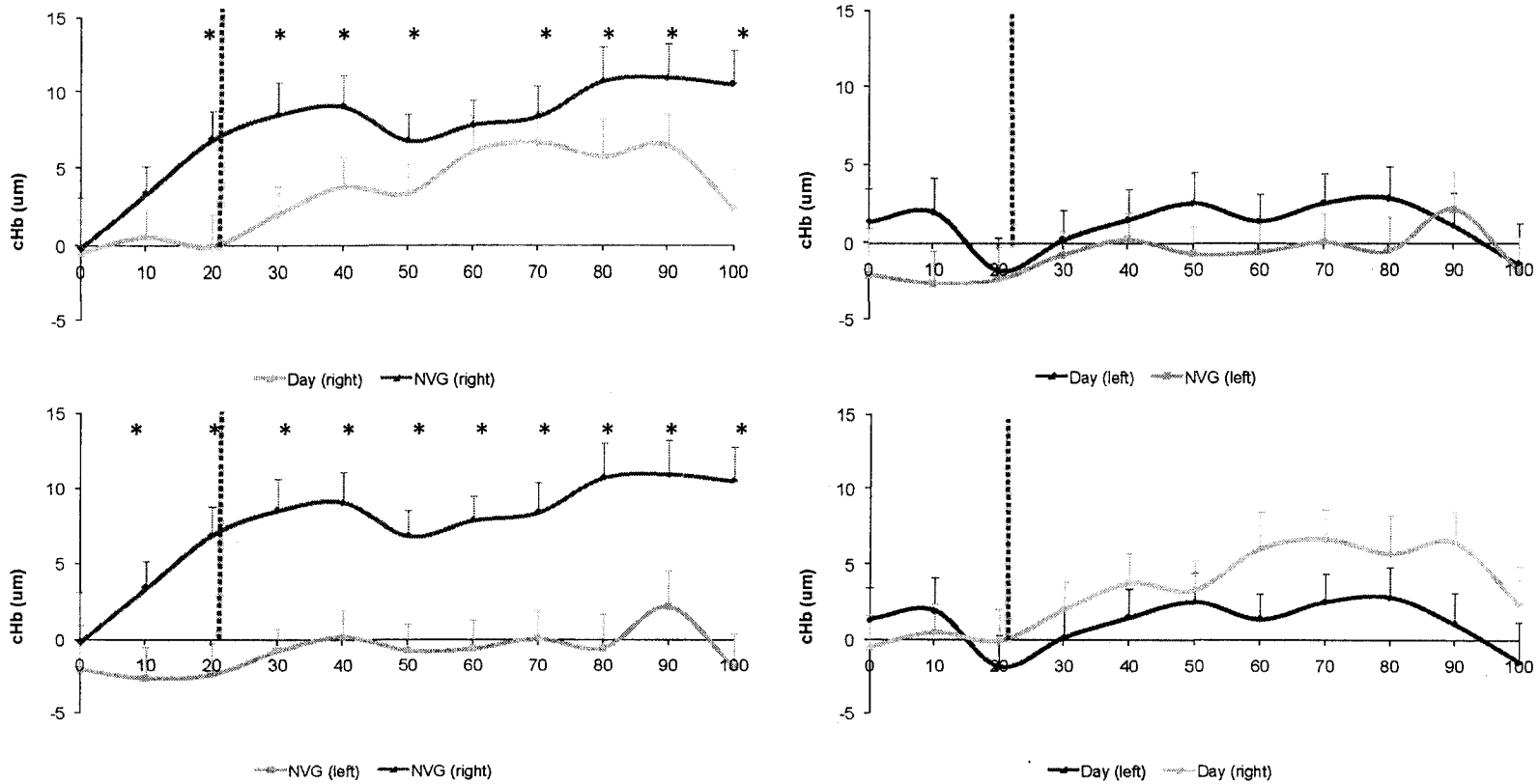


Figure 2.1: tHb responses (N=31) of the right and left trapezius muscles for Day & NVG missions, illustrating change over standardized during flight. Results are presented as comparison to side and condition. Dashed line represents time at which preflight tasks ceased and in-flight tasks commenced.

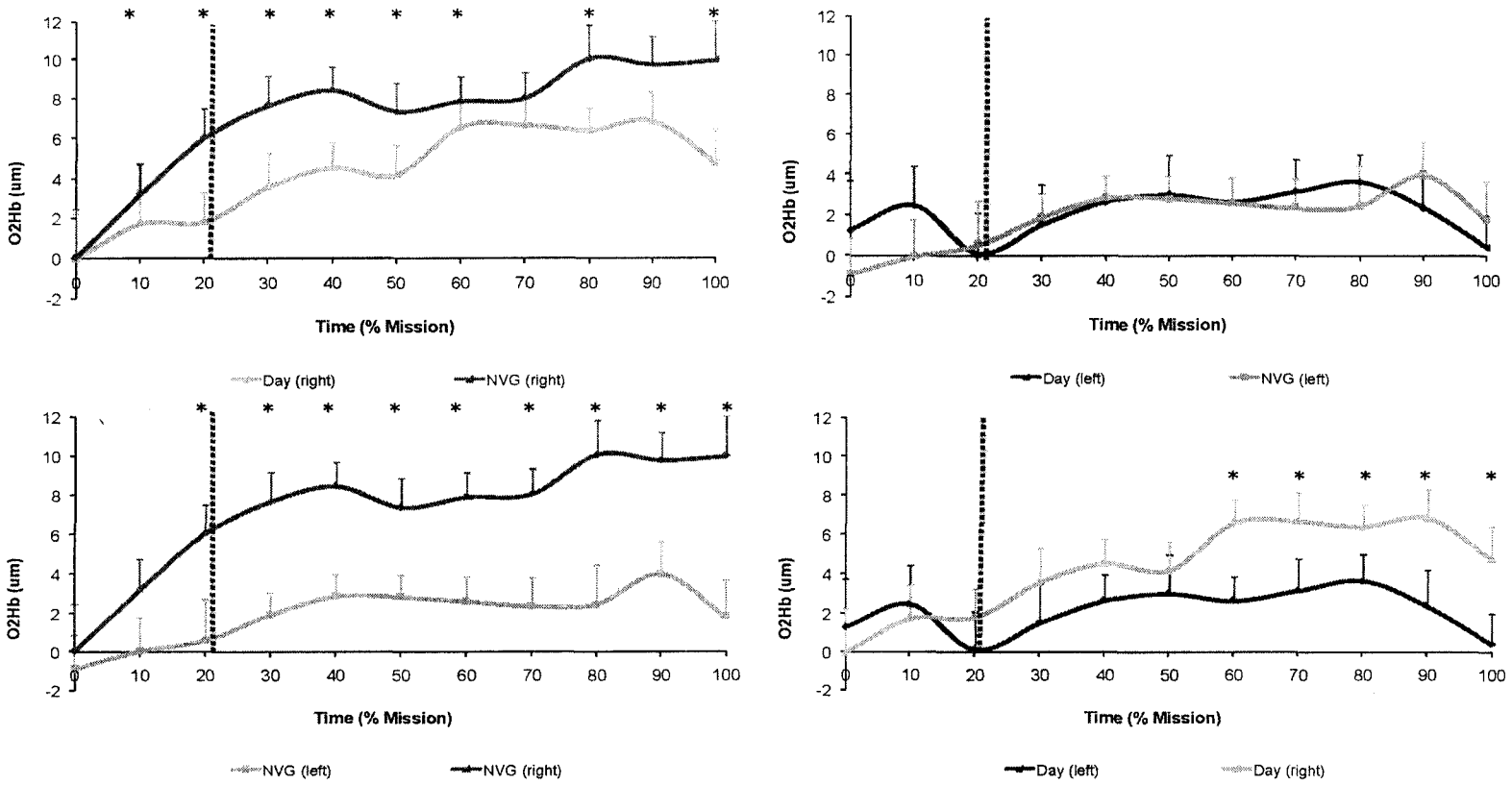


Figure 2.2: HbO_2 responses (N=31) of the right and left trapezius muscles for Day & NVG missions, illustrating change over standardized during flight. Results are presented as comparison to side and condition. Dashed line represents time at which preflight tasks ceased and in-flight tasks commenced.

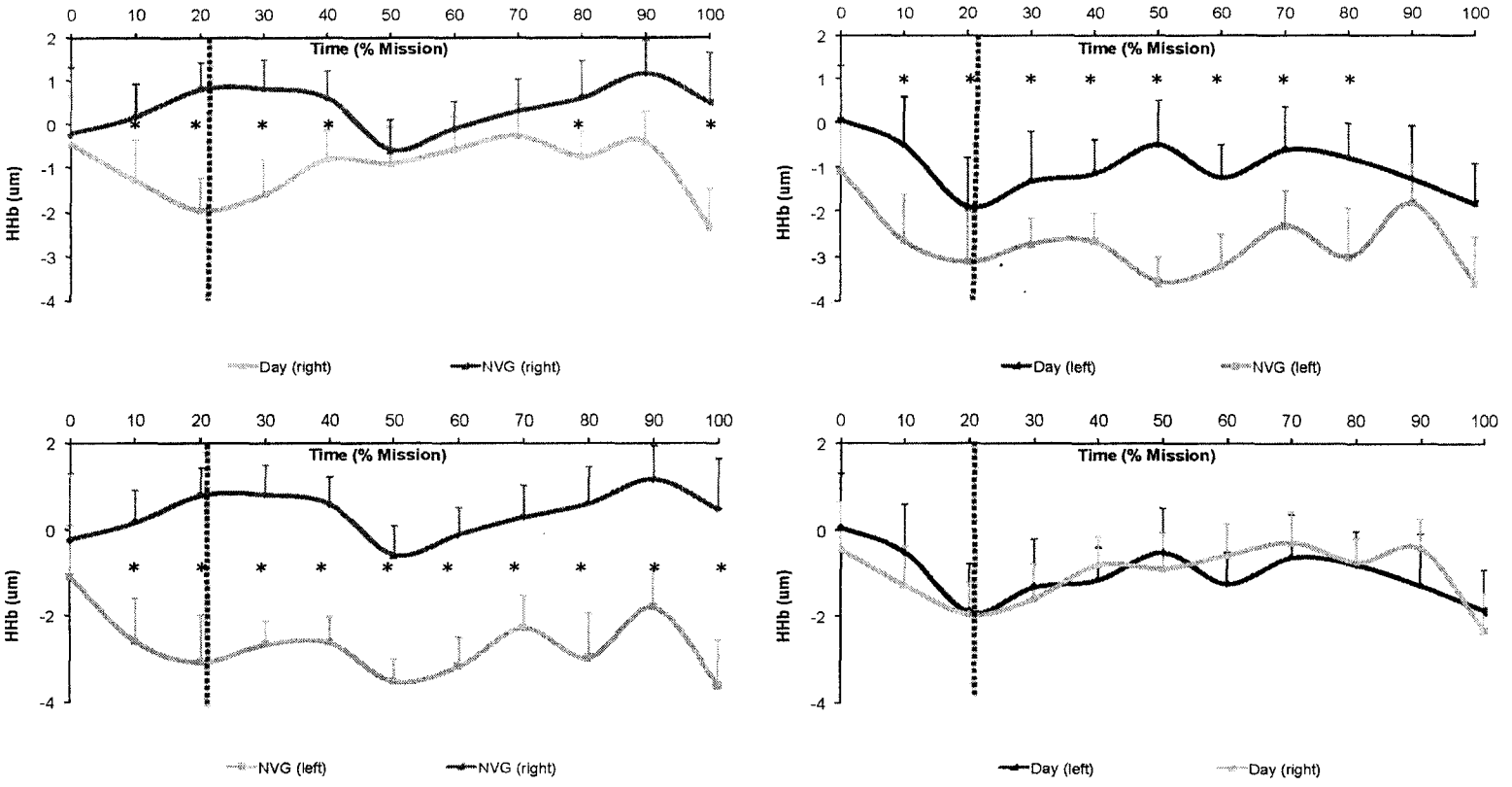


Figure 2.3: HHb responses (N=31) of the right and left trapezius muscles for Day & NVG missions, illustrating change over standardized during flight. Results are presented as comparison to side and condition. Dashed line represents time at which preflight tasks ceased and in-flight tasks commenced.

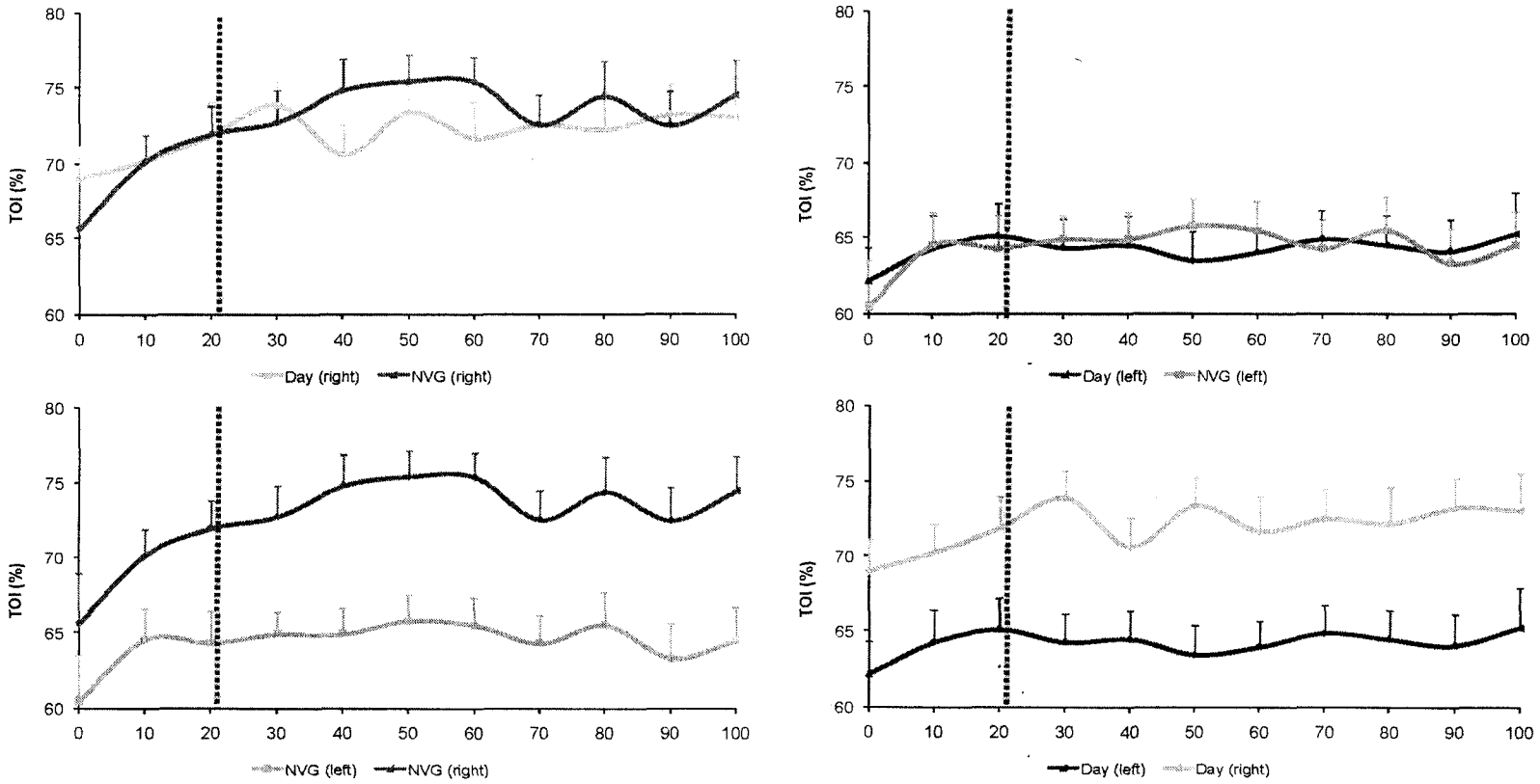


Figure 2.4: TOI responses (N=31) of the right and left trapezius muscles for Day & NVG missions, illustrating change over standardized during flight. Results are presented as comparison to side and condition. Dashed line represents time at which preflight tasks ceased and in-flight tasks commenced.

tHb. Significant differences were noted between Day and NVG simulations for tHb and between the left and right trapezius during these missions. Most importantly, while the tHb response was found to stabilize or illustrated a gradual decrease as the mission length increased during the Day simulation, no such stabilization was evident in the NVG simulations. tHb values during the NVG simulation continued to increase for the mission duration.

HbO_2. HbO_2 results showed a similar trend to those of tHb. Significant differences for the HbO_2 response were noted between the Day and NVG simulations as well as between the left and right trapezius. The Day simulation results showed a stable response for the majority of the simulated mission with a slight decrease noted towards completion. However, the NVG simulations showed a continual and gradual increase for the duration of the mission ($p<0.05$).

HHb. HHb results were not found to be statistically significant over the course of the missions between the different conditions, but significant differences were observed between the left and the right trapezius muscle during both Day and NVG missions.

TOI. TOI values were observed to increase from baseline levels at the start of the simulations during both operational conditions but significant differences were not found between Day and NVG values over the course of the missions. A plateau in TOI values was reached quickly and remained stable thereafter with no statistically significant differences noted between Day and NVG simulations. Additionally, significant

differences between the left and right trapezius were not found for TOI values in either Day or NVG missions.

DISCUSSION

It was hypothesized that NIRS would be capable of identifying oxygenation and haemodynamic changes in the trapezius muscle of CF pilots between NVG and non-NVG simulator flights to provide some indication of whether metabolic differences exist between these conditions. The results of this study showed that significant differences were found between Day and Night missions in muscle blood volume (tHb) and oxygenation (HbO_2) using NIRS. Furthermore, it was found that right and left muscle side differences were statistically significant for tHb, HHb and HbO_2.

The results of this study are consistent with investigations in exercise settings where NIRS has been used for evaluating changes in muscle metabolism during activity in numerous studies. Chance et al. (1992) pioneered the use of NIRS during exercise to evaluate O_2 desaturation in working muscle and used the length of time associated with the resaturation period to comment on approaches to improve performance on an individual basis. Since then, NIRS has been used to evaluate muscle metabolism in numerous sporting and work tasks such as walking and running (Hiroyuki et al., 2004; Bhambhani et al., 1998), cycling (Neary et al., 2002), and isometric exercise (Albert et al., 2004) to name a few.

This study, to the author's knowledge, was the first time that NIRS technology has been successfully used to monitor metabolic and haemodynamic trends within

cervical skeletal muscle of helicopter pilots during training missions. The results demonstrated that significant differences exist in trapezius muscle oxygenation (HbO_2) and total hemoglobin (tHb) during the NVG (night) simulation mission as compared to the non-NVG (day) mission. Our results indicate that the additional weight of the NVG equipment increased the metabolic stress of the trapezius muscles. Muscle oxygenation, as measured by TOI, HbO_2 and tHb, increased to a greater degree from baseline in the NVG setting as compared to the Day mission. This indicates that the trapezius muscles were performing submaximal contractions that did not cause occlusion that would restrict oxygen supply but rather created a stress that required and received an increase in oxygen to meet the increased metabolic demands.

In a diagnostic setting, NIRS has been used to assess oxygenation and metabolism in numerous pathologies ranging from chronic fatigue patients (Neary et al., 2008) to whiplash patients (Leavins et al., 2005a; Leavins et al., 2005b). In whiplash patients, NIRS was used to identify differences in the resting and exercise oxygen saturation in the trapezius muscles of whiplash patients as compared to healthy controls (Leavins et al., 2005a; Leavins et al., 2005b). The conclusion drawn was that this diminished muscle oxygenation at rest and during isometric contraction to fatigue in the injured individuals is likely a contributing factor to the pathology of the injury. Given the observed differences between the left and right trapezius muscles of the subject pilots in the current study, it would appear that these conclusions are potentially relevant to the work proposed by this document with CH-146 aircrew utilizing NVG equipment. Further work should be performed to investigate side differences in levels of pain and symptoms as reported by those that suffer from NVG-induced neck strain. This further work should

continue to investigate specific locations of pain as it relates to NIRS responses or, in further work, the measurement of neuromuscular functioning with electromyography. That said, it is likely that whiplash incidents are more similar to injuries in jet aircrew as compared to the more insidious nature of injury proposed for helicopter aircrew (Hamalainen & Vanharanta, 1992; Hamalainen et al., 1998; Adam, 2004; Fraser, 2007).

The lack of statistically significant differences in TOI values between the left and right trapezius and the Day and NVG missions is of particular interest when coupled with the significant differences found in the other three variables. TOI (which reflects muscle O_2 saturation percentage), is a relative unit of measurement that is dependent upon the ratio of HbO_2 to tHb (i.e., HbO_2 + HHb) values. Both of these variables were seen to increase over the course of the NVG missions yet not disproportionately. Therefore, TOI values were not significantly different between Day or NVG missions on either the left or the right side. The increase in both Day and NVG values over the course of the mission indicates that the trapezius became saturated with O_2 above baseline values. However, these results do not suggest occlusion of blood flow to the region.

Furthermore, the present results indicate that the trapezius muscles are able to achieve a level of metabolic homeostasis during the Day simulations (Figure 2.1). While the addition of the flight helmet to the mass supported by the musculature of the neck in a vibrating environment does increase observable haemodynamic stresses, this homeostasis is achieved early in the mission and maintained for the duration. However, in the NVG simulations, this homeostasis was not observable for the tHb or the HbO_2 responses. tHb showed a constant increase over the entire duration of the NVG missions for both the left and the right trapezius. While the tHb values for the right trapezius were consistently

higher than those for the left trapezius, a plateau was not observed throughout the duration of the mission. It is not possible from the present results to hypothesis whether or not a plateau would be observed prior to fatigue or even when such a plateau would occur. The average length of the NVG simulations in this study was 97.7(16.1) min, whereas these subjects reported the length of their average NVG mission in the airborne helicopter to be approximately 120(24) min. Participants in this study also reported their longest airborne NVG mission to be approximately 210(84) min. Therefore, this data does allow for supposition that the tHb values would increase during the average NVG flight with an operational squadron, and that any of the effects would be more pronounced during these longer sorties. Thus, further research is needed to identify when such a plateau would occur, and this information would be of assistance to the military personnel responsible for setting guidelines and operational thresholds with respect to NVG use.

The increase in tHb is coupled with an increase in oxygenation (HbO_2) to the trapezius muscles during NVG simulations as compared to the Day missions. Again, the values associated with the right trapezius were consistently and significantly higher than the values for the left trapezius for the duration of the NVG mission, indicating a greater haemodynamic response, and, based on HHb, a greater metabolic cost in the right side. HHb reflects the difference between oxygen consumption and utilization (i.e., extraction) in the local tissue and is essentially insensitive of blood-volume changes (Ferrari et al., 1997).

While significant, these findings are not unique with respect to the neuromuscular research on the spinal column and helicopter pilots. Previous research has reported

similar differences between the responses of the right and left sides in the lumbar spine musculature monitored with EMG during the piloting of aircraft similar in cockpit design and layout to the CH-146 (Lopez-Lopez et al., 2001). These authors found the erector spinae muscles to the right of the spinal column to be functioning to a greater degree than the contralateral muscles. The explanation offered by these authors addresses the issue of cockpit ergonomics. This work by Lopez-Lopez et al. (2001) using aircraft models similar in cockpit layout as the CH-146 as operational platforms was similar to our research in which all sorties were flown right handed. The cyclic control is located between the pilot's legs and is designed to be manipulated by the right hand. The collective control is located to the lower left side of the pilot's seat and is manipulated by the left hand. Lopez-Lopez et al. (2001) described the piloting posture as being "slightly forward and rotated to the left because the right arm and shoulder are brought forward to hold the cyclic pitch control of the helicopter" and similar posture was described by Pelham et al. (2005). In the present study, pilots anecdotally reported assuming a similar posture during simulated operations. However, further research needs to use quantitative methods to assess the biomechanical and simultaneous physiological measurements during NVG flight and the associated functional implications.

Given the description of the posture provided by Lopez-Lopez et al. (2001) and Pelham et al. (2005), it is likely that the lumbar deviation requires the pilot to rotate and extend the cervical vertebrae to be able to look forward through the windscreen or at the instrument panel. Additional neck strain would be caused by cockpit design and the necessity of the pilot to be capable of looking at control panels located above and below their horizontal line of sight in the middle consoles (i.e. central display unit for the

navigation system and the fuse panel). The location of the vital display units on the instrument panels is not reported to be a large concern to the pilots during day flights without NVG equipment attached to their helmets. During daylight operations, pilots are able to rely upon their peripheral vision to glance quickly at the required dials and switches with only minor movements of the cervical spinal column. However, night missions severely limit the available light sources and NVG equipment result in the decrease of peripheral sight by approximately 80% (Craig et al., 1997). Therefore, the pilot is required to function with a field of view of approximately 40° when using NVG's. This means the pilot must either t be looking almost directly at a particular instrument to be able to observe the information it is providing him or her or must support the head with the neck in an extended position in order to look under the goggles at the instrument. Obviously, this requires that the cervical spinal column be more mobile and that the musculature responsible for this range of motion be more active.

The event markers imbedded in the collected data that indicated the end of the period in which the pilots were primarily focused on programming the proper radio frequencies and the proper navigational information into the central display unit (CDU) did not correspond with the peak in the data with respect to the near infrared spectroscopy. This stage of the mission did correspond with the initial increases in the metabolic response during the NVG mission but this increase surprisingly continued after the commencement of "head's up" flight. The use of the CDU has been proposed as a link to neck strain by other researchers specifically focused on the Canadian Forces (Adam, 2004; Fraser, 2007). Our findings suggest that while the location of the CDU may be a contributing factor to the neck strain reported by this aircrew, it is not the sole

contributor. For further analyses of this data set with respect to cockpit seat side and counterweight use, please refer to Appendix A and Appendix B. In these chapters, further evidence is provided that the CDU is not the only explanation – the metabolic responses of the right upper trapezius when the pilot was seated in the right cockpit seat was statistically similar to the response of the right upper trapezius when the pilot was seated in the left cockpit seat. The same can be said for the left upper trapezius metabolic response. If the central location of the CDU was the only cause, one would hypothesize that the alternating of seat sides would provide relief to the upper trapezius of the opposite side. These factors influenced the work by Forde et al. (2008), in which analysis of the same missions was undertaken with a focus on cumulative loading analysis linked with posture. Their results were significant with respect to the use of NVG equipment having a profound impact upon the biomechanical stresses experienced by the cervical neck.

The implications of the findings of this research need to be emphasized as the author believes they provide criterion validity. While aspects of the environment within the simulator may have some differences to the environment in the airborne CH-146 Griffon, there are some very important similarities that make the full motion simulator an excellent training tool and an equally effective research tool for the collection of information related to NVG-induced neck strain. First, the simulator cockpit's layout is identical to the layout of the cockpit in the actual aircraft. The tasks performed by the pilots were identical to those that they would normally perform during a wide range of in-flight sorties. Second, the vibrations experienced by the aircrew during training operations in the full-motion flight simulator were designed to be similar to those

experienced by the crew members in the actual aircraft, between 2 and 20Hz (Lopez-Lopez et al., 2001). The environmental differences between the simulator and the actual aircraft include the introduction of temperature (i.e., cold exposure has been anecdotally reported by CF pilots to the research team as an area of concern), +Gz forces, and a larger cervical range of motion to allow the pilot to look out the rear door windows. However, it is unlikely that any of these differences between simulated and actual flight missions would result in the in-flight reduction of the metabolic stress responses that were observed in the simulator. Rather, it would be reasonable to assume that these factors would augment the stress responses observed during airborne operations, making our simulation observations of oxygenation and haemodynamics conservatively valid. Future research in the actual aircraft will allow for the quantification of the effects of these differences on the physiological (and neuromuscular) responses of the trapezius.

A potential limitation of our research was that the day flight was performed before the night (NVG) flight. Although confident that these results were not affected by this, a true experimental design would include a counter-balanced design by having the pilots randomly assigned or complete another trial where the NVG mission was performed first. However, due to the importance of the training sessions, the CF training schedules did not have flexibility to accommodate this counterbalance design.

The present study's results indicate that there is a significant concern related to certain factors or stimuli that were not considered at the outset of the project. First, it appears that muscular endurance should be an area of emphasis in any research involving the physical training or rehabilitative treatment of any aircrew member at risk of or currently suffering from NVG-induced neck strain. The issue of a physical training

regime to address neck strain from both a rehabilitative and preventative standpoint has been discussed previously in the literature (Bronfort et al., 2001; Hamalainen et al., 1994; Hamalainen et al., 1998). Hamalainen et al. (1998) provided preliminary results suggesting that neck specific strength training decreases the number of operational loss of +Gz restrictions among fighter pilots. However, they also recognize that their research had a number of limitations including the lack of a control group, and a small sample size in each group (n=10), thereby restricting their ability to make firm conclusions on the effects of the training methods.

Second, physiological data related to the recovery period for these subjects are not available to present. Because of the applied nature of this study, the collection of NIRS data post-mission was not possible so that any disruption to the training tasks performed by the individual crews was avoided and the intent was to not be disruptive to the overall training schedule and efficiency of the simulator unit. Thus, given the operational timelines and scheduling (missions debriefing, attending ground-school sessions, etc.) at the simulation unit, recovery data was not collected.

Third, the issue of neck strain in helicopter pilots is in its infancy and more research is needed by applied scientists responsible for equipment design and for medical support tasks related to flight operations. Helmet mounted devices are not limited strictly to NVG equipment that will only be worn during selected operations. Military helicopter pilots are dependent upon efficient methods of vital information provision to be successful in today's operational environment and heads-up display (HUD) units are becoming standard equipment for the helicopter pilot. Further work in this area will be required as technology continues to develop. Currently, panoramic NVG (pNVG) are

being developed to address the issue of peripheral vision. The current model of NVG used by many global militaries limit the light intensified field of view to approximately 40°; pNVG will provide a 160% increase in the light intensified field of view to 100° (Craig et al., 1997).

Finally, these reported results are from an all male sample population. While the CF helicopter community is predominantly male, experienced and operational female pilots and flight engineers do work within the CF. Previous research has found that female weight tolerances for helmet mounted devices is similar to male tolerances within the US Army helicopter community (Barazanji & Alem, 2000). Future work will seek confirmation of this finding using NIRS and simulated missions.

CONCLUSIONS

These results indicate the usefulness of NIRS technology to assess the metabolic and haemodynamic effects of NVG equipment on the neck musculature of military helicopter pilots. It was found that the NVG equipment augmented the metabolic stresses experienced by the musculature of the cervical vertebrae in helicopter pilots and found NIRS to be an effective means of quantifying the effects. Future research should focus on investigating the effects of prolonged NVG exposure, such as monitoring for an entire in-flight mission or repeated in-flight missions. To the author's knowledge, this is the first study that has monitored the metabolic responses of the neck musculature during Day and NVG simulated missions that were tactically similar.

ACKNOWLEDGEMENTS

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CHAPTER 3

MEASURING NEUROMUSCULAR FATIGUE IN CERVICAL SPINAL MUSCULATURE OF MILITARY HELICOPTER AIRCREW

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INSERT: This paper, in press with Military Medicine (2009), represents a follow-up to the previous chapter in which NIRS technology was shown to be useful in quantifying metabolic stress during flight operations. In this second study, electromyography (EMG) was used with submaximal isometric contractions to volitional fatigue to evaluate the relationship between 3 superficial muscle groups of the cervical neck.

INTRODUCTION

The issues of muscular fatigue, particularly in the workplace, and the effects of prolonged postures, repeated movements or poorly designed workplace ergonomics have frequently been the motivation for other published studies (Falla, 2004; Vieira & Kumar,

2004). Within the helicopter community of global military forces, these factors have been described as causative factors to "a number of occupation ailments" and "these disorders occur primarily in the neck, lumbar, and upper sacral regions" (Hodgon et al., 1997). A review of literature with aircrew specific publications indicates a growing concern with regards to cervical spinal injuries and pain in helicopter community that may have a prevalence of between 29-90%, depending upon the operational definitions and stratification of results within the individual paper (Thomae et al., 1998; Adam, 2004; Wickes et al., 2005). Regardless of the exact prevalence, flight-induced neck pain results in lost working days from highly trained aircrew that negatively impacts operational schedules and capabilities.

The military rotary-wing working environment contains many specifics that are quite likely to induce muscular fatigue within the aircrew. Large-amplitude, low-frequency vibrations are inherent within helicopter flight by design (Hart, 1988). The mass of the flight helmet plus helmet mounted devices such as night vision goggles (NVG) can be as much as 3.7kg (Harrison et al., 2007b) and survival equipment, depending upon the season, can have a mass of approximately 6.4kg (Weirstra, 2001). Average length of an NVG flight as reported by Canadian aircrew was 2.0 hrs/sortie with the length of maximum NVG-mission (on average) reported as being nearly double that value - 3.5hrs (Harrison et al., 2007c). Pelham et al. (2005) described the typical postures adopted by necessity by helicopter aircrew and the resulting detrimental lumbar kyphosis, thoracic kyphosis, and cervical lordosis. While the focus of that paper was specific to low back pain, the authors do make note of the "multi-segmental mechanical relationship of the spine" and indicate that the spine does operate as a whole unit. These postures are

linked to lengthening and fatiguing of the spinal musculature with a decreased capacity for force development and the nearly constant extension of the cervical spine is thought to result in muscular fatigue of the muscles responsible for maintaining such a position. This would be especially true in an NVG mission due to the additional head supported mass presented by this specialized mission enabling equipment.

Electromyography (EMG) has been used to monitor workplace activity of the neck muscles in low-load or semi-static environments on numerous occasions (Sommerich et al., 2000; Kumar et al, 2001; Kumar et al., 2002; Strimpakos et al., 2005) and, more relevant to this paper, specifically with military aircrew (Ang et al., 2005; Thuresson, 2005a & 2005b; Netto & Burnett, 2006). For the most part, EMG has been used to monitor activity during high +Gz maneuvers or during missions with head supported mass such as NVG. However, EMG was used to evaluate the median frequency response during fatiguing submaximal isometric neck contractions in extension, flexion and left and right lateral flexion. Similar work has been done with erector spinae in the lumbar spine (Albert et al., 2004) and the slope of the median frequency as the individual approaches ultimate fatigue has been shown to differentiate those who experience pain from those who do not (McKeon et al., 2006). However, the EMG related research with military helicopter aircrew and spinal muscles, specifically the cervical spinal muscles, have not highlighted which muscle(s) are most likely to fatigue during prolonged cervical spinal force-producing movements and limit an aircrew member's ability to perform that movement under load (Thuresson et al., 2005b). In the CF helicopter community, this information may prove very important in preventing, mitigating, and rehabilitating the documented epidemic of flight-induced neck strain.

(Adam, 2004). Specifically, this information would provide focus for a physical training program aimed at achieving success in combating flight-induced neck strain with the tactical helicopter community.

Therefore, the purpose was to use EMG in a laboratory setting to evaluate muscle fatigue in the cervical musculature to determine which muscle(s) were most susceptible to fatigue in military helicopter aircrew. Furthermore, this research intended to determine which muscle(s), when fatigued, were the primary influence(s) in limiting force production in cervical spinal movements in military helicopter aircrew. It was hypothesized that key muscles, in particular those primarily responsible for the specific isometric contraction, would display the greatest rate of fatigue as measured with EMG.

METHODS

Subjects. Ethical approval was obtained from the university Review of Ethics Board and subjects were provided verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Forty CF helicopter aircrew (35M, 5 F; Age = 35.3(5.6) yrs; Height = 177.3(8.2)cm; Weight = 82.7(11.9)kg) representing all Canadian tactical helicopter squadrons and levels of experience agreed to participate in the study and signed informed consent (Table 3.1). Subject fitness scores were compiled post hoc from the most recent aircrew fitness result on file. This included a step-test to predict maximal aerobic power (VO_2max).

Characteristic	Mean(SD)
Sex	35 M; 5 F
Handedness	36 R; 4 L
Age (yrs)	35.3 (5.6)
Height (m)	1.77 (0.08)
Weight (kg)	82.7 (11.9)
Resting HR (bpm)	68.0 (8.6)
Resting systolic BP (mmHg)	114.8 (12.5)
Resting diastolic BP (mmHg)	71.8 (9.1)
VO_2Max ($ml \cdot kg^{-1} \cdot min^{-1}$)	40.9 (5.1)
Helicopter Experience (yrs)	4.9 (3.5)
Helicopter Experience (hrs)	1159.7 (945.7)
NVG experience (hrs)	136.1 (120.4)

Table 3.1 – Summary of Subject Characteristics

Electromyography Monitoring. Electrode placement was determined using predetermined measurements based on anatomical landmarks (Seniam, Enschede, the Netherlands). Six EMG channels were collected with a commercially available 8-channel system (Bortec Biomedical Ltd., Calgary, Alberta, Canada). Silver-silver-chloride Red Dot (3M, St. Paul, Minnesota, USA) electrodes were placed in a bipolar arrangement over the right and left splenius capitis (SpC_R & SpC_L), right and left sternocleidomastoid (SCM_R & SCM_L), and right and left upper trapezius (T_R & T_L) muscles with an electrode spacing of 20mm. The placement sites were cleaned with 70% alcohol swab and lightly abraded with fine sandpaper to insure a clean signal. A reference electrode was affixed over the bony protuberance of C7 and signal quality was visually assessed with custom oscilloscope software (US Army Aeromedical Research Laboratory (USAARL), Ft Rucker, Alabama) through the subject's performance of a series of test movements such as neck flexion and extension and shoulder shrugs.

The collected EMG signal was converted to a digital signal through a custom A/D board (USAARL, Ft Rucker, Alabama) and a 16-channel open-ended data acquisition card (Techmatron Instruments Inc., Montreal, Quebec, Canada). Collection frequency was 1024Hz. The digital signal was compiled and stored with custom data acquisition software designed for the testing sequences (USAARL, Ft Rucker, Alabama). Analysis of the EMG data was performed with custom MATLAB (The MathWorks, Natick, Massachusetts, USA) program software (University of New Brunswick, Fredericton, NB, Canada; USAARL, Ft Rucker, Alabama).

For each channel of the EMG signal during each submaximal endurance trial, the median frequency was calculated for every half-second and then normalized. Maximum normalized median frequency values at the start and the end of each trial (Δt=15s) for each muscle were calculated.

Isometric Testing. Isometric testing was performed with subjects seated in a standard CH-146 helicopter cockpit seat with the appropriate safety harness tightened and secured to minimize trunk movements during testing. The description of the custom built isometric testing apparatus supplied by the United States Army Aeromedical Research Laboratory (USAARL) is included in the following section. USAARL researchers have previously used this apparatus and testing methodology in their published research (Fraser, Alem & Chancey, 2006). Calibration of the force transducer and software program was performed using a slope-intercept method with two different calibration weights (11 kg and 55kg) at the start of every data collection trial. A 2" webbing strap was secured around the subject's head and attached to a SSM-AJ-100 force transducer (Interface, Scottsdale, Arizona, USA) that was attached at head level to a 1" square steel pole (Figure 3.1). The pole could be moved to 4 locations around the cockpit seat to allow the subject to perform neck flexion, extension, and left and right lateral flexion contractions (Harrison et al., 2007a).



Figure 3.1. Apparatus for testing isometric cervical flexion

Maximal Voluntary Contractions (MVC). Subjects were instructed to cross their arms on their chest to prevent them from generating additional leverage by grabbing the arm rests of the cockpit chair during the isometric contractions. During the MVC testing sequence, subjects were provided verbal instructions to "gradually ramp" their force up to maximal force production (approximately 2 sec) to avoid an injury of the neck muscles through a "jerking" movement and the large rate of force development and application related to those types of movements. Subjects were provided with a practice trial of the extension contraction as this was deemed to be the movement pattern that was least susceptible to fatigue.

MVC testing order was determined randomly prior to the subject's arrival at the laboratory. Each subject performed three 5s isometric contractions in each direction with a two-minute rest period between contractions. Following each MVC trial, subjects were asked to rate their perceived effort on a modified (0-10) Borg scale (1998), which has been previously used in isometric research (Hunter & Enoka, 2001). The MVC trial of the greatest magnitude was saved as the true MVC score and it was this value that was used to calculate the subsequent 70% MVC submaximal target forces for the other stages of the testing. Borg values during the MVC phase were averaged to provide an aggregate score.

Submaximal Endurance. Subjects performed one submaximal endurance trial for each of the isometric movements and trial order was again the same as the MVC testing. Target force was 70% of the MVC and subjects were instructed to maintain this force for

as long as they could, to a maximum of 180s. Subjects were provided a five-minute rest period between trials. Subjects were again asked to rate their perceived exertion on the Borg scale at the completion of each submaximal isometric endurance contraction.

Data Analysis. Statistical analysis was performed with SPSS V16.0 (SPSS Inc., Chicago, USA). MVC strength values and times to exhaustion used paired t-tests to analyze for differences between paired movement directions (i.e. flexion vs. extension or right flexion vs. left flexion) to identify the presence of any statistically significant differences or imbalances. The normalized median frequency values of each submaximal fatigue trial were analyzed for differences between the starting and end peak values using paired t-tests. All correlations between the paired samples were statistically significant and are reported in conjunction with the p-values and t-scores below. Differences in RPE values were also investigated with a GLM repeated measures ANOVA. Significance was set at $p<0.05$.

RESULTS

Paired t-tests.

Extension. MVC strength for the extension movement was 229.7(50.6)N and the reported RPE was 8.4(1.5) on a scale of 10. Time to exhaustion for the submaximal endurance trial was 55.6(34.5)s and the associated RPE was 9.0(1.2). The SpC_R and SpC_L muscles were the only muscles to exhibit a change in normalized median frequency that

was statistically significant (Left $p<0.01$, $t=6.69$, $R^2=0.58$, $df=33$; Right $p<0.01$, $t=7.74$, $R^2=0.64$, $df=33$) (Figure 3.2). The mean decrease in normalized median frequency in left and right normalized median frequency was 21.2(22.6)% and 22.5(18.1)%, respectively.

Flexion. MVC strength for the flexion movement was 169.6(44.5)N with an RPE value of 8.2(1.3). Time to exhaustion for the submaximal endurance trial was 40.6(15.6)s and the reported RPE was 8.9(1.2). Significant differences in start and end normalized median frequencies for the SCM_R, SCM_L, T_R, and T_L muscles were observed (Figure 3.3). The SCM_R ($p<0.01$, $t=4.00$, $R^2=0.81$, $df=34$) and SCM_L ($p<0.01$, $t=6.40$, $R^2=0.81$, $df=34$) muscles displayed significant decreases in normalized median frequencies (Left 14.8(18.3)%; Right 11.5(16.0)%), indicative of fatigue. The trapezius muscles, T_R ($p<0.01$, $t=-6.82$, $R^2=0.48$, $df=32$) and T_L ($p<0.01$, $t=-6.24$, $R^2=0.52$, $df=33$), displayed increases in normalized median frequencies (T_R 94.1(145.6)%; T_L 119.9(142.6)%), indicative of increased activity as the trial progressed.

Right Flexion. MVC strength for the flexion movement was 176.8(43.9)N with an RPE of 8.4(1.4). Time to exhaustion for the submaximal endurance trial was 62.4(31.2)s and the associated RPE was 8.4(1.4). Only the SCM_R displayed a significant change ($p<0.01$, $t=5.03$, $R^2=0.44$, $df=35$) in normalized median EMG frequency during the submaximal endurance trial (Figure 3.4). The normalized median EMG frequency for the SCM_R decreased (14.7(15.7)%), indicating the onset of fatigue.

Left Flexion. MVC strength for the flexion movement was 186.1(43.3)N and RPE was 8.6(1.4). Time to exhaustion for the submaximal endurance trial was 65.3(33.5)s and subjects reported an RPE of 8.9(1.3). The prime agonists, the SpC_L (7.3(29.5)%) and the SCM_L (11.2(13.2)%), both demonstrated statistically significant decreases in normalized median EMG frequency ($p=0.02$, $t=2.92$, $R^2=0.79$, $df=35$; and $p=0.03$, $t=4.87$, $R^2=0.78$, $df=34$, respectively) (Figure 3.5).

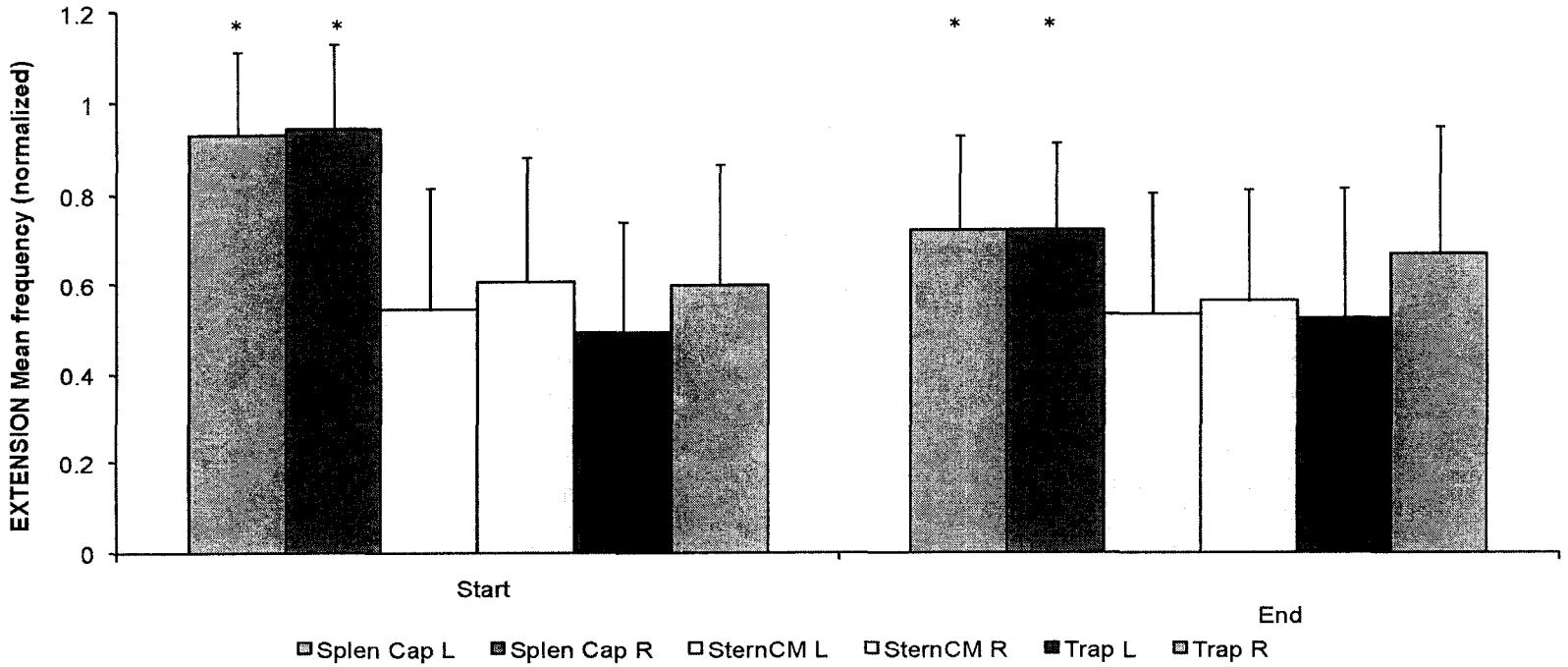


Figure 3.2 – Changes in Normalized Median EMG Frequency of Cervical Muscles During Isometric Extension Submaximal Endurance Trial (* denotes significant differences from baseline; $p<0.05$).

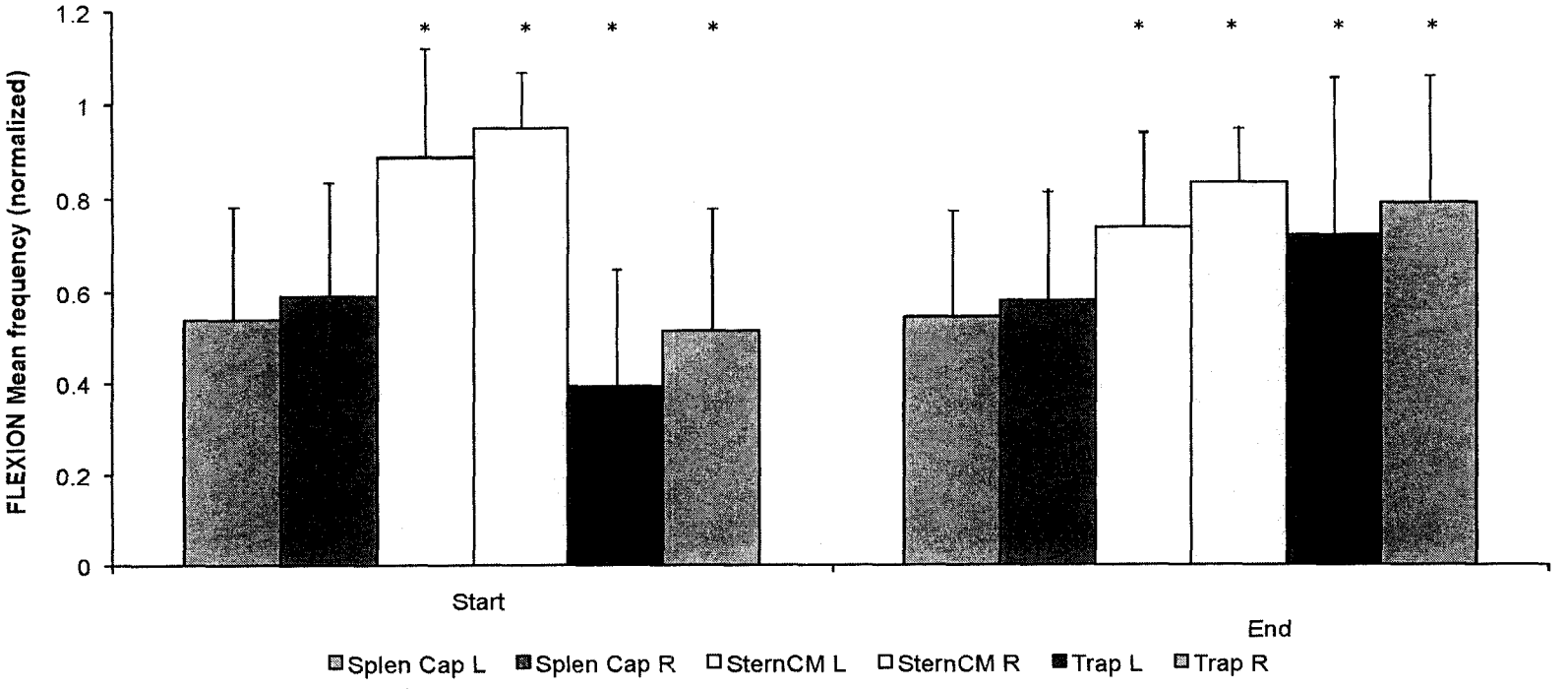


Figure 3.3 – Changes in Normalized Median EMG Frequency of Cervical Muscles During Isometric Flexion Submaximal Endurance Trial (* denotes significant differences from baseline; $p<0.05$).

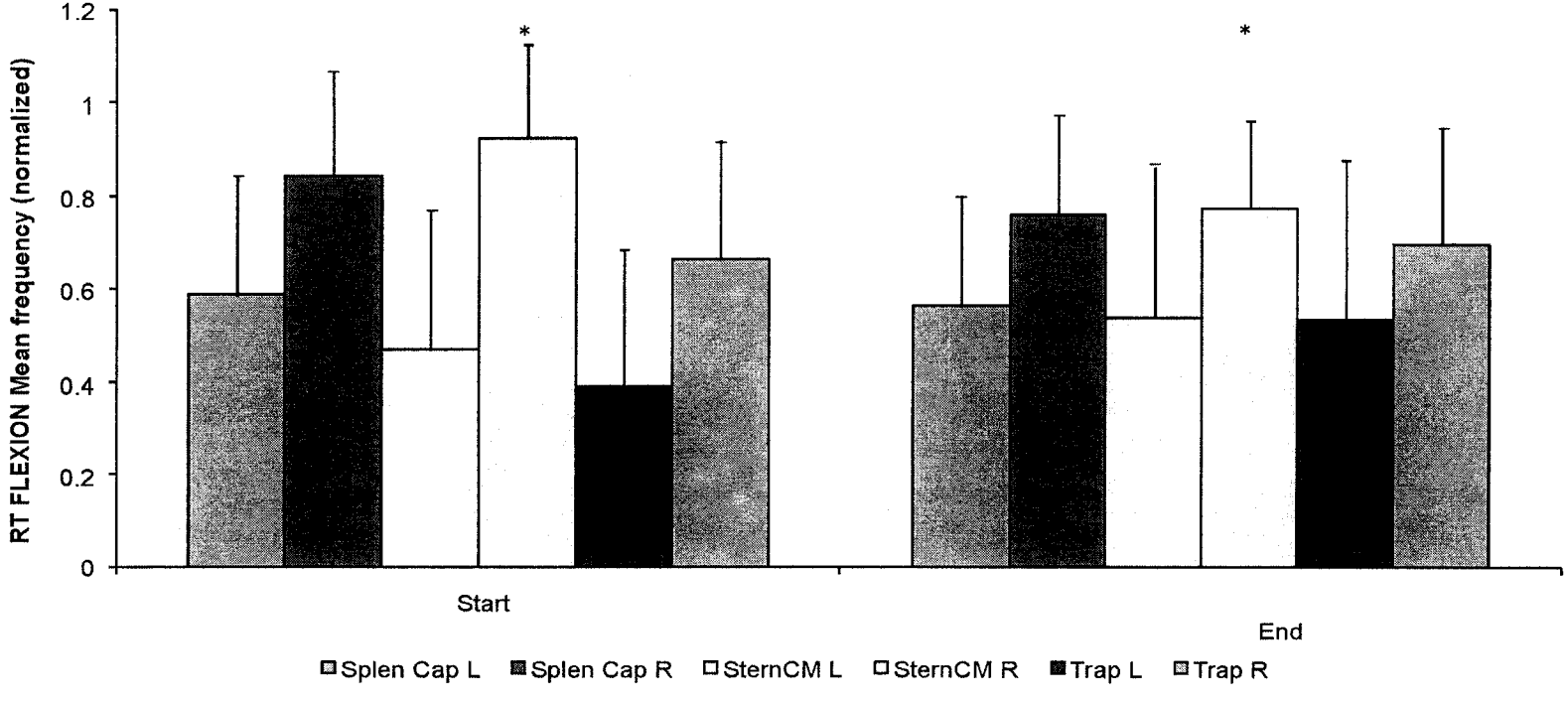


Figure 3.4 – Changes in Normalized Median EMG Frequency of Cervical Muscles During Isometric Right Flexion Submaximal Endurance Trial (* denotes significant differences from baseline; $p<0.05$).

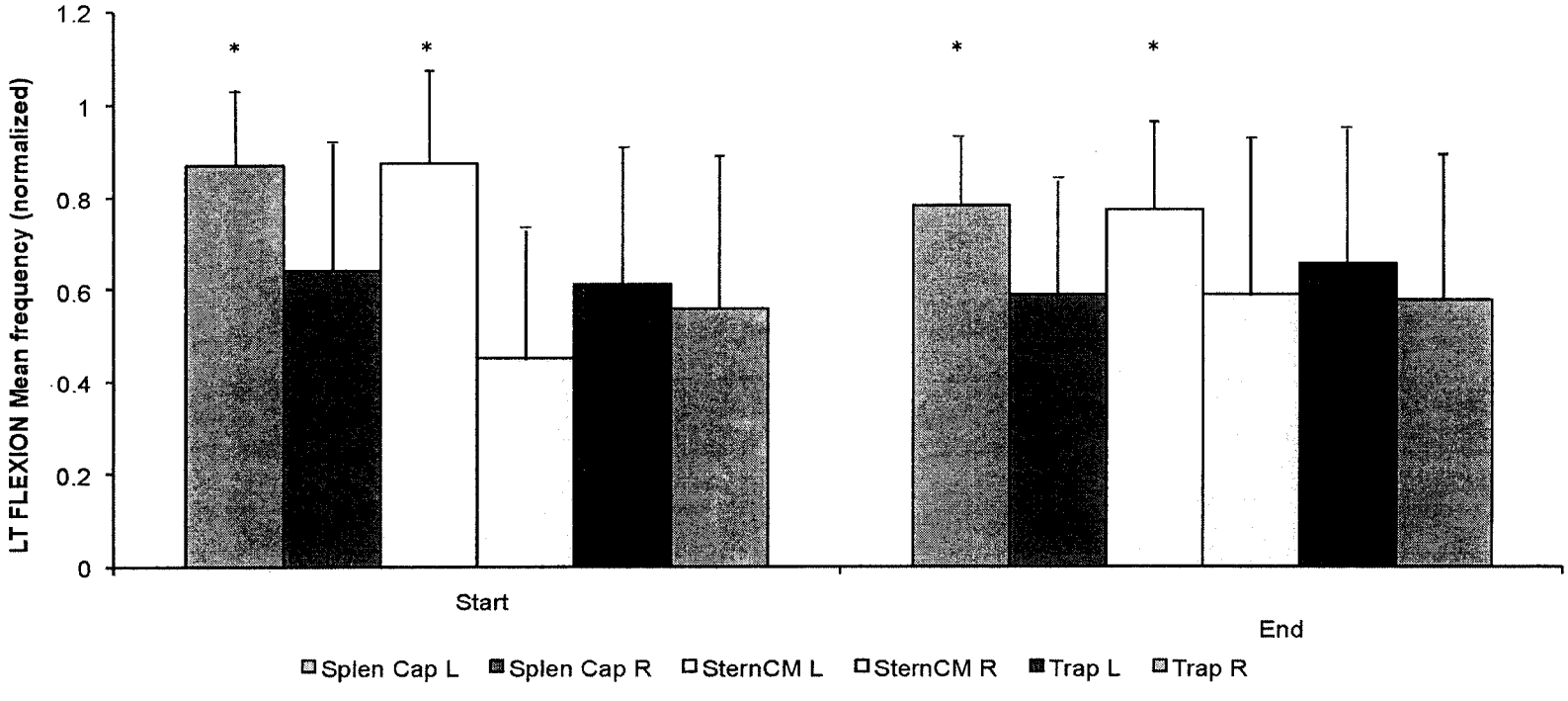


Figure 3.5 – Changes in Normalized Median EMG Frequency of Cervical Muscles During Isometric Left Flexion Submaximal Endurance Trial (* denotes significant differences from baseline; $p<0.05$).

PAIRED T-TESTS BETWEEN ISOMETRIC MOVEMENTS.

Flexion vs. Extension. The extension MVC values were significantly greater than the flexion MVC values for the 40 subjects ($p<0.01$, $t=-11.31$, $R^2=0.751$, $df=38$). Additionally, the time to fatigue for the extension isometric movement was significantly greater than the time to fatigue for the flexion isometric movement ($p<0.01$, $t=-2.83$, $R^2=0.34$, $df=37$). No statistical difference was found in the RPE scores for the MVC isometric movements ($p=0.11$, $t=-1.63$, $R^2=0.88$, $df=37$) or for the submaximal endurance trials ($p=0.65$, $t=-0.46$, $R^2=0.73$, $df=36$).

Right vs. Left Flexion. The left flexion MVC values were significantly greater than the right flexion MVC values for the 40 subjects ($p<0.01$, $t=-2.87$, $R^2=0.89$, $df=38$). However, no statistical difference was found between the times for the submaximal fatigue trials for right and left flexion ($p=0.26$, $t=-1.14$, $R^2=0.89$, $df=37$). Furthermore, no statistical difference was found in the RPE scores for the MVC isometric movements ($p=0.07$, $t=-1.89$, $R^2=0.90$, $df=37$) or for the submaximal endurance trials ($p=0.81$, $t=-0.24$, $R^2=0.85$, $df=36$).

RPE. No statistically significant differences were found between RPE for the 4 isometric MVC trials ($p=0.81$, $F=7.34$, $df=37$) or for the 4 isometric submaximal fatigue trials ($p=0.92$, $F=9.12$, $df=37$). The reported values do indicate a maximal effort for all trials.

DISCUSSION

This study showed that normalized median frequency in an EMG signal, collected during an isometric endurance trial, was an effective method of measuring neuromuscular fatigue in the cervical muscles of military helicopter aircrew. The use of bilateral EMG monitoring of these 6 muscles (splenius capitis, sternocleidomastoid, upper trapezius) during the submaximal fatigue trial has provided insight into which muscles, particularly the smaller agonists, fatigue to a greater degree as compared to the other agonist muscles. This insight will be of benefit to those who are responsible for the design of fitness or rehabilitation programs for those aircrew members who may develop or who have already developed neck pain related to flight.

Surprisingly, based on our previous NIRS neuromuscular fatigue was not observed in the upper trapezius muscle during the submaximal fatigue trials (Harrison et al., 2007a). Alternatively, the trapezius was the largest of the muscles monitored during this protocol and it would seem logical that the smaller muscles (splenius capitis or sternocleidomastoid) would be first to fatigue, particularly during the extension or the lateral flexion trials. The development of fitness training programs aimed at improving the strength and endurance of the cervical muscles in military helicopter aircrew should, according to these results, consider improving the function of the smaller cervical muscles.

Previous studies aiming to improve function of the neck and shoulder among neck pain sufferers, while including neck specific training, have also specifically targeted the trapezius muscles and sometimes other large muscles of the body, and/or the associated

strength capacities of those muscles (Conley et al., 1997a; Conley et al., 1997b; Ylinen et al., 2003; Ylinen et al., 2006). While "exercise interventions are deemed essential for the effective management of patients with neck pain" (Falla, 2004), the current approach aimed at improving strength and inducing hypertrophy may not provide benefits that are transferable to the military aircrew population (although this remains to be tested prospectively with a well-controlled and designed training program; see discussion below). The MVC force results exceeded the force requirements of the cervical musculature dictated by head mounted equipment masses and the forces they produce in a low +Gz operating environment (Weirstra, 2001). Furthermore, at no point in the testing did the trapezius function decrease significantly during any of the submaximal endurance trials. This difference in behaviour in the trapezius muscles as compared to the splenius capitis and sternocleidomastoid muscles is consistent with other findings from isometric testing protocols (Kumar et al. 2003). Therefore, it would seem appropriate to target the muscles that did display fatigue over the course of the submaximal isometric trials as it would appear that the onset of fatigue in these muscles had a negative impact on force production or maintenance during the trial. That is, for the extension movement, both the left and right splenius capitis muscles should be targeted; for the flexion movements, the left and right sternocleidomastoid muscles should be the focus; coincidentally, the targeting of these muscles for both extension and flexion movements will address the needs of the lateral flexion trials.

Our previous work reported in Chapter 2, discussed the upper trapezius muscles as it related to simulated in-flight monitoring of the cervical region of Canadian Forces aircrew. The reasons for this, as discussed in the preceding chapter, were due to practical

considerations. The NIRS, in pilot work, did not provide a clean signal that lent itself to analysis when the probes were applied to the region superficial to the splenius capitis and the erector spinae. As such, this is the first investigation from our laboratory to investigate the effects of reasonably strenuous and sustained exertion on the smaller muscle groups. While our previous work had quantified an increase in the metabolic response of the upper trapezius, a larger muscle group by comparison, this work indicates that this increase may have been accompanied by a decrease in capacity for force production of the smaller muscle groups such as the sternocleidomastoid and the splenius capitis.

As for whether or not strength or endurance capacities of the neck musculature should be the focus of a training intervention for military aircrew, some provisional conclusions can be drawn from these results. Compared to a healthy non-military population, our aircrew population exhibited greater MVC isometric strength results (Seng et al., 2002). Compared to Ang et al. (2005) and their work with military helicopter pilots, reporting both neck pain and healthy data, the MVC results for these aircrew, when units converted to force (N) from work (Nm) by estimation, appear to have been much higher than their reported flexion and extension results. Furthermore, 18 (45%) of the current subjects did report neck pain that was either mild or very mild at the time of testing. These subjects provided average MVC results that were greater than those averages reported by Ang et al. (2005) for either the healthy control or the neck pain group of helicopter pilots. In fact, the single individual which reported the greatest neck pain in our study performed an extension MVC of 297.1N and a flexion MVC of 200.9N, compared to a population average extension of 229.7(50.6)N and a population average

flexion of 169.6(44.5)N. This would suggest that any increase in strength alone of the cervical musculature will be insufficient to fully address the issue of neck pain. McKeon et al. (2006) found differences between time to exhaustion while performing isometric testing of the lumbar region between subjects with pain as compared to subjects without pain (increased performance time, relative to the pain group). Roy et al. (1990) also linked EMG signal and fatigue with the occurrence of pain in the low back of elite rowers. These two references lend further support to the hypothesis of pain reduction in the cervical spinal region through the improvement of the muscular endurance capacity.

Analysis of a military helicopter aircrews' working environment would lend further support to incorporating an element of muscular endurance to any neck related training program aimed at preventing or rehabilitating flight-induced neck pain. As reported by Weirstra (2001), the forces experienced by a helicopter aircrew member are far less in magnitude as compared to the forces produced during the MVC trials. Even when the +Gz forces experienced by military helicopter aircrew are considered, the forces produced during the MVC trials by the present subjects are still much greater in magnitude than those experienced at the neck during flight. Therefore, if any improvement in muscular performance is going to be of benefit to aircrew, this data would suggest that it might be the muscular endurance component.

Only one side difference was found to be significant following the statistical analyses. The MVC result for left lateral flexion was significantly greater than the MVC result for right lateral flexion. The submaximal trial times to exhaustion and the normalized EMG results did not provide any evidence of a left or right deficit. However, the MVC result was used to determine the workload for the submaximal endurance trials.

Therefore, it would appear that this deficit was present in the submaximal endurance trials because the right side did not perform longer during the lateral flexion trials as compared to the left despite having a lower calculated workload (i.e. 70% of the right MVC would be less than 70% of the left MVC).

The greater strength on the left side was a surprise finding. These subjects were primarily right handed. Western society is primarily a right handed society in terms of interacting with technology and everyday items. It would seem logical to expect that the right lateral flexion MVC would be greater than the left lateral flexion MVC. However, this was not the case. The literature, though, does provide a potential explanation. Previous work with spinal musculature EMG in helicopter pilots reported greater rates of activation on the right side of the spine in the lumbar spinal muscles (Lopez-Lopez et al., 2001). Previous work in a flight simulator with military helicopter pilots in which the trapezius muscles were monitored with near infrared spectroscopy (NIRS) also reported greater metabolic activity (i.e., muscle oxygenation and blood volume) on the right side during day missions and night missions in which the added mass of the night vision goggles was present (Harrison et al., 2007b, 2007c, 2007d). Therefore, while it might seem logical to expect a predominantly right handed population of regular members of society to demonstrate greater strength in the musculature of the right side of their body, this may be another factor that does not transfer to a military helicopter aircrew population. The members of the military helicopter aircrew population may be experiencing some level of sustained fatigue in the musculature on the right of the spinal column as a result of their duties and job demands. The author is hopeful these results

may influence the design of appropriate training programs to address this issue in the near future.

CONCLUSION

Normalized median EMG frequency of the cervical neck musculature during submaximal isometric endurance contractions can be used in military helicopter aircrew to identify those muscles that fatigue quickest and effect force production and maintenance. This approach will be beneficial in applying and assessing muscular training programs aimed at preventing and rehabilitating flight-induced neck pain among this population.

ACKNOWLEDGEMENTS

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CHAPTER 4

NECK PAIN AND PHYSIOLOGICAL CAPACITIES IN HELICOPTER AIRCREW

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INSERT: This paper, submitted to Aviation Space & Environmental Medicine represents a detailed description of the capacities and characteristics of current Canadian Forces aircrew populations. This collection of this comprehensive data set was performed to gather the data that would be necessary for the subsequent two chapters in which more advanced statistical methods were applied to determine methods of describing neck pain and differentiating the symptomatic from the asymptomatic aircrew.

INTRODUCTION

The prevalence of neck pain in the adult population is described as being as widespread as 66.7% with nearly 5% of the population reporting disabling neck pain (Cote et al., 1998). The topic of flight-induced neck strain among military aircrew is a topic of interest and major concern in the literature, and much has been written about fixed wing and fast jet aircrew and the incidence of neck strain among them (Ang et al., 2005; Ang et al., 2006). However, differences in +Gz forces between these aircrew suggest that +Gz alone are not responsible for neck pain and disability in rotary wing aircrew community. Recently, research has been conducted to investigate the incidence rates and factors related to flight-induced neck strain among military helicopter aircrew to fill this void (Adam, 2004; Thuresson, 2005).

Reports of injuries among helicopter aircrew vary in the available literature. Thomae et al. (1998) report an incidence rate of 29% in Australian helicopter aircrew but do not provide further details of the injuries. Details of injuries among Canadian Forces (CF) helicopter aircrew are available but do not come from one source. Physiotherapy ergonomic reports from one training squadron indicate that the rear operators, the flight engineers, of the CH-146 Griffon are experiencing neck strain issues as a result of their working environment (Weirstra, 2001). This includes NVG use. Detailed survey reports of the forward operators, the pilots, provide some alarming statistics (Adam, 2004). For example, of the pilots who have more than 150 flight hours logged with NVG, over 90% report neck pain. In total, nearly 80% of respondents had experienced flight-related neck

pain and NVG use was the most common factor attributed to causing this neck pain by respondents (Adam, 2004). Other factors include vibration, the use of other helmet mounted devices, and cumulative flight hours.

In other reports, physiological variables have been obtained as they related to the cervical musculature of helicopter aircrew. In-flight evaluation and laboratory isometric strength testing results with EMG have been used to evaluate aircrew reporting and not reporting neck pain (Ang et al., 2005; ,de Oliveira & Nadal, 2004). NIRS has been used in previous research with pilots in a full-motion flight simulator (Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c). However, to the best of the author's knowledge, no study has yet evaluated the physiological characteristics of the cervical spine musculature with EMG, NIRS, and isometric force capacity collected simultaneously and combined these results with detailed chronicles of flight experience and neck pain reports. Such an approach would provide extremely detailed information of the physiological response in terms of muscle activation and muscle metabolism to force production and maintenance in the sagittal plane, flexion and extension, as well as the frontal plane, left and right flexion of the neck. Through collaborative efforts, the author has designed an assessment protocol for the cervical spine musculature to determine maximal force production as well as the maximum duration to perform an isometric endurance test while simultaneously monitoring the appropriate muscles with EMG and NIRS. This will allow us to identify the muscles most susceptible to fatigue during each isometric contraction. This paper is the first time in which a detailed analysis of questionnaire information as well as physiological capacities has been included for the same subjects in the same publication. The research purposes here are to provide details

of CF helicopter aircrew and to identify differences, if any exist, among different strata of the CF helicopter aircrew population related to neck pain with respect to flight experience, anthropometrics, neuromuscular function, force production or metabolic activity. It was hypothesized that the incidence of neck pain reports would increase as flight experience increased. It was also hypothesized we would see an increase in neck pain reports among FEs as compared to pilots due to the differences in job demands (Weirstra, 2001). Finally, it was hypothesized that differences would exist within the physiological variables as a result of these differences, should any exist.

METHODS

Subjects. Ethical approval was obtained from the university Review of Ethics Board. All subjects were volunteers who provided informed consent after receiving verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Forty aircrew (22 pilots: 18M & 4F; 18 flight engineers: 17M & 1F) volunteers participated in the research.

Protocol. Subjects were provided our customized questionnaire that inquired as to their basic characteristics, their personal flight experience history, fitness history, as well as their neck pain history (Appendix E). Personal characteristic information gathered through the questionnaires included aircrew position, counterweight usage, dominant hand, and anthropometrics such as height and weight. Inquisitions as to the subject's flight experience dealt with fixed wing and rotary wing experience as a function of years

and hours, total NVG experience in hours, average NVG mission length in hours, and longest NVG mission length in hours. Subjects confirmed the accuracy of these results with the information contained within their personal flight log books. Fitness history included the results of their most recent annual military aircrew fitness test and personal fitness training reports (within 12 months). These results were gathered by various fitness testers at various bases using standardized procedures. Summary result reports were gathered at the completion of the study by this research team. Predicted VO_2max results were obtained from a predictive step-test (Canadian Society for Exercise Physiology, 2003). Hand grip results were obtained using a hand grip dynamometer while pushups were performed to exhaustion and sit-ups were performed within a 60s timeframe as per the standards outlined by the Canadian Society for Exercise Physiology (2003).

Electromyography Monitoring. Electrode placement was determined using anatomical landmarks (Seniam, Enschede, Netherlands). Six EMG channels were collected with a commercially available 8-channel system (Bortec Biomedical Ltd., Calgary, Alberta). Silver-silver-chloride Red Dot (3M, London, Ontario) electrodes were placed in a bipolar arrangement over the right and left splenius capitis, right and left sternocleidomastoid, and right and left upper trapezius muscles with an electrode spacing of 20mm. The placement sites were cleaned with 70% alcohol swab and lightly abraded with fine sandpaper to insure a clean signal. A reference electrode was affixed over the bony protuberance of C7 and signal quality was visually assessed with custom oscilloscope software (US Army Aeromedical Research Laboratory (USAARL), Ft

Rucker, Alabama) through the subject's performance of a series of test movements such as neck flexion and extension and shoulder shrugs.

The collected EMG signal was converted to a digital signal through a custom A/D board (USAARL, Ft Rucker, Alabama), and a 16-channel open-ended data acquisition card (Techmatron Instruments Inc., Montreal, Quebec). Collection frequency was 1024Hz. The digital signal was compiled and stored with custom data acquisition software designed for the testing sequences (USAARL, Ft Rucker, Alabama). Analysis of the EMG data was performed with custom MATLAB (The MathWorks, Natick, Massachusetts) program software (University of New Brunswick, Fredericton, New Brunswick; USAARL, Ft Rucker, Alabama).

For each channel of the EMG signal during each submaximal endurance trial, the median frequency was calculated for every half-second and then normalized. Maximum normalized median frequency values at the start and the end of each trial (Δt=15s) for each muscle were calculated. Results are then reported as normalized percent change (Δ%) for the EMG signal from each of the muscles during each isometric submaximal endurance trial.

Near Infrared Spectroscopy Monitoring. The trapezii monitoring was performed bilaterally using the NIRO-300 Spatially Resolved Spectroscopic (SRS) oximeter (Hamamatsu Photonics KK, Hamamatsu City, Japan). NIRS is an optical tool used to measure quantitative changes in muscle oxygenation (HbO_2), deoxygenation (HHb) and blood volume (tHb) during rest and physical activity, and has been used previously in the laboratory to monitor the erector spinae muscles (Albert et al., 2004; McKeon et al.,

2006). The technical description and operation of the NIRO-300 has been described in detail elsewhere (Suzuki et al., 1999), and previous work has demonstrated that NIRS is reliable between testing sessions (Bhambhani et al., 1998). Briefly, the NIRO-300 uses different laser diodes with 4 wavelengths (776, 826, 845, 905nm) as the light source. Utilization of four lasers increases the sensitivity of instrument measurement as more chromophores can be detected in the localized region of light interrogation. The muscles are penetrated with near infrared light (700-1000nm) and the attenuation of this light is measured. The absorbency is calculated using a modified Beer-Lambert law (Delpy & Cope, 1997).

The NIRS probes were placed bilaterally at the C7 level of the upper trapezius. Placement location was selected as a result of pilot work in the simulator (Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c), discussions with CF medical personnel, and previous work in the laboratory with whiplash sufferers (Leavins et al., 2005a, Leavins et al., 2005b). A standardized one minute of baseline data was collected at the start of each testing session with the subject seated in the cockpit seat with their head in neutral position.

The NIRS variables collected were oxyhemoglobin (HbO_2), de-oxyhemoglobin (HHb), total hemoglobin (tHb), cytochrome oxidase-c (CytOx), and tissue oxygenation index (TOI). Results are reported as a percent change ($\Delta\%$) for each of the variables from the baseline measurement at the start of the submaximal endurance trial to the final measurement the coincided with volitional fatigue and the cessation of force production.

Isometric Testing. Isometric testing was performed with a modified version of the methodology developed at USAARL. Subjects were seated in a standard CH-146 cockpit seat with the appropriate 4-point safety harness tightened and secured to minimize trunk movements during testing. A 2-inch webbing strap was secured with Velcro^{TM} around the subject's head and attached to a SSM-AJ-100 force transducer (Interface, Scottsdale, Arizona) that was attached at head level to a 1-inch square steel pole. The pole could be moved to 4 locations around the cockpit seat to allow the subject to perform neck flexion, extension, and left and right lateral flexion contractions.

Maximal Voluntary Contractions (MVC). Subjects were instructed to cross their arms on their chest to prevent them from generating additional leverage by grabbing the arm rests of the cockpit chair during the isometric contractions. During the MVC testing protocol, subjects were provided verbal instructions to "gradually ramp" their force up to maximal force production to avoid an injury of the neck muscles through a "jerking" movement and the large rate of force development/application related to those types of movements. Subjects were provided a familiarization practice trial of the extension contraction as this was deemed to be the movement pattern that was least susceptible to fatigue.

MVC testing order was determined randomly prior to the subject's arrival at the laboratory. Each subject performed three maximal 5s isometric contractions in each direction with a two-minute rest period between contractions. Following each MVC trial, subjects were asked to rate their perceived effort on a modified Borg scale (Borg, 1998). The MVC trial of the greatest magnitude was saved as the true MVC score and it was this

value that was used to calculate the subsequent 70% submaximal target forces for the other stages of the testing. The coefficient of variation for the MVC trials was 7.1%. Borg scale values during the MVC phase were averaged to provide an aggregate score.

Submaximal Endurance. Subjects performed one submaximal endurance trial for each of the isometric (flexion, extension, left and right lateral flexion) movements and trial order was again the same as the MVC testing. Target force was 70% of the MVC and subjects were instructed to maintain this force for as long as they could, to a maximum of 180s. Subjects performed only one endurance trial per isometric contraction and were provided a five-minute rest period between trials. Subjects were again asked to rate their perceived exertion (RPE) on the Borg scale at the completion of each submaximal isometric endurance contraction.

Statistical Analyses. Data was compiled and analyzed with SPSS V16.0 software (SPSS Inc., Chicago, USA). Means and standard deviations of flight experience and neck pain reports are presented as an aircrew population (N=40) and as a function of aircrew position, pilot (N=22) and flight engineer (N=18), and as a function of level of rotary wing flight and NVG experience. One-way ANOVA was used to identify differences between populations in terms of aircrew position or NVG experience. MVC strength values and times to exhaustion used paired t-tests to analyze for differences between paired movement directions (i.e. flexion vs. extension or right flexion vs. left flexion) to identify the presence of any statistically significant differences or imbalances. The normalized median frequency values of each submaximal fatigue trial were also analyzed

for differences between the starting and end peak values using paired t-tests. Differences in RPE values were also investigated with a General Linear Model (GLM) repeated measures ANOVA. Significance was set at $p<0.05$.

RESULTS

Subject Characteristics. Subject characteristics pertaining to flight experience are presented as a population value and as a comparison of pilot and flight engineer population results in Table 4.1. Subject characteristics related to annual aircrew fitness test results are also presented as a population value and as a comparison of pilot and flight engineer population results in Table 4.2. All results are reported as means (standard deviations). No significant differences were noted between the aircrew positions with respect to any fitness test measurements ($p<0.05$).

Characteristic	Units	All Aircrew (N=40)	Pilots (n=22)	Flight Engineers (n=18)
Age	yrs	35.3 (5.6)	33.2 (5.2)*	37.6 (5.2)*
Height	m	1.77 (0.08)	1.77 (0.08)	1.77 (0.09)
Weight	kg	82.7 (11.9)	78.9 (10.7)*	75.3 (12.0)*
Male, Female	M,F	35, 5	18, 4	17, 1
Dominant Hand	R,L	36, 4	20, 2	16, 2
Regular Smokers		5	0*	5*
Total Flight Experience	yrs	6.5 (4.7)	8.5 (6.3)	5.1 (2.7)
Total Flight Experience	hrs	1318.5 (1128.5)	1652.7 (1437.8)	1236.4 (691.8)
Helicopter Flight Experience	yrs	4.5 (3.5)	5.2 (4.3)	4.6 (2.8)
Helicopter Flight Experience	hrs	1047.2 (943.5)	1240.1 (1128.6)	1084.4 (766.4)
NVG Experience	hrs	123.2 (119.1)	108.5 (107.7)	160.3 (128.8)
Average NVG Mission Length	hrs	1.7 (0.8)	1.5 (1.0)	1.8 (0.7)
Longest NVG Mission Length	hrs	3.1 (1.8)	2.6 (1.8)	3.6 (2.0)
Ratio of Helicopter Flight to NVG Flight Hours		10.9 (10.3)	15.2 (13.2)*	6.6 (1.7)*
Regular Counterweight Users		38	20	18
Neck Pain Reporters		21 (53%)	10 (47%)	11 (61%)

* denotes significant difference between Pilots & Flight Engineers

Table 4.1 – Flight Experience Characteristics of All Subjects

Fitness Value	Units	All Aircrew (N=40)	Pilots (n=22)	Flight Engineers (n=18)
Resting HR	bpm	68 (8.6)	69.6 (6.9)	67.6 (9.5)
Systolic BP	mmHg	114.8 (12.5)	109.6 (11.6)	118.8 (12.7)
Dyastolic BP	mmHg	71.8 (9.1)	67.1 (6.6)	74.4 (9.1)
Predicted VO_2	ml/kg/min	40.9 (5.1)	42.9 (6.2)	39.7 (4.2)
Right Hand Grip Strength	kg	25.8 (5.0)	23.0 (3.8)	27.5 (5.0)
Left Hand Grip Strength	kg	25.0 (5.2)	21.2 (3.0)	27.1 (5.0)
Push Ups		23.8 (9.1)	29.2 (9.0)	20.9 (7.6)
Sit Ups		31.0 (10.0)	33.9 (10.1)	29.6 (9.6)

Table 4.2 – Fitness Characteristics of All Subjects

Maximal Voluntary Contractions Isometric Force & RPE. The MVC results are summarized in Table 4.3. As indicated, significant differences were found between paired contractions (i.e. flexion and extension; right and left flexion) for the population as a whole and for the FE subcomponent of the population. The only significant difference between the pilot and FE subpopulations was found for left flexion, with the FE population displaying a greater capacity for force generation in this plane. This also indicated that this FE population displayed a left/right side imbalance with a significantly greater capacity for force generation evident during left flexion. The currently studied pilot population displayed left/right side balance during the lateral flexion MVC protocols. No significant differences were found between isometric contractions, either as a whole sample population or when stratified by aircrew position. No significant differences were observed for MVC results between asymptomatic and symptomatic aircrew. No significant differences were found between RPE values for any of the isometric contractions, either within groups or between groups. Results for the TTE trials are presented in Table 4.4.

MVC Strength	All Aircrew (N=40)	Pilots (n=22)	Flight Engineers (n=18)
Flexion (N)	169.6 (44.5) ¤	157.3 (58.2) ¤	175.7 (41.0) ¤
Extension (N)	229.7 (50.6) ¤	209.7 (67.7) ¤	241.9 (50.0) ¤
Right Flexion (N)	176.8 (43.9) ¥	158.1 (51.6)	190.6 (47.2) ¥
Left Flexion (N)	186.1 (43.3) ¥	165.3 (53.2) *	202.2 (43.7) ¥*
Flexion RPE	8.2 (1.3)	8.5 (1.1)	7.8 (1.5)
Extension RPE	8.4 (1.5)	8.8 (1.4)	7.9 (1.7)
Right Flexion RPE	8.4 (1.4)	8.6 (1.2)	8.0 (1.7)
Left Flexion RPE	8.6 (1.4)	8.7 (1.3)	8.3 (1.6)

¤ denotes significant difference between flexion and extension within population
¥ denotes significant difference between right & left flexion within population
* denotes significant difference between Pilots & Flight Engineers

Table 4.3 – Summary of MVC Force & RPE for Entire Sample, Pilots, and Flight Engineers. Significance was determined as $p<0.05$.

Time To Fatigue (70%MVC)	All Aircrew (N=40) Mean (SD)	Pilots (n=22) Mean (SD)	Flight Engineers (n=18) Mean (SD)
Flexion (s)	40.6 (15.6) ¤	39.1 (17.2) ¤	43.1 (14.0)
Extension (s)	55.6 (34.5) ¤	54.6 (36.5) ¤	57.6 (35.2)
Right Flexion (s)	62.4 (31.2)	59.6 (21.1)	64.4 (40.4)
Left Flexion (s)	65.3 (33.5)	69.8 (40.8)	58.2 (19.8)
Flexion RPE	8.9 (1.2)§	8.9 (1.1) §	8.9 (1.4) §
Extension RPE	8.9 (1.2) §	9.2 (0.9) §	8.6 (1.6) §
Right Flexion RPE	8.8 (1.1) §	8.9 (1.1)	8.8 (1.3) §
Left Flexion RPE	8.9 (1.3) §	9.0 (1.1) §	8.7 (1.6) §

¤ denotes significant difference between flexion and extension within population
¥ denotes significant difference between right & left flexion within population
* denotes significant difference between Pilots & Flight Engineers
§ denotes significant difference between Time To Fatigue RPE and MVC RPE

Table 4.4 – Summary of Time To Exhaustion of Submaximal Endurance Trial & RPE for Entire Sample, Pilots, and Flight Engineers. Significance was determined as $p<0.05$.

Submaximal Endurance Isometric Time to Exhaustion & RPE. The submaximal endurance results are summarized in Table 4.3. No significant differences were found between isometric contractions, either as a whole sample population or when stratified by aircrew position. No significant differences were found between RPE values for any of the isometric contractions, either within groups or between groups. However, when the RPE values for the MVC and submaximal endurance trials were compared, significant differences were found between the RPE associated with the MVC, and the RPE associated with the specific endurance trial for each of the isometric contractions for the population as a whole and for each subpopulation. The only non-significant result for this comparison was found in the pilot subpopulation for the right flexion RPE values.

NIRS & EMG Results During Submaximal Endurance Protocol. No significant differences were found for any of the NIRS and EMG results between the population subgroups, pilots and FEs. Therefore, for ease of presentation, results are presented in the single group format (N=40). NIRS (Fig 4.1 to 4.5) and EMG (Fig 4.6 to 4.9) results are discussed with respect to the each separate isometric submaximal endurance trial.

Flexion (Figure 4.6). The EMG signal for the sternocleidomastoid, bilaterally displayed statistically significant decreases in normalized median frequency (left: -14.8(18.3)%; $p<0.01$, $t=6.40$, $R^2=0.81$, $df=34$; right: -2.5(55.8)%; $p<0.01$, $t=4.00$, $R^2=0.81$, $df=34$). The trapezius EMG signal, bilaterally, increased during the submaximal fatigue trial (left: 119.9(142.6)%; $p<0.01$, $t=-6.24$, $R^2=0.52$, $df=33$; right: 96.8(146.8)%; $p<0.01$, $t=-6.82$,

R^2=0.48, df=32). Coincident with the increase in EMG signal in the left and right trapezius, there was a statistically significant change in some of the NIRS variables. Bilateral changes in HbO_2 (left: -0.2(111.5)%, p<0.01; right: 123.3(322.6)%,p<0.01), in HHb (left: 132.7(672.7)%, p<0.01; right: -139.8(691.8)%; p<0.01) (Fig 4.1), in tHb (left: 5.2(232.8)%, p<0.01; right: 200.3(625.7)%, p<0.01) (Fig 4.3) occurred. A decrease in CytOx was observed in the left trapezius during flexion (-71.2(228.4)%; p<0.01) (Fig 4.5) while the right trapezius displayed a decrease in TOI during the flexion trial (-8.7(11.2)%; p<0.01) (Fig 4.4).

Extension (Figure 4.7). The only statistically significant change to the EMG signals for the extension trial occurred with bilateral decreases in the splenius capitis muscles (left: -21.2(22.6)%; p<0.01, t=6.69, R^2=0.58, df=33; right: -22.5(18.1)%; p<0.01, t=7.74, R^2=0.64, df=33), the smallest of the prime agonists. Additionally, statistically significant changes in HbO_2 (left: 40.2(243.5)%, p<0.01; right: 91.3(302.0)%, p<0.01) (Fig 4.1), tHb (left: 13.9(288.1)%; p<0.01, right: -27.0(250.1)%; p<0.01)(Fig 4.3) were observed bilaterally during extension. There was also a decrease in TOI during the extension isometric contraction (-6.8(16.8)%; p=0.02)(Fig 4.4) in the left trapezius.

Right Flexion (Figure 4.8). Again, fatigue as measured by EMG was observed in the smallest prime agonist. The right sternocleidomastoid muscle displayed a decrease in normalized median frequency (-11.8(23.5)%; p<0.01, t=5.03, R^2=0.44, df=35) over the course of the submaximal endurance trial. Also on the right side, the trapezius displayed an increase in HbO_2 (16.1(251.2)%, p<0.01) (Fig 4.1) and tHb (63.8(190.8)%; p<0.01)

(Fig 4.3), but decrease in HHb (-52.6(362.3)%; p=0.02)(Fig 4.2). However, in the left trapezius, an increase occurred in HHb (12.4(258.5)%; p<0.01)(Fig 4.2) while a decrease occurred for HbO_2 (-6.4(134.2)%; p<0.01)(Fig 4.1) and tHb (-31.0(306.4)%; p<0.01)(Fig 4.3).

Left Flexion (Figure 4.9). For both of the smaller prime movers, the EMG signal displayed decreases in normalized median frequency during the submaximal endurance trial. The left splenius capitis (-7.3(29.5)%; p=0.02, t=2.92, R^2=0.79, df=35) and sternocleidomastoid (-11.2(13.2)%; p=0.03, t=4.87, R^2=0.78, df=34) both displayed signals indicating the occurrence of fatigue. For the NIRS variables in the left trapezius, increases in HbO_2 (17.9(157.6)%, p<0.01) (Fig 4.1), HHb (41.2(843.9)%; p=0.01)(Fig 4.2), and tHb (189.1(649.1)%; p<0.01)(Fig 4.3) were found to occur. For CytOx, decreases were observed in the left (-16.8(725.7)%; p<0.01) and right (-68.2(172.9)%; p=0.05) (Fig 4.5) trapezius during the left flexion trials. In the right trapezius, a decrease in TOI (-8.2(14.6)%; p<0.01) (Fig 4.4) was also observed.

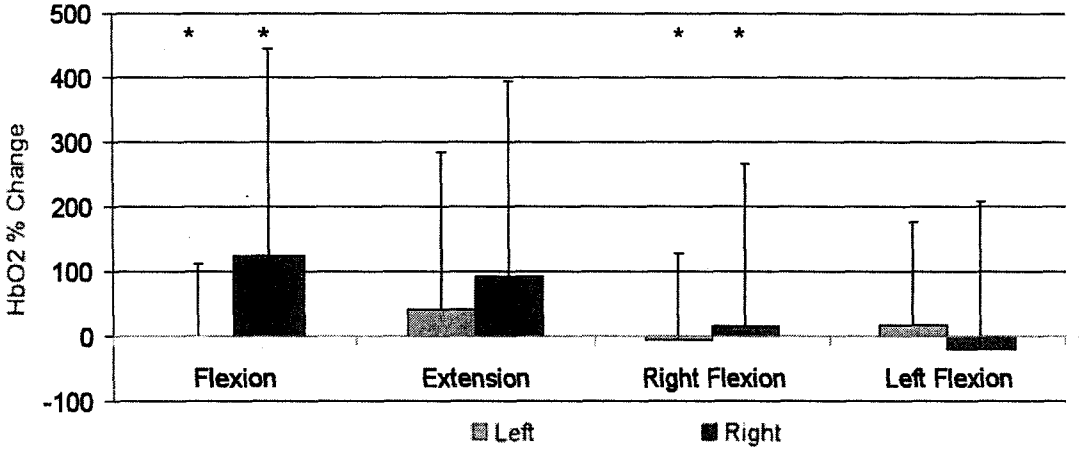


Figure 4.1– Trapezius Muscle Oxyhemoglobin (HbO_2) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Flexion, Extension, and Right and Left Flexion.

*denotes significant differences from baseline ($p<0.05$).

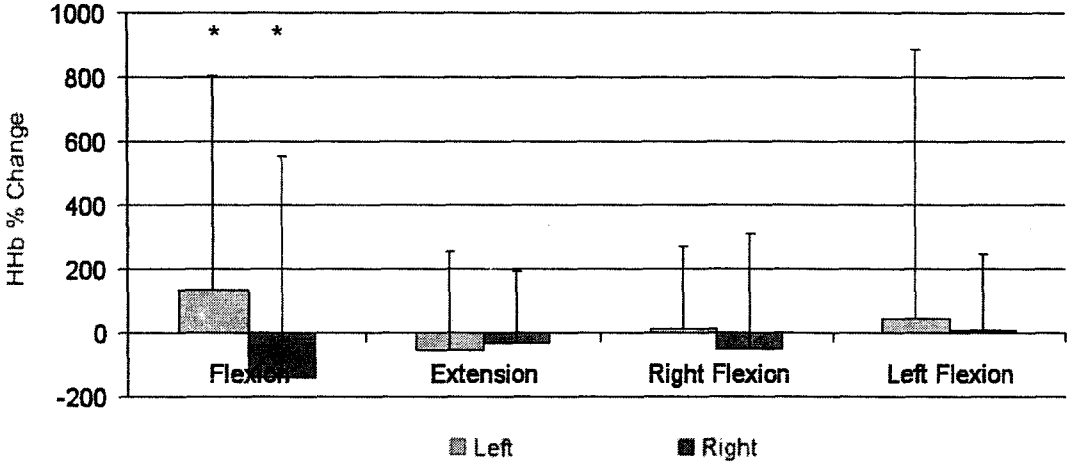


Figure 4.2– Trapezius Muscle Deoxyhemoglobin (HHb) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Flexion, Extension, and Right and Left Flexion.

*denotes significant differences from baseline ($p<0.05$).

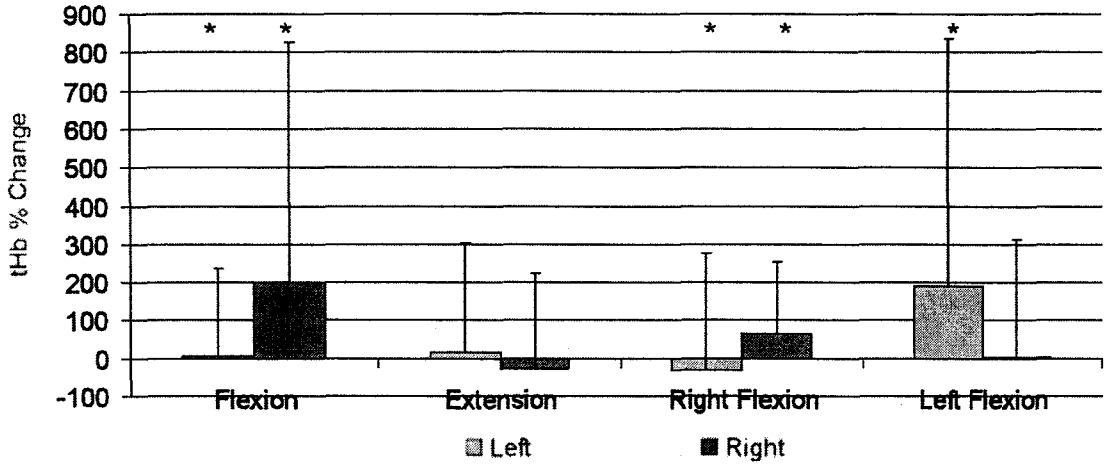


Figure 4.3– Trapezius Muscle Combined Hemoglobin (tHb) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Flexion, Extension, and Right and Left Flexion.

*denotes significant differences from baseline ($p<0.05$).

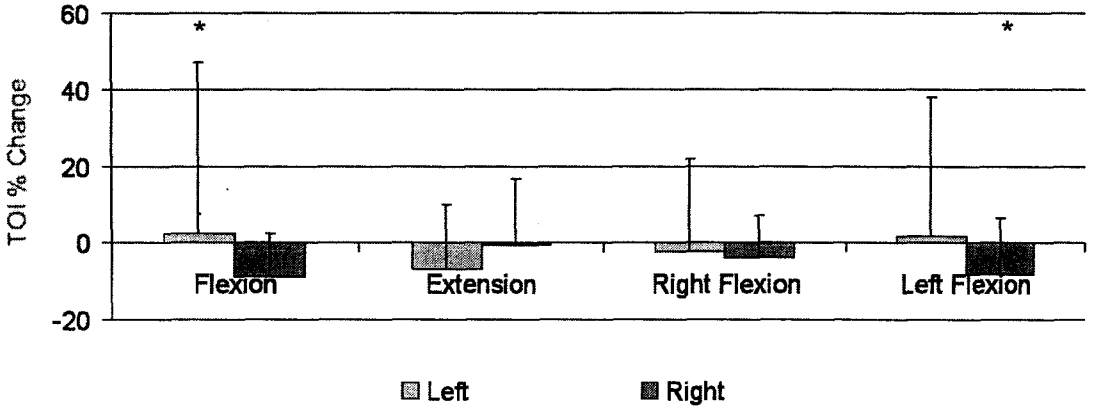


Figure 4.4– Trapezius Muscle Tissue Oxygenation Index (TOI) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Flexion, Extension, and Right and Left Flexion.

*denotes significant differences from baseline ($p<0.05$).

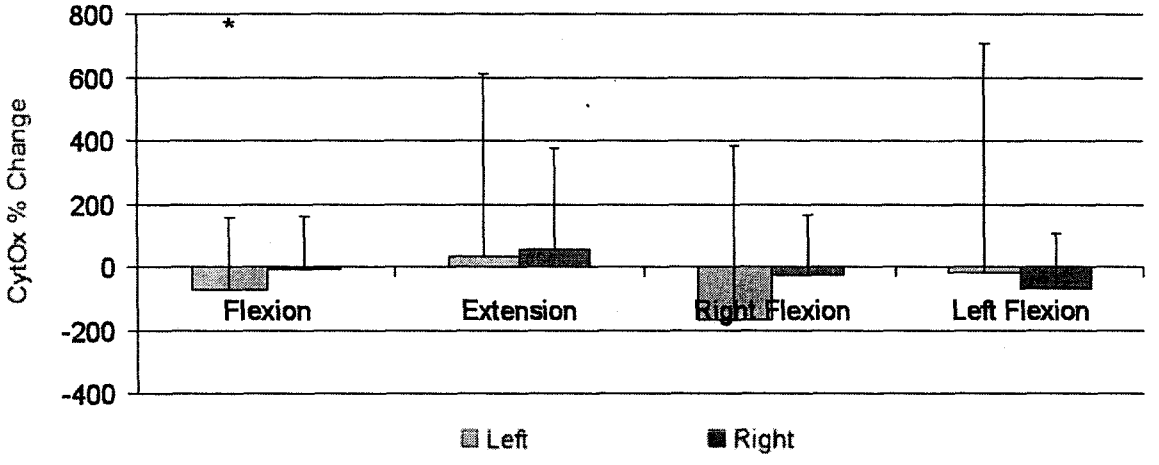


Figure 4.5– Trapezius Muscle Cytochrome Oxidase (CytOx) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Flexion, Extension, and Right and Left Flexion.

* denotes significant differences from baseline ($p<0.05$).

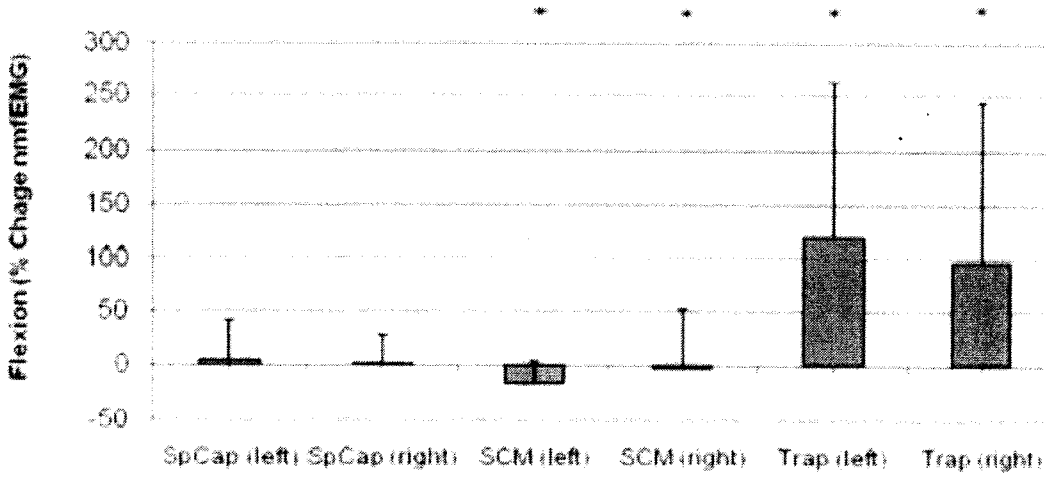


Figure 4.6 – Normalized Median Frequency EMG (nmfEMG) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Flexion.

* denotes significant differences from baseline ($p<0.05$)

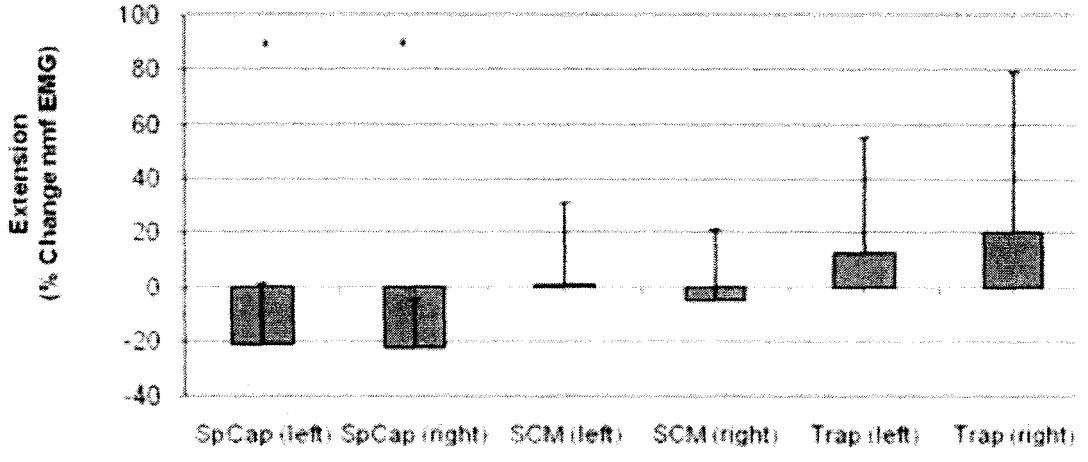


Figure 4.7 – Normalized Median Frequency EMG (nmfEMG) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Extension.

* denotes significant differences from baseline ($p<0.05$)

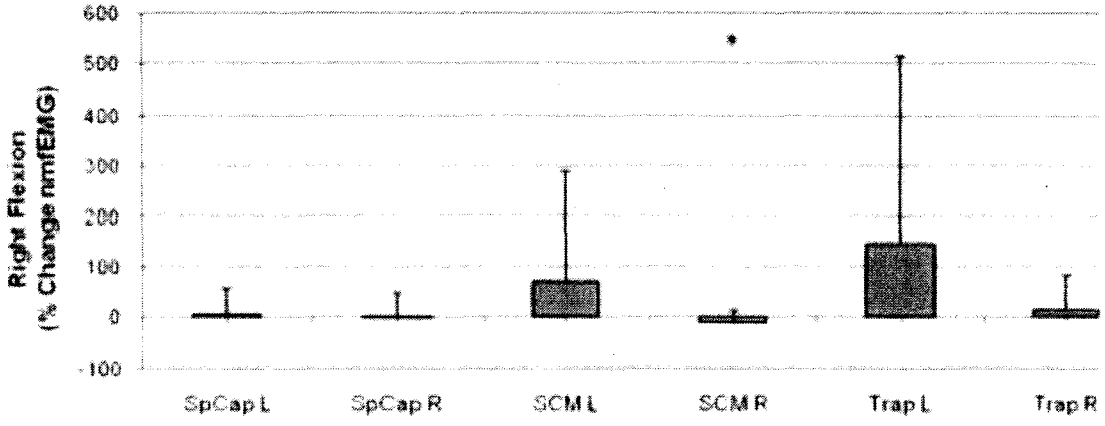


Figure 4.8 – Normalized median frequency EMG (nmfEMG) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Right Flexion.

* denotes significant differences from baseline ($p<0.05$).

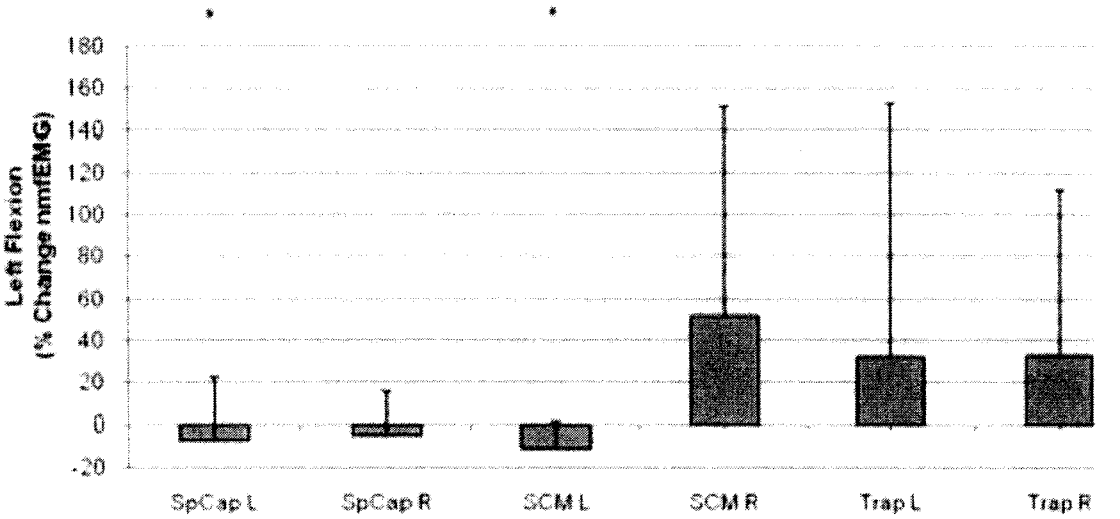


Figure 4.9 – Normalized Median Frequency EMG (nmfEMG) Responses as Percent Change for Sample Population for Submaximal Endurance Trial in Left Flexion.

* denotes significant differences from baseline ($p<0.05$)

DISCUSSION

This study demonstrated a number of novel findings and inclusions not previously observed in my previous NVG research. The subject characteristics including fitness, neck pain, and flight experience of the present study's subjects have been documented and detailed. While these items have been previously reported in the literature, this paper is the first to provide a comprehensive analysis of all these aspects on the same subject pool. Furthermore, this research has included simultaneous EMG and NIRS results during fatiguing submaximal isometric testing of the cervical musculature that also indicated which muscles displayed the greatest changes in median frequency, as determined with EMG, as the onset of fatigue occurred. To the author's knowledge, this is the first study of its kind with respect to the issue of neck strain in helicopter aircrew. This paper presents remarkable similarities between helicopter pilot and flight engineer aircrew with respect to subject characteristics as they relate specifically to neck-strain. Very few differences exist in the presently reported sample population between pilots and flight engineer subjects in any category. Statistical differences were observed for only four of the twenty-four variables assessed. The flight engineer population was older ($p=0.014$), was more likely to smoke tobacco ($p<0.01$), had a smaller body mass ($p=0.04$), and had a greater ratio of logged helicopter flight hours per NVG flight hour logged ($p=0.02$) than the pilot population.

The author believes a further important and novel finding from this data set is the remarkable similarity between pilots and FEs with respect to strength and endurance capacities (i.e. isometric performances) of the cervical spine musculature, including

metabolic (i.e. NIRS) and neuromuscular (i.e. EMG) activity. Some may regard this as a comparison of central and peripheral factors but this will remain a hypothesis while research techniques continue to use the current procedures. The NIRS variables can be considered the measurement of local factors related to muscular metabolism, in this case sustain exertion until volitional fatigue. The EMG variables can be considered a measure of the central drive with respect to force production and maintenance during the sustained exertion until volitional fatigue. EMG measures muscle activity using quantification of the electrical signal. In skeletal muscle during contraction, the magnitude of this activity is determined by metabolites localized in the muscle as well as the neural drive from the motor cortex to the muscle. Although it was not the focus of this study, one way to differentiate between the central and peripheral aspects of the EMG signal and associated capacity for force production is through the use of the interpolated twitch technique (Gandevia, 2001). However, the application of this technique to the musculature of the cervical spine during isometric contractions is not practical due to the extreme risk of severe injury that would be associated with this method in this situation.

The only significant difference between the pilots and flight engineers was found during MVC testing for left flexion (Table 4.3). FEs displayed greater strength as compared to the pilots during this isometric test, but both groups of subjects displayed results that would exceed the demands placed upon the cervical musculature in supporting a 3.7kg flight helmet with NVG and counterweight equipment in the low +Gz (between 1.0 and 2.0Gz) environment of the helicopter cockpit (Hamalainen, 1993). According to the work of Sovelius et al. (2008), a helmet and NVG set-up with a mass of 2.3kg under +Gz loadings of 1.0 to 4.0Gz elicited a muscular response, as measured with

EMG, did not exceed activity that corresponded with 30% of MVC. Other researchers have also demonstrated the impact of both mass and center of gravity has upon the muscular activity of the neck (Gallagher et al., 2007). Those unfamiliar with NIRS research may be concerned with reports of significant differences despite the presence of large standard deviations. These large standard deviations are not unusual in NIRS results (Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c) and the results have been standardized to percentages from the original units. This is to address both individual differences in baseline values that occur at the start of the testing and at the start of each particular trial (flexion, extension, left or right flexion). This study used a randomized trial order and provided rest periods but was not able to fully return the hemodynamic variables to baseline prior to the commencement of each trial. The author feels confident in these results given the research design (randomized trial order, large sample size, periods of rest between trials, etc.) and data analysis techniques (standardization of values for each trial, and statistical methods used).

Left and right side differences are again reported specific to CF helicopter aircrew with respect to physiological measurements. The left flexion MVC results were significantly greater as compared to the right flexion results for all subjects. The left flexion submaximal endurance protocol also elicited a fatigue response as measured with EMG and showed significant decreases in median frequency from both smaller agonists, the splenius capitis (~7%) and the sternocleidomastoid (~11%), as compared to the fatigue response elicited from just the sternocleidomastoid (~12%) during the right flexion trial. Previously published results from applied research in a full motion flight simulator also indicated significantly different metabolic reactions in the right trapezius

of pilots during simulated NVG missions (Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c). Thuresson et al. (2005) reported significantly different neuromuscular responses assessed with EMG during left rotation as compared to right flexion while investigating the effects of using helmets with different masses. Further review of the literature indicated that left and right side differences with respect to muscle activity and metabolism are becoming the norm in helicopter pilots flying tandem seat aircraft (de Oliveira & Nadal, 2004; Lopez-Lopez, 2001). Using a helicopter with a cockpit very similar to our work, Lopez-Lopez (2001) describe side differences in EMG assessment of lumbar erector spinae during flight, and Pelham et al. (2005) provide evidence that postural and ergonomic factors contribute to these results in helicopter pilots.

Given the greater magnitude of the neck-strain issue in the CF that has been reported by others (Adam, 2004; Fraser et al., 2006) as compared to the findings of those performing similar research with the militaries of other nations (Thomae et al., 1998), this finding has provided further evidence that CF neck-strain is being influenced by a factor or factors unique to the CF but common to both CF pilots and FEs. Common factors include duty schedules, operational climate and survival gear, and lifestyle. However, the most obvious common factor to both of these aircrew positions is the airframe itself. Recent work has demonstrated the potentially damaging vibration spectrum of the CH-146 and further work should investigate this matter in greater detail (Chen et al., 2007).

The relationship between muscular fatigue and pain in the associated region has been the focus of a large number of research studies. McKeon et al. (2006) noted increased time to exhaustion in subjects without low back pain performing the Biering-

Sorenson test as compared to age-matched subjects with low back pain. Armstrong (1990) suggests that muscular fatigue and particularly the localized depletion of intracellular Ca^{++} ions may lead to the release of enzymes, such as phospholipase A, that could be harmful to the sarcoplasmic reticulum. The most obvious presenting symptom of these events would be the report of pain in that region. Roy et al. (1990) also linked fatigue, as measured with EMG, to back pain in collegiate rowers and strongly suggested that neuromuscular fatigue measured by EMG could be used as a means of investigating pain in an objectively quantifiable manner.

The discrepancy with respect to age of the two sub-populations is easily explained. In the CF, very few members start immediately as flight engineers. For the most part, individuals "remuster" or retrain to transfer to the position of a flight engineer after some time in the CF. On the other hand, pilot is a direct entry position in which an individual may enter the CF as a pilot without prior experience in another military trade. As a result, an inexperienced flight engineer is likely to be older than an inexperienced pilot. This may introduce a factor that does warrant further investigation as increasing age has consistently been associated with increased rates of sarcopenia; this sarcopenia has also been determined to begin in approximately the 4th and 5th decades of life and continue for the duration of the individual's adult life (Rosenberg, 1997). Future investigators may wish to consider a longitudinal study to investigate this phenomenon as it relates to neck pain in aircrew over the duration of their flying career.

However, these few differences did not influence the results with respect to the research hypothesis. No differences were found between pilots and flight engineers for any of the fitness results, for any of the raw experience variables, or for number of

aircrew reporting neck pain (p=0.423). This last result is promising in terms of designing fitness regimens to address prevention and mitigation of flight-induced neck strain within the CF's tactical helicopter community. While job descriptions and duties vary greatly between pilots and flight engineers (Weirstra, 2001), the prevalence of neck pain does not. It would seem logical then to hypothesize that a common cause rather than specific job duties and working environment is the primary cause of flight induced neck pain. Some potential common causes may include type of aircraft used by the CF, type of NVG equipment used by the CF, number of hours logged in helicopter flight and/or with NVG's, some physiological trait of the musculature of the cervical spine, or a general fitness result.

This study dichotomized the neck pain results from this data set to address whether or not the subject was a sufferer. The present research did not evaluate the subjective results of neck pain severity within the population for two reasons. First, we used the dichotomized approach due to concerns that inter-subject differences in describing the neck pain were present. Pain thresholds differ from person to person. Second, and related to the first reason, it was felt that an issue of "under-reporting" the severity of the neck pain may have been present. While subjects were comfortable in identifying themselves as a neck pain sufferer, some were hesitant to describe the precise severity of their discomfort due to fears that the results may end up in the possession of a flight surgeon and lead to their temporary or permanent grounding from flight duties. For this study's purposes, the dichotomized results are sufficient.

The use of dichotomized pain results with respect the helicopter aircrew of the Canadian Forces has been reported previously. Fraser (2007) has previously analyzed a

number of variables with respect to flight experience in years and hours, flight crew position, helmet type, neck circumference and occurrence of neck pain. In all analyses with respect to pain, whether it was "pain during flight", "pain after flight" or the more general "pain related to flight", a simple YES or NO was used. These results, using a logarithmic odds calculation, were compared to similar results from the United States Army. Fraser (2007) concluded that the Canadian Forces' aircrew are at an increased risk of neck injury.

Thuresson (2005) presents data in which pain-free helicopter pilots were used as subjects for a series of studies but does not present characteristics with respect to anthropometrics, flight experience, or fitness. The results with respect to age, height and weight of helicopter aircrew were similar to the results presented by other studies (Ang et al., 2005; Ang et al., 2006; Landau et al., 2006). However, the helicopter subjects used by all of these studies were found to have more total helicopter flight experience. Interstingly, Landau et al. (2006) did not receive a single report of neck pain from their helicopter pilots despite MRI evidence of cervical disc degeneration in 50% of these subjects. This wide range of pain reports strongly suggests that obtaining accurate pain reports, both in aircrew members accurately identifying themselves as a sufferer and in accurately reporting the severity of the pain, remains a challenge for military medical specialists as well as researchers in this field. The present study did not have access to an MRI but based on the results of Landau et al. (2006), it may be helpful in future studies to correlate with the subjective nature of questionnaire data.

Comparison of these results with the most detailed investigation of CF helicopter aircrew presently available indicates that this sample population of 40 aircrew members

is very similar to the much larger sample population of 196 aircrew presented by Adam (2004). This included measurements of age, height, weight, flight experience, and NVG experience. Adam's (2004) survey indicated that nearly two-thirds of pilots and FEs identified NVG missions as the cause of their neck pain. Adam (2004) noted that nearly 80% of the pilots and FEs in his sample reported neck pain, as compared to the 53% reported by this study. While this may seem like a large discrepancy, two factors can quite easily explain this difference.

First, the present research design included questionnaire results coupled with participation in isometric testing of the musculature of the cervical spine. Because of the nature of this study, the subjects were known to the researcher and the questionnaire results could be matched to physiological data. While subject confidentiality was maintained in this research study, subject anonymity with respect to the research team was not possible. In the case of Adam (2004), large groups completed and returned their surveys en masse and as such subject confidentiality and anonymity were aspects of the study. As a result, subjects were likely to be more honest in reporting and describing their symptoms without fear of being removed from flight status by the researcher.

Second, potential volunteer subjects were given detailed descriptions of the requirements of the testing session. This included descriptions of isometric maximal voluntary contraction testing of the cervical spinal musculature. While data was collected from some aircrew members who were reporting neck pain of varying levels of severity, it is likely that some neck pain sufferers self-selected to exclude themselves from participation by not volunteering. Therefore, it is likely that had these individuals

provided questionnaire data, the neck pain results reported here would be higher and therefore more similar to those reported from CF subjects by Adam (2004).

The final major difference between this study and those currently available in the literature is the inclusion of the results from the most recent aircrew fitness tests of each of these subjects. Very little has been published to date with respect to physical fitness of helicopter aircrew members and fitness testing results with quantification of aerobic capacity or other fitness results and neck pain prevalence is not readily available in the literature beyond the present study. Compared to the available Canadian population normative data tables, these subjects appear to be, at the very least, in reasonable physical condition (Canadian Society of Exercise Physiology, 2003). Considering that these aircrew results are conservatively similar to other CF reports with respect to neck pain reports (Adam, 2004), it would appear that the CF has a unique magnitude with respect to the neck-strain issue.

CONCLUSION

This study did not find statistically significant differences between flight engineers and pilots with respect to reports of neck pain, flight experience or physical fitness results. Furthermore, this research demonstrates that the metabolic and neuromuscular characteristics specific to the musculature of the cervical spine is also statistically similar between CF pilot and flight engineer aircrew members. This strongly suggests that neck pain among CF helicopter aircrew is caused by a common factor among the front seat and back seat aircrew that is yet to be identified. Regardless, the CF

has a concern with respect to flight-induced neck strain among helicopter aircrew that presents a uniquely high proportion of injured aircrew.

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Chapter 5

A Predictive Logistic Regression Equation for Neck Pain in Helicopter Aircrew

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INSERT: This paper, submitted to Aviation Space & Environmental Medicine represents the application of the data presented in the previous paper towards one of the end goals of this thesis – predicting neck pain as a function of the flight history, fitness results, and physiological responses to standardized testing procedures. As this chapter describes, the results produced an extremely robust logistic regression equation that can be used to predict the likelihood of an aircrew member to report neck pain.

INTRODUCTION

This is the follow-up to the previous paper in which results were presented from questionnaire data and physiological assessment of Canadian Forces (CF) helicopter aircrew as they relate to the issue of neck strain (Harrison et al., Submitted). Previous papers have discussed in detail the results from the published literature on the topic of neck strain related to military flight duties, specifically in rotary wing aircrew (Harrison et al., Submitted). Previously reported results have shown that there are very few differences between pilot and flight engineer members of the CF tactical (CH-146

Griffon) helicopter community in regard to physiological capacities and responses as well as physical anthropometrics, flight and NVG experience histories, and neck pain reports. Therefore, it seems the causative factors are more general in nature rather than specific to a certain aircrew position. However, identifying the factors that distinguish those that suffer from neck pain from those that do not report neck pain will require a focused statistical approach with the rigorous analysis.

Neck strain related to the duties of military helicopter flight continues to become well-documented in the available literature. Multiple efforts have been made to characterize this neck strain in physiological (Ang et al., 2006; Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c; Thuresson, 2005) and epidemiological (Adam, 2004; Ang et al., 2006; Fraser et al., 2006; Thomae et al., 1998; Wickes et al., 2005) papers. A rather detailed summary of the job demands of the flight engineer aircrew in the CF has already been presented as it relates to potential causes of neck injury (Weirstra, 2001). However, there has been little cross-over of investigative disciplines within the same study.

The epidemiological results indicate CF aircrew is at an increased risk for injury as compared to their counterparts in the militaries around the world (Fraser et al., 2007; Thomae et al., 1998). Adam (2004) found a threshold value of 150hrs of NVG flight time after which 90% of CF helicopter aircrew begin to report neck pain. Other reports (Weirstra, 2001; Wickes et al., 2005) identify NVG as a perceived cause of neck strain among aircrew. Nearly two-thirds of their subjects identified regular helicopter flight profile with NVG equipment as the cause of their discomfort and NVG usage has been labeled as a causative factor in numerous other neck strain research endeavours (Butler,

1992; Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c; Weirstra, 2001; Wickes et al., 2005).

Evaluation of the physiological characteristics and capacities of the muscles of the cervical spine has been conducted with electromyography (EMG) (Ang et al., 2005; Thuresson, 2005), isometric strength test results (Ang et al., 2006) and with near infrared spectroscopy (NIRS) (Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c). While some studies have used two of these methods of simultaneously evaluating the cervical musculature (Ang et al., 2006; Gallagher et al., 2007; Harrison et al., submitted), linking all three evaluation tools simultaneously has not yet occurred.

In the case of this data set that was the basis for the first paper in this series, the author has collected a great deal of quantitative information using numerous research tools and methods. To further analyze this wealth of information to obtain the factors that differ statistically between the healthy and injured aircrew populations, logistic regression analysis was used.

In this data set, the author has collected information related to the muscular strength and endurance in multiple anatomical planes and monitored the cervical muscles responsible for the movements with EMG and NIRS. Second, this study also performed the evaluation of the flight experience history and self-reported history of neck pain of these subjects while collecting important anthropometric measures. This resulted in a detailed summary of each of the subjects in the databank.

It was hypothesized that logistic regression analysis, using dichotomized pain results, would highlight the statistically significant variables within this expansive data set that differentiated the symptomatic from the asymptomatic aircrew (Harrison et al.,

submitted). We hypothesized that this would allow us to identify an EMG and/or NIRS variable or trend that would differentiate the aircrew with pain from their asymptomatic counterparts. The intent was to allow the results of the logistic regression drive our further work in the area of prevention and rehabilitation through prescribed and supervised physical fitness regimes, using this finding.

METHODS

Subjects. Ethical approval was obtained from the university Review of Ethics Board, and subjects were provided verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Forty aircrew (22 pilots: 18M & 4F; 18 flight engineers: 17M & 1F) subjects representing all Canadian tactical helicopter squadrons and levels of experience volunteered to participate in the study and signed informed consent. Details of subject characteristics can be found in the previous chapter (Harrison et al., submitted).

Protocol. Subjects were provided our customized questionnaire that inquired as to their basic characteristics, their personal flight experience history, fitness history, as well as their neck pain history (Appendix E). Personal characteristic information gathered through the questionnaires included aircrew position, counterweight usage, dominant hand, and anthropometrics such as height and weight. Inquisitions as to the subject's flight experience dealt with fixed wing and rotary wing experience as a function of years and hours, total NVG experience in hours, average NVG mission length in hours, and longest NVG mission length in hours. Subjects confirmed the accuracy of these results

with the information contained within their personal flight log books. Fitness history included the results of their most recent military aircrew fitness test and personal fitness training reports (within the previous 12 months). These results were gathered by various fitness testers at various bases using standardized protocols and procedures adopted by the Canadian Forces. Summary result reports were gathered at the completion of the study by this research team. VO_2max results were obtained from a predictive step-test (Canadian Society for Exercise Physiology, 2003). Hand grip results were obtained using a hand grip dynamometer while pushups were performed to exhaustion and sit-ups were performed within a 60s timeframe as per the standards outlined by the Canadian Society of Exercise Physiology (2003).

Electromyography Monitoring. Electrode placement was determined using anatomical landmarks (Seniam, Enschede, the Netherlands). Six EMG channels were collected with a commercially available 8-channel system (Bortec Biomedical Ltd., Calgary, Alberta). Silver-silver-chloride Red Dot (3M, London, Ontario) electrodes were placed in a bipolar arrangement over the right and left splenius capitis, right and left sternocleidomastoid, and right and left upper trapezius muscles with an electrode spacing of 20mm (21). The placement sites were cleaned with 70% alcohol swab and lightly abraded with fine sandpaper to insure a clean signal. A reference electrode was affixed over the bony protuberance of C7 and signal quality was visually assessed with custom oscilloscope software (US Army Aeromedical Research Laboratory (USAARL), Ft Rucker, Alabama) through the subject's performance of a series of test movements such as neck flexion and extension and shoulder shrugs.

The collected EMG signal was converted to a digital signal through a custom A/D board (USAARL, Ft Rucker, Alabama), and a 16-channel open-ended data acquisition card (Techmatron Instruments Inc., Montreal, Quebec). Collection frequency was 1024Hz. The digital signal was compiled and stored with custom data acquisition software designed for the testing sequences (USAARL, Ft Rucker, Alabama). Analysis of the EMG data was performed with custom MATLAB (The MathWorks, Natick, Massachusetts) program software (University of New Brunswick, Fredericton, New Brunswick; USAARL, Ft Rucker, Alabama).

For each channel of the EMG signal during each submaximal endurance trial, the median frequency was calculated for every half-second and then normalized. Maximum normalized median frequency values at the start and the end of each trial (Δt=15s) for each muscle were calculated. Results are then reported as normalized percent change (Δ%) for the EMG signal from each of the muscles during each isometric submaximal endurance trial.

Near Infrared Spectroscopy Monitoring. The trapezius muscle monitoring was performed bilaterally using the NIRO-300 Spatially Resolved Spectroscopic (SRS) oximeter (Hamamatsu Photonics KK, Hamamatsu City, Japan). NIRS is an optical tool used to measure quantitative changes in muscle oxygenation (HbO_2), deoxygenation (HHb) and blood volume (tHb) during rest and physical activity, and has been used previously in the laboratory to monitor the erector spinae muscles (Albert et al., 2004; McKeon et al., 2006). The technical description and operation of the NIRO-300 has been described in detail elsewhere (Suzuki et al., 1999), and previous work has demonstrated

that NIRS is reliable between testing sessions (Bhambhani et al., 1998). Briefly, the NIRO-300 uses different laser diodes with 4 wavelengths (776, 826, 845, 905nm) as the light source. Utilization of four lasers increases the sensitivity of instrument measurement as more chromophores can be detected in the localized region of light interrogation. The muscles are penetrated with near infrared light (700-1000nm) and the attenuation of this light is measured. The absorbency is calculated using a modified Beer-Lambert law.

The NIRS probes were placed bilaterally at the C7 level of the upper trapezius. Placement location was selected as a result of pilot work in the simulator (Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c), discussions with CF medical personnel, and previous work in the laboratory with whiplash sufferers (Leavins et al., 2005a; Leavins et al., 2005b). A standardized one minute of baseline data was collected at the start of each testing session with the subject seated in the cockpit seat with their head in neutral position.

The NIRS variables collected were oxyhemoglobin (HbO_2), de-oxyhemoglobin (HHb), total hemoglobin (tHb), cytochrome oxidase-c (CytOx), and tissue oxygenation index (TOI %; a measure of saturation). Results are reported as a percent change ($\Delta\%$) for each of the variables from the baseline measurement at the start of the submaximal endurance trial to the final measurement the coincided with volitional fatigue and the cessation of force production.

Isometric Testing. Isometric testing was performed with a modified version of the methodology developed at USAARL. Subjects were seated in a standard CH-146 cockpit seat with the appropriate 4-point safety harness tightened and secured to minimize trunk

movements during testing. A 2-inch webbing strap was secured with Velcro™ around the subject's head and attached to a SSM-AJ-100 force transducer (Interface, Scottsdale, Arizona) that was attached at head level to a 1-inch square steel pole. The pole could be moved to 4 locations around the cockpit seat to allow the subject to perform neck flexion, extension, and left and right lateral flexion contractions.

Maximal Voluntary Contractions (MVC). Subjects were instructed to cross their arms on their chest to prevent them from generating additional leverage by grabbing the arm rests of the cockpit chair during the isometric contractions. During the MVC testing protocol, subjects were provided verbal instructions to "gradually ramp" their force up to maximal force production to avoid an injury of the neck muscles through a "jerking" movement and the large rate of force development/application related to those types of movements. Subjects were provided a familiarization practice trial of the extension contraction as this was deemed to be the movement pattern that was least susceptible to fatigue.

MVC testing order was determined randomly prior to the subject's arrival at the laboratory. Each subject performed three maximal 5s isometric contractions in each direction with a two-minute rest period between contractions. Following each MVC trial, subjects were asked to rate their perceived effort on a modified Borg scale (Borg, 2003). The MVC trial of the greatest magnitude was saved as the true MVC score and it was this value that was used to calculate the subsequent 70% submaximal target forces for the other stages of the testing. The coefficient of variation for the MVC trials was 7.1%. Borg values during the MVC phase were averaged to provide an aggregate score.

Submaximal Endurance. Subjects performed one submaximal endurance trial for each of the isometric (flexion, extension, left and right lateral flexion) movements and trial order was again the same as the MVC testing. Target force was 70% of the MVC and subjects were instructed to maintain this force for as long as they could, to a maximum of 180s. Subjects performed only one endurance trial per isometric contraction and were provided a five-minute rest period between trials. Subjects were again asked to rate their perceived exertion (RPE) on the Borg scale at the completion of each submaximal isometric endurance contraction.

Statistical Analyses. Details of the statistical analyses of the variables collected in this series of studies can be found in preceding publications (Harrison et al., submitted). Logisitic regression analysis was performed with SPSS V16.0 (SPSS Inc., Chicago, USA). Pain reports were dichotomized into aircrew reporting pain (P) and aircrew reporting no pain (NP). The results for all collected with questionnaires (anthropometrics, flight experience, neck pain history) and the isometric testing (MVC, EMG, NIRS) were stratified into quartiles (Q_1: 0-24%; Q_2: 25-49%; Q_3: 50-74%; Q_4:75-100%). These quartile variables and dichotomized variables were used in the initial analysis, one variable at a time, using logistic regression to identify relationships between response results from related research (Harrison et al., submitted) results. As per the rules governing logistic regression, any empty cells contained within the dataset were omitted for that particular regression analysis. Significance was set at $p<0.05$. Upon determining which of the variables were significantly different between the symptomatic and

asymptomatic groups, the raw values for those significant variables were analyzed to produce a logistic regression equation that contained as few variables as possible while explaining the greatest proportion of cases of reported neck pain.

RESULTS

Anthropometric, Fitness & Flight History Results. Only height was found to be statistically different between the symptomatic (1.74(0.07)m) and asymptomatic (1.81(0.04)m) groups ($p=0.04$). Using our logistic regression equation, age, weight, handedness, tobacco use and sex were not found to be statistically linked with the occurrence of neck pain among the aircrew. Additionally, the majority of the fitness measurements did not provide any statistically significant insight into predicting neck pain with logistic regression. The only significant difference between the neck pain group and the non-pain group occurred with systolic blood pressure. The neck pain group displayed a lower systolic blood pressure (106.8(11.5) mmHg) as compared to the asymptomatic group (118(11.0) mmHg) ($p<0.01$).

Flight history provided more significant findings where greater total flight experience (P=1507.5(1121.4)hrs; NP=915.0(1094.2)hrs; $p<0.01$), helicopter flight experience (P=1234.1(857.6)hrs; NP=766.8(1027.4); $p=0.01$), NVG experience (P= 149.3(127.4)hrs; NP= 82.3(94.8)hrs; $p=0.03$) and a history of longer NVG mission experience (P= 3.4(1.9)hrs; NP= 2.5(1.7)hrs; $p=0.01$) were all associated with increased incidence of reported neck pain. Greater length of average NVG mission was

approaching significance in terms of displaying a relationship with neck pain reports (P=1.8(0.8)hrs; NP= 1.6(0.9)hrs; p=0.05) as assessed by logistic regression.

Isometric Strength & Endurance Results. No significant differences were found with the logistic regression analysis between any of the isometric maximal voluntary contraction (MVC) or isometric submaximal endurance trial results with respect to resultant force, time-to-exhaustion (TTE), or rating of perceived exertion (RPE) in any plane (e.g. flexion, extension, right and left lateral flexion) as pertaining to neck pain.

EMG Results. Two of the normalized median frequency changes during the isometric submaximal endurance protocol as assessed with EMG returned significant results as a function of Pain and No Pain groups with logistic regression. During the extension trial, the right splenius capitis displayed a lesser decrease in normalized median frequency (p=0.04) for P (-20.8(11.6)%) as compared to the NP (-27.1(23.8)%) group. During the left flexion fatigue trial, the right sternocleidomastoid displayed an increase in activation as assessed with normalized median EMG frequency (p=0.035) in the P (46.3(51.1)%) group as compared to the minute decrease in activation displayed by the NP (-3.1(33.7)%) group.

NIRS Results. During the extension fatigue trial, two significant differences were found in the right trapezius with NIRS. For the CytOx-c response, NP displayed an increase (p=0.01) in the redox state (161.8(311.6)%) as compared to P (-28.7(353.7)%). For tHb, the P group experienced a 30.5(270.0)% increase as compared to a

-118.4(242.0)% change in the NP group (p=0.03). During right flexion, further differences in the right trapezius (HbO_2: P= -3.5(32.5)%; NP= 3.6(8.2)%; p<0.01) and left trapezius (TOI: P= -5.9(108.7)%; NP= 119.7(185.6)%; p=0.02) were evident. During left flexion, group differences in the right trapezius (HbO_2: P= -80.5(225.9)%; NP= 64.8(245.0)%; p=0.02) occurred. For the flexion trial, differences in HbO_2 (P= -26.8(30.2)%; NP= 47.4(45.9)%; p=0.03) were observed in the left trapezius muscle. Graphical views of these findings are available in Figure 4.1 and Figure 4.2 of this dissertation.

Logistic Regression Equation. Using the variables deemed significant through the preliminary analysis, a logistic regression equation was generated that contained as few variables as possible while still being extremely predictive. Of the 15 variables entered into the logistic regression, only height (Ht) and longer single NVG mission experience (NVGmax; hrs) were the only two to be used in the final equation below:

$$\textbf{Neck Pain} = [e^{-(27.12 - 18.36Ht + 1.37NVGmax)}] / [1 + e^{-(27.12 - 18.36Ht + 1.37NVGmax)}] \quad \textbf{(5.1)}$$

Using a cut-point of 0.3 to generate a cautious predictive equation, the resultant predictive value was 84.2% with Nagelkerke R^2 = 0.692 and Hosmer-Lemeshow Chi Squared = 5.012 (p=0.76).

Cross Validation. Using the 19 subjects from the original pool for which the author had complete data points for the variables that were determined to be significant

for the loading into the logistic regression analysis, the equation above was generated. Following that derivation, a cross-validation analysis was performed using the 21 subjects who had been removed from the original sample used to derive the logistic regression equation. The author had values for Ht and for NVGmax for all subjects. Using the remaining subjects (n=21) who were excluded from the generation of the equation and a blinded procedure, it was possible to correctly classify 15 subjects with respect to neck pain (71.4%). Overall, for the total sample size (N=40), the equation correctly classified 31 of the respondents with respect to neck pain (77.5%).

DISCUSSION

This paper has provided a predictive equation that uses the total body height measurement and the longest single NVG mission in an aircrew member's logbook to predict neck pain. These data suggest that neck pain can be predicted with 78% accuracy. However, at this time, the equation merely suggests this possibility and does not provide a concrete means by which to do so. With more investigation, it might be possible to develop a mathematical formula for the aircrew population in general with respect to risk factor exposure during NVG operations.

A number of variables were not included in the final regression equation due to the lack of any additional information with respect to the logistic regression equation for which other variables did not account. That is to say, the inclusion of these variables did not improve the accuracy or robustness of the derived logistic regression equation.

Isometric testing results and aircrew fitness tests were not sufficient in providing any information, even in the initial analysis with quartile data that differentiated symptomatic aircrew from their asymptomatic colleagues. It is interesting to note that while variables assessed with EMG and NIRS were deemed worthy of inclusion for the final regression analysis, they did not play a factor in the resulting predictive equation. NIRS was likely excluded from the inclusion in the final equation due to the presence of extremely large standard deviation values. These large standard deviation values are not uncommon to research reports using NIRS as a measurement tool and are not an area of concern – rather, they are an explanation for why not one of the NIRS variables was utilized in the logistic regression equation. The lack of significant findings in the isometric force, endurance, and perceived exertion results as they relate to symptomatic and asymptomatic aircrew is in keeping with the available literature for this subject population. Ang et al. (2005) report differences in isometric strength exist in the cervical muscles of fast-jet aircrew when injured individuals are compared to non-injured counterparts. However, the same study did not find statistically significant differences between helicopter pilots reporting and not reporting neck pain. In fact, the only difference between healthy and injured helicopter pilots reported by Ang et al. (2005) was a shallower slope in the EMG median frequency decline for the cervical flexors of the injured pilots during fatigue testing.

Height provides an interesting topic for discussion. As demonstrated by the logistic regression equation, increasing height was negatively associated with the reporting of neck pain (i.e. an increase in height represented a decreased risk of neck pain) as determined by the provided equation. This may be because taller subjects also

tended to be the larger subjects who would have a greater degree of lean muscle mass as compared to the small subjects. For these subjects, an 8lbs flight helmet would represent a smaller mass as compared relatively to their body mass. In this case, it would be reasonable to expect the magnitude of the stress would be decreased. The unpublished work of Forde et al. (2008) indicated that the cumulative load in simulated flight, as calculated with video analysis of posture and time, was significantly increased with the use of NVG. Perhaps further investigation in this avenue could be of benefit to explaining these results. Further detailed biomechanical studies could investigate for differences with respect to technique in performing the particular aircrew member's in-flight duties with respect to biomechanical stress and reported neck pain.

With respect to flight experience and other literature, Wickes et al. (2005) found significant differences in total flight hours between helicopter pilots who reported neck pain and pilots who did not report any pain. In their study, the more experienced pilots were more likely to report neck pain than those with fewer hours. In comparison to fast jet aircrew, the authors found that significant differences in flight hours only occurred in helicopter aircrew with respect to neck pain reports. Similarly, these logistic regression results of a heterogeneous population of aircrew, both pilots and FEs, suggest that an increasing level of experience with helicopter flight and particularly NVG flight is linked to neck pain. It is interesting to note that the average number of hours of NVG flight in the logbooks of the aircrew subjects who reported neck pain was 149.3 (127.4)hrs. This is very similar to the threshold presented by Adam (2004) at which 90% of CH-146 aircrew report neck pain. Adam found that threshold to be 150hrs of NVG flight.

However, as discussed in the previous paper (Chapter 4), discrepancies in number of aircrew members reporting neck pain exist between CF aircrew and their British counterparts (Harrison et al., submitted). But, whatever the unique characteristic(s) may be, commonality exists in that both CF pilots and flight engineers are adversely affected. The most obvious commonality is that the NVG equipment used by the CF is common to pilot and FE positions. But, similar equipment is currently in use by US aircrew and their rates of injury are reportedly less than those of the CF (Fraser et al., 2006). The second commonality to both CF aircrew positions is the airframe itself, the CH-146 Griffon and its associated vibration spectrum.

Periods of vibration experienced by the human body result in specific physiological responses. Maikala and Bhambhani (2004) describe periods of whole body vibration as causing decreases in muscle oxygenation and blood volume as assessed with NIRS in erector spinae muscles. Specific to the neck when loaded with a flight helmet and NVG, research has described the additional load as an aggravating factor with respect to musculoskeletal stress (Barazanji & Alem, 2000; Butler, 1992). EMG analysis of the cervical musculature by Thuresson (2005) demonstrated that muscle activity peaks at lower frequencies such as 4-5Hz at an acceleration of $2m/s^2$.

The vibration spectrum of the CH-146 has recently been investigated (Chen et al., 2007). The findings indicated that a primary harmonic of the CH-146 is approximately 5Hz, mirroring one of the frequencies used by Thuresson (2005) to investigate the impact helmet mass and seated posture had upon the musculature of the cervical spine. The results of Chen et al. (2007) indicate this frequency results in the helmeted head of the aviator experiencing a displacement in the Z-axis that is twice the magnitude of the Z-

axis displacement of the airframe's floor in all flight conditions. Other studies have long found that the maximum energy transfer to the human spine during whole body vibration occurs at approximately 4.5-5Hz (Maikala & Bhambhani, 2004). While the head is loaded with the additional mass of a helmet and NVG, this amplification of displacement could be a mechanism of injury, especially during prolonged or repeated exposure. Under vibration similar to those experienced by CH-146 aircrew, Thuresson (2005) noted that the right trapezius and lower neck muscles were more affected by vibration than the left side. Our research and that of others have documented side differences either in simulated flight or in laboratory settings with both metabolic and neuromuscular variables (Lopez-Lopez et al., 2001 Harrison et al., 2007a; Harrison et al., 2007b; Harrison et al., 2007c; Harrison et al., Submitted). Additional research should focus on quantifying the vibration spectrum of the CH-146 in comparison to other airframes used around the world and, with fully-loaded flight helmets, in comparison to the head tolerances described by Barazanji & Alem (2000) and Alem et al. (1995).

While research studies continue to investigate the specific factor(s) resulting in neck pain among CF helicopter aircrew, these logistic regression results have provided a simple approach for units and chains of command to monitor their pilots. The clinical use of this equation must be addressed with care as more research is required to confirm these results. While a "cause and effect" relationship cannot be determined from these results (i.e., being of height x, one cannot exceed y hours of NVG flight without developing neck pain), they provide a starting point to investigate some of the factors that may contribute to neck pain. Our data set incorporated approximately 15% of the total number of active

aircrew currently serving on this aircraft type in the Canadian Forces and further data should be collected to increase the robustness of the equation.

CONCLUSION

This data must be interpreted with caution as this is the first study to my knowledge that has predicted neck pain using a simple logistic regression equation by using standing height combined with NVG flight hours (i.e., the ability to predict the likelihood of an aircrew member reporting neck pain as a result of exposure to NVG operations). The end product of such future investigations could provide an equation that could quite possibly aid in the prevention of neck pain among helicopter aircrew by permitting the calculation of guideline thresholds for maximal allowable single mission NVG exposure. It can be reasonably hypothesized that further investigation with a larger sample size and a lengthier research design would provide an equation by which each specific helicopter unit that employs NVG in the CF could employ for aircrew monitoring in the derivation of flight schedules and monitoring the risk associated with NVG flight.

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CHAPTER 6

THESIS CONCLUSION

MAJOR FINDINGS

This thesis has provided a number of significant findings to the area of neck strain research specific to CF helicopter aircrew. **Paper 1** provided evidence that the mass of the NVG equipment significantly augmented the metabolic stress associated with simulated operations as compared to a day-time helmet configuration. To the author's knowledge, this is the first time the stress of the musculature has been measured during settings that mimic the work-load and time-span of military NVG missions. These results provided evidence that the metabolic activity in the upper trapezius muscles is augmented to such a degree as to be measured as a result of the circumstances associated with NVG use.

One of the most notable findings from **Paper 1** (Trapezius Muscle Oxidative Metabolism Measured With NIRS in Helicopter Pilots Flying a Simulator) was the presence of asymmetry with respect to the metabolic responses to NVG missions. The metabolic stress measured in the upper trapezius on the right side was significantly greater than the metabolic stress from the left upper trapezius, despite comparable baseline measurements between the sides at the outset of the simulated missions. **Appendix A** (Helicopter Cockpit Seat Side and Trapezius Muscle Metabolism with Night Vision Goggles) further investigated this matter and it was found that this asymmetry

occurred regardless of whether the pilot was seated in the left or the right cockpit seat. The musculature of the right side of the cervical spine displayed an increased metabolic stress as the mission progressed in duration. In terms of mitigating this metabolic stress, **Appendix B** (Physiological Effects of Night Vision Goggle Counterweights on Neck Musculature of Military Helicopter Pilots) provided preliminary evidence that the use of counterweights aided in decreasing the metabolic stress measured in the pilots.

Paper 2 (Measuring Neuromuscular Fatigue in Cervical Spinal Musculature of Military Helicopter Aircrew) investigated the capacities of the superficial cervical muscles with respect to their fatigue response during submaximal isometric contractions to volitional fatigue. It was found that the smallest of the prime agonists, for example the splenius capitis, bilaterally, during extension displayed a greater neuromuscular fatigue as assessed with EMG normalized median frequency as compared to the upper trapezius muscles, bilaterally. Links between muscle fatigue as measured with EMG and pain have been established (McKeon et al. 2006; Armstrong, 1990; Roy et al., 1990).

Paper 3 (Neck Pain and Physiological Capacities in Helicopter Aircrew) provided a very detailed summary of the characteristics of a representative sample of the CF's tactical aviation community. Beyond the void in the literature that this information filled, the paper provided strong evidence for the similarity between the subpopulations that compose the pilot and flight engineer communities. With respect to anthropometrics, flight history, and physiological variables, these two subpopulations are remarkably similar with very few significant differences to be found. Such a finding would suggest that any remedy to neck strain in the form of strength and conditioning programs could treat the populations similarly.

Paper 4 (A Predictive Logistic Regression Equation for Neck Pain in Helicopter Aircrew) and **Appendix D** (Differentiation of Physiological Measures of Neck Myalgia by Principal Component Analysis) used an advanced statistical analyses to look at my data from another perspective in an attempt to identify individuals who reported neck pain. This preliminary assessment showed that height of the aircrew member and length of the longest NVG mission in the aircrew member's logbook are two variables that must be considered. This finding in which a lengthy NVG mission plays a role in the reporting of neck pain is supported by the previous results reported by Adam (2004) related to CF aircrew. However, we were surprised that the physiological variables assessed did not play as an important role in this analysis, but was probably related to the large variability (SD) between subjects in the tissue oxygenation and neuromuscular parameters commonly assessed in this type of research. It is well documented that large individual variation exists when using NIRS data (Albert et al., 2004). This equation does require further investigation before it can be applied to the "boots on the ground." Further investigation involving a lengthier research design with a larger sample size would be required before the equation could be applied to generate guidelines and limits on duration of risk factor exposures. However, these preliminary results strongly suggest that future work in this avenue should occur.

Appendix D took the analysis in a different direction and applied PCA to the EMG waveforms related to the submaximal isometric ramping trials in which force was gradually increased from resting values to 70% of MVC and then gradually decreased to resting values in the same incremental process. The scores generated by the PCA, when analyzed by ANOVA, were found to be statistically different between aircrew with

reported neck pain as compared to their reportedly asymptomatic counterparts. This is, to the author's knowledge, the first time in which physiological differences have been detected between those two populations of aircrew, those with reported pain and those without reported pain. Even more importantly, these differences are detectable using a submaximal isometric procedure that would potentially permit even aircrew with the most severe of neck pain to participate in research and testing. It is likely that this fatigue measured with EMG and these differences in neuromuscular activity have contributed to the pain experienced by these aircrew members. When this is coupled with the findings of altered metabolism as assessed with the NIRS, the evidence suggests a physiological mechanism of injury that is focused on soft tissue such as the skeletal muscle. Further investigations into the realm of neck pain as it relates to CF helicopter aircrew should pursue this avenue in addressing the issue.

DIRECTIONS FOR FUTURE RESEARCH

Concurrent to this thesis, the research group has been investigating the efficacy of physical training programs specifically aimed at the musculature of the trunk and neck to alleviate neck pain among military helicopter aircrew. The results are expected in the near future and they will provide insight into an additional method of combating the issue. It is anticipated that the program for which the most potential is scientifically demonstrated will be made available to the Canadian Forces helicopter community as a whole as well as to the CF's global allies. However, this is likely the last of the "easy" methods – if improvements in physical fitness specific to the cervical region cannot

adequately address this matter, further investigations will likely have to focus on cockpit ergonomics and equipment design. Such work would be slower in delivering results as well as more costly in terms of achieving operational deliverables. Furthermore, the time elapsed during that design would allow the issue of NVG-induced neck strain to continue to negatively impact the aircrew. That said, it would not be prudent to ignore cockpit ergonomics regardless of the level of success of the physical training programs. As discussed in a subsequent portion of this section related to future research, the research group has already begun the preliminary investigations into this vein.

Further physiological research should return the focus to addressing the issue of pain and its physiology. This thesis has quantified pain in general terms – aircrew members were either self-reported symptomatic or self-reported asymptomatic. Specific quantification of the physiology of aircrew focused on capacities and function – MVC, submaximal endurance, neuromuscular measures of performance and fatigue, and metabolic function. Work recently commenced in associated laboratories has focused on quantifying the degree of pain using pressure algometers and specific locations of the cervical anatomy. This method will allow for the inclusion of more subjects within the data set as well as more accurately gauge the effectiveness of any training program with respect to levels of pain in the aircrew.

This is not to say that the measurement of capacity and function of the musculature of the cervical spinal region should cease. In fact, it should continue to collect data in its broadest form. Future research using EMG should consider using more in-depth techniques such as electrode arrays to identify "hot spots" within the muscles during actual and simulated flights and with different helmet configurations.

Investigations of workplace demands and ergonomics should continue to make efforts to gather data from airborne operations. This will allow for the collection of job-specific data from the aircrew, particularly the FE population. However, the research design should be cognizant of the inherent risk with respect to military flight. Safety standards such as emergency egress schedules and electromagnetic interference (EMI) levels from the research equipment to the crucial flight instruments and controls are not negotiable. As such, they will take some time to address.

In the mean time, associated laboratories have also begun to investigate ergonomic issues such as differences in cumulative loading of the cervical spine during day and NVG operations in the flight simulator located at 403 (Helicopter) Operational Training Squadron ((H)OTS) in Gagetown, NB. The decreased field of view during day operations ($\sim 100^0$) as compared to NVG operations ($\sim 40^0$) coupled with the increased mass of the helmet during NVG use is likely to cause significantly larger cumulative loads during the NVG missions (Forde, 2008). This line of research will provide the cornerstone for further ergonomic assessments and adjustments to the equipment and workspaces of helicopter aircrew. The effect of the movement of any flight control or instrument can be compared to the current arrangement to provide indisputable evidence to support or reject any hypothesized ergonomic improvement. Changes in any aspect of the helmet and configuration of helmet mounted equipment can also be compared to this benchmark to the same end.

Finally, the most important avenue of investigation should address the effects of the whole body vibration (WBV) experienced by helicopter aircrew in the Canadian Forces. Chen et al. (2007) have presented findings that strongly suggest the frequency of

vibration and the associated frequency of the harmonics during a wide range of the flight profiles of the CH-146 match the resonant frequency of the human spine almost perfectly. This supports research presented by Maikala and Bhambhani (2004). As a result, the magnitude of vertical displacement of the head of the pilot from WBV measured during operations is larger by a factor of 2 as compared to the displacement of the floor or even the seat of the aircraft. It is easy to hypothesize that chronic injuries are likely to occur as a result of this WBV while the cervical spine of the aircrew member is loaded with a 3.68kg helmet and in an awkward assumed posture for an extended period of time. I am aware that researchers at the National Research Council (NRC) and at the Royal Military College (RMC) have begun to focus their work specifically on the vibration issue and how it relates to military flight. This will provide important information in the future.

To date, the work contained within this doctoral dissertation has spawned two additional projects. Under co-operative efforts between the University of Regina (Dr. J. Patrick Neary) and the University of New Brunswick (Dr. Wayne J. Albert), two Masters theses have further investigated this topic. At Regina, Ms. Danielle Salmon has completed the data collection associated with a physical strength and endurance training program aimed at increasing the capacities of the cervical spinal musculature. In New Brunswick, Mrs. Kelsey Forde has completed an ergonomic assessment of the biomechanical variables associated with simulated flight during day and night conditions. Her cumulative load assessment has produced significant results and publications are pending in the near future.

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APPENDIX A

HELICOPTER COCKPIT SEAT SIDE AND TRAPEZIUS MUSCLE METABOLISM WITH NIGHT VISION GOGGLES

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INSERT: This paper, published in Aviation Space and Environmental Medicine (2007; 78:995-8) represents work that was performed in the flight simulator at 403 (Helicopter) Operational Training Squadron at Canadian Forces Base Gagetown, NB. This work utilized near infrared spectroscopy technology to investigate the effects of different cockpit seat sides on the metabolic stress as a result of the use of night vision goggles during flight operations.

INTRODUCTION

Helicopter pilots' reporting of low back pain has been well-documented but only recently has cervical pain become the focus (Butler, 1992; Thuresson, 2005; Wickes et

al., 2005). Canadian Forces (CF) sources indicate that more than 50% of their helicopter aircrew members experience neck strain and 5 of 7 tactical squadrons reported that more than 70% of pilots were reporting neck pain (Adam, 2004). US Army documents indicate that NVG missions are more stressful than daylight flights for helicopter pilots but are vague in describing their methods for quantifying this stress (Hart, 1988). CF documents indicate that 90% of all pilots exceeding 150hrs of NVG experience report neck pain of unknown origin (Adam, 2004). These reports of neck pain have been linked to a number of factors specific to military helicopter flight including: vibration, helmet mounted devices such as night vision goggles (NVG), in-flight postures, cumulative flight hours, location of critical flight instruments, and +Gz forces (Thuresson, 2005; Weirstra, 2001; Wickes et al., 2005).

Various means of mitigating aircrew neck pain have been suggested or implemented. Exercise programs focusing on muscular strength are suggested for fast jet aircrew (Alricsson et al., 2004). In helicopter aircrew, aerobic exercise training decreases the rates of reported neck pain (Wickes et al., 2005) and counterweights decrease trapezius muscle metabolic stress during NVG missions (Harrison et al., 2007a).

Helicopter pilots' low back muscular activity, monitored with electromyography (EMG), has been shown to be increased on the right side of the spinal column as compared to the left side under normal flight conditions (Lopez-Lopez et al., 2001). Also, muscle metabolism has been shown to be significantly greater on the right side of the cervical spine as compared to the left side (Harrison et al., 2007a). These differences were noted to be significantly augmented during NVG operations during which aircrew

would be required to support larger helmet masses with the muscles of the cervical spine (Harrison et al., 2007a).

The proposed explanation for this side-specific phenomenon involves cockpit ergonomics. Pilots are required to adopt a twisted and hunched posture to manipulate the collective control with the left hand and the cyclic control with the right hand (Lopez-Lopez et al., 2001; Pelham et al., 2005). This may be of significance for militaries using helicopters that seat the pilots side-by-side to work together to perform the necessary duties related to the mission. In these aircraft, the position and location of the collective and cyclic controls are identical, regardless of whether or not the pilot is in the right or left seat and pilots can alternate between sitting in either cockpit seat. However, investigations have not connected the physiological stresses of the neck musculature with cockpit seating side.

While the posture adopted to manipulate the flight controls is the same regardless of whether the pilot is in the right or the left seat, the location of some crucial flight instruments is different. While not all helicopters have the same instrument panel configuration or the same instrumentation, the CH-146 Griffon does have a central display unit (CDU) that has been anecdotally linked to NVG-induced neck strain for pilots. The CH-146 CDU is located between the pilots; that is, to the left of the pilot in the right seat, and to the right of the pilot in the left seat. The CDU location below the horizontal has been identified as a possible cause of NVG-induced neck strain among Canadian Forces (CF) pilots (Weirstra, 2001). Therefore, we hypothesized that differences in the right and left trapezius responses to NVG simulator flights would differ in magnitude depending upon the cockpit seating side used by the pilot in the course of a

simulated mission as a result of this relative change in location of crucial instruments. The purpose of this study was to compare the differences between seat sides with respect to the metabolic changes bilaterally in the trapezius muscles in helicopter pilots during simulated NVG missions.

METHODS

Subjects. Ethical approval was obtained from the University of New Brunswick's Research Ethics Board. Subjects were provided verbal and written summaries of the project by CF personnel and the research team. Twenty-five male pilots representing all Canadian tactical helicopter squadrons representing all levels of flight and NVG experience signed written informed consent.

Protocol. Pilots were monitored during simulated flights with helmet-mounted NVG equipment. Flights (97.7 (16.1) min) were performed in the full-motion CH-146 Flight Simulator (CAE, Montreal, Canada) at Gagetown, New Brunswick. Mission design was determined by CF guidelines and was not interrupted or altered in any way by this research. Mission tasks were varied to address the annual recurrency training requirements of the CF tactical helicopter population, including emergency procedures, instrument flight segments, and precision navigation tasks. These tasks required teamwork between the aircrew members in a stressful environment and demanded that attention be paid to all cockpit instruments. Pilots wore their own helmets (approximate day mission mass =1.6kg; approximate NVG mission mass = 3.7kg) in their normal

configuration and sat in the same seat position for their monitored flights. Fourteen pilots sat in the left side seat while seventeen sat in the right side seat during the missions.

Near Infrared Spectroscopy (NIRS). The trapezius muscles were monitored bilaterally using a NIRO-300 oximeter (Hamamatsu Photonics KK, Hamamatsu City, Japan). NIRS is an optical tool that measures quantitative changes in muscle oxygenation, deoxygenation, and blood volume in vivo. In addition, NIRS has the ability to monitor % oxyhaemoglobin saturation. The technical description and operation of the NIRO-300 has been detailed elsewhere (Suzuki et al., 1999). Briefly, the NIRO-300 uses laser diodes with 4 wavelengths (776, 826, 845, 905nm) as the light source. Utilization of 4 lasers increases the instrument's measurement sensitivity as more chromophores can be detected in the region of interrogation. The tissue is penetrated with light, the attenuation is measured, and absorbency is calculated using a modified Beer-Lambert law.

NIRS probes were placed bilaterally on the upper trapezius at the C7 level. Placement location was the result of pilot work in the flight simulator, discussions with CF medical personnel, and previous work in the laboratory with whiplash sufferers (Leavins et al., 2005a; Leavins et al., 2005b).

The source-receiver distance of the probes was set at 5.0cm for all subjects, allowing for a penetration depth of approximately 2.5cm. The concentration changes from baseline are reported as a delta value in $\mu M \cdot cm^{-1}$ for total haemoglobin (tHb; HbO_2 + HHb), oxyhaemoglobin (HbO_2), deoxyhaemoglobin (HHb); as well, the tissue oxygenation index (TOI) was monitored to reflect changes in the percent of Hb saturation with oxygen. Sampling speed was set at 6 Hz to account for rapid dependent variable changes during the missions.

One minute of baseline data was collected at the start of each mission with the pilots wearing their helmet while seated in the cockpit of the simulator. During all simulated night missions, the pilots wore their helmets with helmet-mounted NVG devices. For all pilots used in this analysis, counterweight equipment was present. Counterweight equipment is a 600g lead mass attached with Velcro^{TM} to the back of the helmet to act as a balance for the mass of the NVG. CF command does not dictate that this equipment is mandatory for its aircrew and its usage has been left to the discretion of each individual.

Statistical Analyses. The mission times were standardized to percentages to account for minor and non-significant differences in mission duration between pilots. Data was compiled as running averages during two minute periods at every tenth percentile (i.e. 0, 10, 20...100% mission time) and analyzed using SPSS V16.0 (SPSS, Chicago USA). Visual inspection for outliers was performed to identify values that were greater than two standard deviations from the mean. A factorial split-plot analysis of variance (ANOVA) was conducted to determine significant differences between the metabolic responses of the trapezius muscles as a function of left and right seating and of mission condition. Significance was set at $p<0.05$.

RESULTS

Twenty-five male subjects (37.8 (7.3) yrs; 1.78 (0.04) m; 81.8 (10.4) kg) using counterweights volunteered to be monitored during their training sessions in the flight simulator. The subjects had 8.1 (6.6) yrs experience as CF helicopter pilots logging

1783.2 (1441.4) flight hrs and 145.3 (141.8) NVG hrs. Data sets of two subjects were removed due to concerns of erroneous measurement as a result of incomplete NIRS-probe adherence to the skin of the trapezius resulting in unreliable data (±3SD's). The remaining 23 datasets (10 left seat, 13 right seat) were analyzed to determine differences between the left and right trapezius muscles during simulated NVG flights. Results will be presented with respect to each of the NIRS variables.

tHb. tHb displayed significant differences when analyzed by left and right trapezius responses during NVG missions. The tHb response in the right trapezius muscle (10.98 (2.34) $\mu M \cdot cm^{-1}$) was significantly ($p<0.05$) greater than the response in the left trapezius muscle (2.19 (2.87) $\mu M \cdot cm^{-1}$) during the NVG flights. However, when analyzed for differences for these variables between the pilots who performed the mission in the left seat as compared to those pilots who performed the mission in the right, no significant differences were found between the left (Left seat 5.76 (7.10) $\mu M \cdot cm^{-1}$; Right seat 4.88 (5.06) $\mu M \cdot cm^{-1}$) and right (Left seat 13.66 (18.33) $\mu M \cdot cm^{-1}$; Right seat 7.81 (14.72) $\mu M \cdot cm^{-1}$) trapezius responses ($p=0.72$).

HbO_2. Similar to tHb, significant differences were found between the left (2.98 (12.35) $\mu M \cdot cm^{-1}$) and right (8.23(14.71) $\mu M \cdot cm^{-1}$) trapezius HbO_2 during NVG simulations ($p<0.01$). However, no significant differences between the left (Left seat 5.68 (4.46) $\mu M \cdot cm^{-1}$; Right seat 5.60 (4.29) $\mu M \cdot cm^{-1}$) and right (Left seat 9.83 (14.02) $\mu M \cdot cm^{-1}$; Right seat 6.98 (15.01) $\mu M \cdot cm^{-1}$) trapezius were noted when analyzed as a function of cockpit seating ($p=0.57$).

TOI. The TOI results did not produce significant differences between mission conditions compared by left (64.17 (8.56)%) and right (73.08 (8.26)%) trapezius results ($p=0.42$) or in the left (Left seat 61.44 (11.19)%; Right seat 67.15 (6.36)%) or right (Left seat 71.77 (5.83)%; Right seat 74.29 (8.90)%) trapezius as a function of cockpit seating ($p=0.72$).

HHb. Significant differences were noted between left (-2.21 (8.03) $\mu M \cdot cm^{-1}$) and right (1.08 (3.62) $\mu M \cdot cm^{-1}$) trapezius HHb responses ($p<0.01$). Similar to the other variables, no significant differences were for the left (Left seat -1.49 (3.89) $\mu M \cdot cm^{-1}$; Right seat -0.97 (4.01) $\mu M \cdot cm^{-1}$) or right (Left seat 1.97 (6.11) $\mu M \cdot cm^{-1}$; Right seat -3.17 (8.52) $\mu M \cdot cm^{-1}$) trapezius when analyzed by seat side ($p=0.21$).

DISCUSSION

This study showed three important findings with respect to NVG-use and cockpit ergonomics in CF helicopter pilots during simulated NVG flights: (1) metabolic differences exist between the right and left trapezius muscles regardless of seat side; (2) no significant differences were observed between any of the metabolic (TOI, HbO_2, tHb, HHb) variables in the trapezius muscles as a function of seat position; (3) during NVG use, the increase in muscle metabolism was due to factors other than left or right seat position.

Significant differences were noted between the metabolic responses of the right and left trapezius muscles during NVG simulated missions. This finding with a specific focus to cockpit seat sides furthers the reported findings of significant differences between trapezius muscle metabolism between day and NVG flight conditions (Harrison et al., 2007a). The right trapezius demonstrated a greater metabolic stress as compared to the left trapezius, 2-fold, in the case of HbO_2, over the duration of the simulated mission despite comparable baseline measurements for the trapezius, bilaterally.

A split within the sample population occurred between those performing the mission tasks in the right seat as compared to those in the left seat. Given the opportunity for duty schedules to be designed to allow rotation between left and right seats for successive flights, it seemed prudent to investigate whether this would provide any relief to the elevated metabolic stress experienced by the right trapezius versus the left trapezius during NVG-missions.

It was hypothesized that metabolic responses would differ as a function of left or right cockpit seating. The flight simulator used for this study and for training of the CF aircrews on an annual basis has been designed to be as realistic as possible. While +Gz forces are not recreated in the simulator, aspects such as vibration and cockpit layout are nearly identical. Furthermore, indications are that the +Gz forces experienced by helicopter aircrew rarely exceed +2.0 Gz and do not exceed of +1.0 Gz for sustained periods as in fast jets, resulting in "low load – long duration" type injuries (13).

The location of some crucial flight instruments with respect to the left and right seats led us to this experimental question. In the CH-146, like other military helicopters, the position of crucial cockpit instruments such as the central display unit (CDU) and the

fuse panel changes relative to the left and right seats for different missions. Both items are located between the pilots' seats; the fuse panel is located on the ceiling panel (i.e. above the horizontal relative to neutral cervical posture) while the CDU is at the same level as the seat bottom (i.e. below the horizontal relative to neutral cervical posture). To view other control displays that are central on the instrument panel, the pilot in the left seat would have to twist their neck slightly to the right while the pilot in the right seat would have to twist their neck slightly to the left. But these results did not support the hypothesis that the metabolic increases between the trapezius muscles of right-seated pilots would be different than the responses of left-seated pilots. In this case, it would appear that the relative lateral position of instruments such as the CDU and fuse panel do not influence the metabolic response of the trapezius muscles.

This is of interest because the location of other crucial flight controls is constant regardless of left or right seating. The position of the two hand-held controls, the cyclic (controls horizontal movement) and the collective (controls blade pitch, thereby changing vertical altitude and horizontal velocity) do not change from the right seat to the left seat. The right hand always controls the cyclic while the left hand always manipulates the collective. The author's previous work and the work of others suggest that it may be these constants that contribute to physiological stress in the musculature of the spinal column of helicopter pilots (Harrison et al., 2007a; Lopez-Lopez et al., 2001; Pelham et al., 2005). EMG has been used to investigate the muscular activity of the musculature of the lumbar spine and found significant increases in muscle activity on the right side of the spinal column as compared to the left side (Lopez-Lopez et al., 2001). These results reported significant differences in the metabolic activity of the right trapezius as

compared to the left trapezius (Harrison et al., 2007a), showing the right trapezius muscle was more metabolically active as compared to the left trapezius - independent of day or NVG mission. Pelham et al. (2005) described the posture adopted by many military helicopter pilots in response to not only the flight controls but the cumbersome survival gear and noted a twisted posture that would increase the stress and strain experienced by the structures on the right side of the spinal column.

Regardless of cockpit seat position, NVG-mass was the same for each subject. This mass on the helmet is identified as a perceived cause of aircrew neck strain (Wickes et al., 2005). Mounting physiological evidence indicates helmet-mounted devices are a leading cause of neck strain or decreased operational performance in helicopter pilots (Butler, 1992; Harrison et al., 2007a; Wickes et al., 2005). This is of concern because current trends indicate that the frequency of aircrew members using helmet mounted devices in the course of their missions is increasing (Thuresson, 2005). NVG currently use a pair of light amplification tubes, limiting the field of view to 40^0 and requiring frequent neck movements to bring items of interest into clear view. Work is currently being done to increase this to 100^0 with panoramic NVG (pNVG) (Craig et al., 1997) as another method to reduce the need for movements of the neck in viewing crucial flight information is the use of head-up display (HUD) units. The HUD projects crucial flight data such as heading, altitude, and velocity onto the image produced by the NVG tubes. These items are displayed on the instrument panel in front of the pilots but below the horizon. Placing the information from these controls onto the field of view displayed by the NVG with HUD may reduce the need for the pilot to flex their neck to see the instrument displays and obtain the required information. However, these items will

increase the helmet mounted mass supported by the neck as well as the moment of inertia as a result of the change in the center of gravity of the head. To the author's knowledge, no study has been conducted from a physiological or biomechanical perspective to indicate whether this increase in mass benefits aircrew by decreasing the need for movements of the cervical spine, particularly in flexion and extension.

While the present study's results suggest lateral positioning of the equipment does not influence metabolic stress experienced by the trapezius muscles, as measured with non-invasive NIRS, further work should be performed to address this issue more completely. This will assist in future ergonomic cockpit designs.

CONCLUSION

No metabolic differences were found in the trapezius muscles of pilots flying a full-motion flight simulator during NVG missions as a function of left or right cockpit seating. These results suggest that the lateral location and position of the navigational equipment does not influence metabolic stress as experienced by helicopter pilots as a function of left or right cockpit seating. As such, rotating pilots from the left seat to the right seat with each successive flight is not likely to mitigate the effects of NVG-induced neck strain in military helicopter aircrew using NVG. Future work will employ a cross-over design (ie. have pilots tested in both seat sides during NVG visions) to better address this issue of left and right cockpit seating.

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Appendix B

Physiological Effects of Night Vision Goggle Counterweights on Neck Musculature of Military Helicopter Pilots

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INSERT: This paper, published in Military Medicine (2007; 172:864-70) represents work that was performed in the flight simulator at 403 (Helicopter) Operational Training Squadron at Canadian Forces Base Gagetown, NB. This work represents a preliminary investigation of the efficacy of counterweight equipment in alleviating some of the metabolic stress resulting from the use of night vision goggles during flight operations.

INTRODUCTION

The issue of neck strain among military aircrews as a result of headborne equipment usage is becoming an area of increasing concern globally. In particular, night vision goggles (NVG) appear to be one of the most significantly and commonly reported

cause of this neck strain (Wickes et al. 2005; Thuresson, 2005; Thuresson et al., 2005). It is quite likely that this is simply due to the fact that NVG are the most commonly used equipment mounted on the helmet. Hostile missions often use the cover of darkness as a mission enabler. Search and rescue missions often occur at night and frequently during less than ideal environmental conditions. Both of these types of missions frequently rely on NVGs to facilitate a successful sortie. The ability to allow aircrew to see their environment clearly while operating in the dark or in poor weather is vital to preventing serious accidents that could cost lives; however, the unfortunate paradox is that the use of the NVG equipment may lead to debilitating musculoskeletal injuries (Wickes et al. 2005; Thuresson, 2005; Thuresson et al., 2005).

Thomae et al. (1998) found that 29% of Australian military helicopter pilots experienced neck pain while Wickes et al. (2005) recently presented data suggesting that approximately 55% of Royal Air Force helicopter pilots experience neck pain. While the numbers fluctuate from study to study and are dependent on a number of criteria that are unique to each study, it is evident that the results of neck pain are negative to both the individual and the military units involved. Currently, NVG-induced neck strain in the Canadian Forces (CF) helicopter community is described as "a major concern" and almost 90% of CF helicopter pilots, with more than 150 NVG flight hours logged, report neck strain to some degree, ranging from "relatively minor to serious impairments" (Thomae et al., 1998). Depending on the severity of the pain, pilots may be seen by medical doctors who can prescribe analgesics, be seen by physiotherapists, or be grounded until symptoms cease to limit performance (Wickes et al. 2005; Thuresson, 2005; Thuresson et al., 2005; Adam, 2004).

NVGs provide additional mass to the flight helmet of aircrew and can alter the center of gravity as a result (Thuresson, 2005; Thuresson et al., 2005;Ashrafiuon et al., 1997; Butler, 1992). The CF recognizes the addition of mass to the front of the helmet as a potential cause of strain to the neck extensors and has made counterweight (CW) equipment optionally available to their pilots and flight engineers. CW is the method of choice currently used by the pilots as a means of mitigating neck strain problems. However, the CW requires that the spinal column support additional weight in the vertical plane when the head is in neutral posture and CW can place additional stress on the musculature when the head posture is not neutral (i.e. neck flexion and/or rotation to better view cockpit instrumentation). Deviations from the netural may cause the mass to cease to act as a balance but to act as a means of increasing the moment that would also increase the metabolic and physiological demands of the cervical muscles. Due to this theoretical possibility, CW equipment is not mandatory and while many aircrew members use it, some do not.

To the author's knowledge, no study has investigated the physiological effects of CW equipment on the musculature of the neck during simulated operations. To date, studies have investigated the effects head and neck posture on the activity of the neck's musculature (9) and the effects of head worn equipment such as NVG and postures during discrete activities rather than an analysis of the interactions between head supported load, posture, and mission demands (Thuresson, 2005; Thuresson et al., 2005; Ang et al., 2005; Harrison et al., 2007a). Theory would dictate that CW equipment would be effective at minimizing the strain placed upon the neck extensors when the head is in a neutral position. However, during flight, helicopter crewmembers do not spend large

periods of time seated in a static posture with the head in a neutral position, and research has shown that crewmembers who spend a great deal of time performing dynamic tasks while wearing helmet mounted devices like NVG report experiencing neck pain more frequently (1). In a helicopter's cockpit, many of the control instruments are located below the horizontal and require the pilot to flex the neck to observe the information being displayed. Additionally, a number of articles have described the stress placed upon the musculature and skeletal structures of the spinal column while piloting a helicopter (Thuresson, 2005; Thuresson et al., 2005; Thomae et al., 1998; Thuresson et al., 2003; Lopez-Lopez et al., 2001;,de Oliveira & Nadal, 2004; Pelham et al., 2005). A considerable amount of information is available related to the prevalence and causes of neck strain among fast jet aircrew members (Oksa et al., 1996; Hamalainen et al., 1994; Hamalainen et al., 1998; Knight & Baber, 2004). However, fast jet crewmembers are more likely to be exposed to high +Gz-forces as compared to their helicopter colleagues and this fact combined with head postures, rather than NVG mass, has been identified as the proposed cause of the injuries in the jet population.. Helicopters, though, are notorious for large amplitude small frequency vibrations that can be fatiguing over time for the musculature of the spine (Hart, 1998). This fatigue could potentially lead to increased force placed on the pain sensitive structures such as the discs, spinal ligaments, or facet joints. For this reason, an assessment of the metabolic and physiological response to extended exposure through a simulated mission may prove enlightening as compared to a series of discrete postures while wearing helmet mounted equipment in various configurations.

Therefore, the purpose of this investigation was to monitor and compare the oxygenation and hemodynamic changes of the trapezius muscles bilaterally in helicopter pilots with and without counterweights added to the helmet. It was hypothesized that the additional counterweight, coupled with tasks required of a helicopter pilot during simulated missions, would place increased stress upon the musculature of the neck extensors (i.e., trapezius muscle).

METHODS

Subjects. Ethical approval was obtained from a duly constituted Review of Ethics Board and subjects were provided verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Thirty-three CF helicopter pilots representing all Canadian tactical helicopter squadrons agreed to participate in the study and signed informed consent (Table 1). All subjects had spent extensive training time in the simulator over the course of their flying time with the CF and were familiar and comfortable with the operating environment. Two data sets were removed from the analysis matrix because they contained data points that were $\geq \pm 3$ SD from the mean. Inspection of the data files indicated that these outlying data points were most likely the result of loss of adhesion of the NIRS probes to the trapezius muscles over the course of the simulated flight.

Characteristic	CW Mean (SD)	nCW Mean (SD)	Combined Mean (SD)
Age (yrs)	38.4 (7.0)	28.7 (1.5)	37.0 (7.4)
Height (m)	1.77 (3.54)	1.79 (6.20)	1.78 (0.39)
Weight (kg)	81.8 (10.4)	70.1 (2.7)	80.6 (10.5)
Resting HR (bpm)	70.3 (12.4)	63.5 (13.3)	68.7 (12.5)
Resting systolic BP (mmHg)	114.3 (14.5)	120.0 (9.4)	115.6 (13.4)
Resting diastolic BP (mmHg)	74.4 (9.4)	68.3 (9/6)	72.9 (9.5)
VO_2Max ($ml \cdot kg^{-1} \cdot min^{-1}$)	40.0 (3.4)	45.4 (7.9)	41.3 (5.1)
Grip strength – right (kg)	26.7 (3.6)	25.1 (1.6)	26.4 (3.3)
Grip strength – left (kg)	25.8 (3.4)	24.7 (2.6)	25.5 (3.2)
Total Flight Experience (yrs)	13.9 (9.0)	6.8 (3.0)	12.7 (8.6)
Total Flight Experience (hrs)	3155.3 (2581.0)	787.5 (356.8)	2743.5 (2511.9)
Helicopter Experience (yrs)	8.1 (6.6)	2.3 (0.5)	7.1 (6.4)
Helicopter Experience (hrs)	1783.2 (1441.4)	496.3 (307.4)	1559.3 (1400.5)
NVG experience (hrs)	151.1 (143.2)	48.8 (22.5)	133.3 (135.7)
Avgerage NVG Flight Duration (hrs)	2.0 (0.4)	2.1 (0.3)	2.0 (0.4)
Maximum NVG Flight Duration (hrs)	3.7 (1.5)	2.9 (0.4)	3.5 (1.4)
Simulated Day (min)	96.0 (17.0)	99.2 (15.8)	96.6 (16.5)
Simulated NVG (min)	98.6 (16.9)	94.3 (13.5)	97.7 (16.1)

Table B.1: Summary of Subject Demographics

Experimental Protocol. The pilots were monitored during two simulated flights – one with NVG equipment mounted on the helmet and one without NVG equipment. Monitored flights occurred at approximately the same time of day on two successive days, with the day flight performed on Day 1, and the Night (NVG) flight performed on Day 2. All missions were performed in the CH-146 Griffon Flight Simulator (CAE, Montreal, Canada) at Base Gagetown, New Brunswick. Mission design was determined by CF recurrency training guidelines and was not interrupted or altered in any way by this research design. The missions were similar in task and duration between conditions and subjects (Table B.1). Pilots wore their standard military issue helmets in their normal configuration and sat in the same seat position for their monitored flights.

Near Infrared Spectroscopy Monitoring. The trapezii monitoring was performed bilaterally using the NIRO-300 Spatially Resolved Spectroscopic (SRS) oximeter (Hamamatsu Photonics KK, Hamamatsu City, Japan). NIRS is an optical tool used to measure quantitative changes in muscle oxygenation (HbO_2), deoxygenation (HHb) and blood volume (tHb) during periods of physical activity and rest that has proved useful in evaluating responses in erector spinae in vivo (Albert et al., 2004). The technical description and operation of the NIRO-300 has been described in detail elsewhere (Suzuki et al., 1999) and previous work has demonstrated that NIRS is highly reliable between testing sessions (Bhambhani et al., 1998). Briefly, the NIRO-300 uses different laser diodes with 4 wavelengths (776, 826, 845, 905nm) as the light source. Utilization of four lasers increases the sensitivity of instrument measurement as more chromophores can be detected in the localized region of light interrogation. The muscles are penetrated

with near infrared light (700-1000nm) and the attenuation of this light is measured. The absorbency is calculated using a modified Beer-Lambert law (Delpy & Cope, 1997).

The NIRS probes were placed bilaterally at the C7 level of the upper trapezius muscles (Figure B.1), with the skin marked to ensure the same location of placement on the subsequent night mission. Placement location was selected as a result of pilot work in the simulator (Harrison et al., 2007a), discussions with CF medical personnel, and previous work in the laboratory with whiplash sufferers (Leavins et al., 2005a; Leavins et al., 2005b). A standardized one minute of baseline data was collected at the start of each mission with the pilots wearing their helmet and seated in the cockpit of the simulator. During simulated day missions, the helmet did not include any helmet mounted devices (i.e., NVGs) or additional weight. During all simulated night missions, the pilots wore their helmets with the appropriate helmet mounted NVG devices. For those pilots whose personal preference dictated their usage, counterweight equipment was present (CW, n=25; nCW, n=6). Counterweight equipment is an approximately 600g lead mass attached with Velcro^{TM} to the back of the pilot's helmet to act as a balance for the mass of the NVG equipment attached to the front of the helmet. CF command does not dictate this equipment to be mandatory for its aircrew and its usage has been left to the discretion of each individual. A CF flight helmet weighs 1.7kg and when loaded with NVG and CW, it has a mass of 3.68kg (Weirstra, 2001). In the more experienced and highly trained pilots (approximately 2400hrs, on average, in CF helicopters, n=10), a heads-up display (HUD) unit was also affixed to the helmet. The HUD provides the pilot with critical flight instrumentation information on the NVG display and adds approximately 200g to the helmet anteriorly.

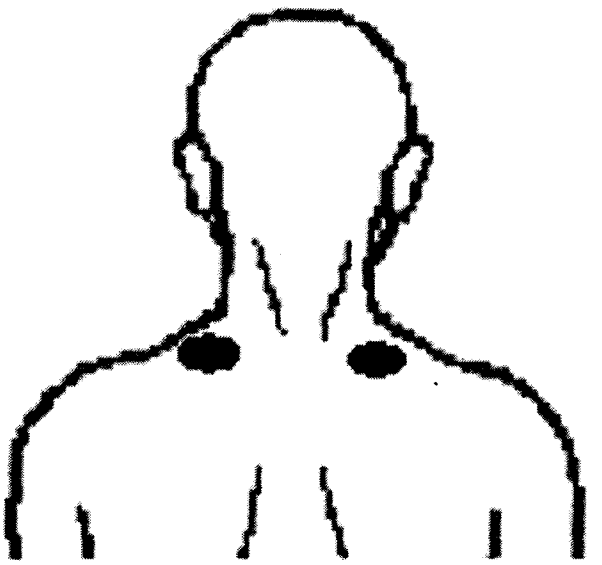


Figure B.1: Placement of NIRS probes on the trapezii muscles during simulated missions

The source-detector distance of the NIRS probe was set at 5.0cm for all subjects, allowing for a penetration depth of approximately 2.5cm (Ferrari et al., 2003). Published values for differential pathlength factor (DPF) are available for the forearm and calf (van der Zee et al., 1992;, Duncan et al., 1995), but are not available for the trapezius muscles. Thus, a DPF was not used in this study, but instead the concentration changes from baseline are reported as a delta value in $\mu M \cdot cm^{-1}$ for total hemoglobin (tHb), oxyhemoglobin (HbO_2), deoxyhemoglobin (HHb) recorded, as well as the tissue oxygenation index (TOI, %) to reflect the percent of Hb saturation being monitored. The sampling speed of the NIRO-300 was set at 6 Hz to account for rapid changes in the dependent variables during the separate missions. Data was collected in real time for the duration of the simulated mission and these files were later analyzed off-line and manipulated to be reported as percentiles of mission length to allow for ease of comparison in the statistical analyses.

Statistical Analyses. Prior to the statistical analyses, the raw data was visually inspected for error messages within the output and appropriate measures were taken on a case by case basis. The mission times were then standardized to percentiles to account for minor (non-significant) differences in mission durations between the individual subjects and crews (Day = 96.6 (16.5) minutes ; NVG = 97.7 (16.1) minutes). Data was compiled and analyzed using SPSS V16.0 (SPSS, Chicago, USA). Visual inspection for outlying measurements was performed to identify values that were greater than two standard deviations from the mean. A split-plot repeated-measures analysis of variance (ANOVA) was conducted to determine left and right side differences, between

conditions, and over the course of each mission for the variables tHb, HbO_2, HHb and TOI. Significance was set at $p<0.05$. Effect sizes were defined as large ($\eta^2>0.80$), medium ($0.79> \eta^2>0.21$) and small ($\eta^2<0.20$). Due the imbalance in the sample populations, analysis using Levene's statistic was conducted to assess the homogeneity of the CW and nCW populations.

RESULTS

Of the four dependant variables measured, only TOI returned a Levene's statistic that was significant, indicating that the sample populations were not homogenous. However, the dependent variables HbO_2, HHb and tHb passed the test for homogeneity and also demonstrated significant differences when CW and nCW results were compared, independent of left and right sides. The effect sizes when the data was analyzed using "subject" as a random factor indicate the results are valid (Table B.2). While significance was found with respect to the analysis of the hemodynamic responses when they were analyzed by left and right sides, the effect size was too small to allow this result to be considered conclusive (Table B.2).

	tHb	HHb	O_2Hb
Fixed Factor(s)	p-value (η^2)	p-value (η^2)	p-value (η^2)
CW vs. nCW	0.859 (0.001)	0.488 (0.017)	0.974 (0.000)
CW vs. nCW by Subject	<0.01 (0.308)	<0.01 (0.315)	<0.01 (0.338)
CW vs. nCW by Side	<0.01 (0.083)	<0.01 (0.058)	<0.01 (0.061)
CW vs. nCW by Mission Time	<0.01 (0.059)	<0.01 (0.014)	<0.01 (0.108)

η^2: small < 0.20; 0.21 < medium < 0.79; large > 0.80

$p < 0.05$

Table B.2: One-way ANOVA Results for tHb, HHb, O_2Hb by Counterweight, Subject, Side and Time (N=31; NVG-CW n=25, NVG n=6)

Hemodynamic variables. Figure B.2 graphs the responses of the hemodynamic variables for which the Levene's statistic was not significant, allowing for comparison of Day and NVG missions under the CW and nCW conditions. Total blood volume (tHb; $\mu M \cdot cm^{-1}$) was significantly greater in the nCW group as compared to the CW group and the response from both the CW and nCW during Day missions. The concentration of deoxygenated hemoglobin (HHb; $\mu M \cdot cm^{-1}$) was found to be significantly different between the CW and the nCW populations. The pilots in the nCW population displayed a decreased HHb as compared to the pilots in the CW population. The HbO_2 observed in the nCW group was significantly greater in comparison to that observed in the CW group and the Day flights.

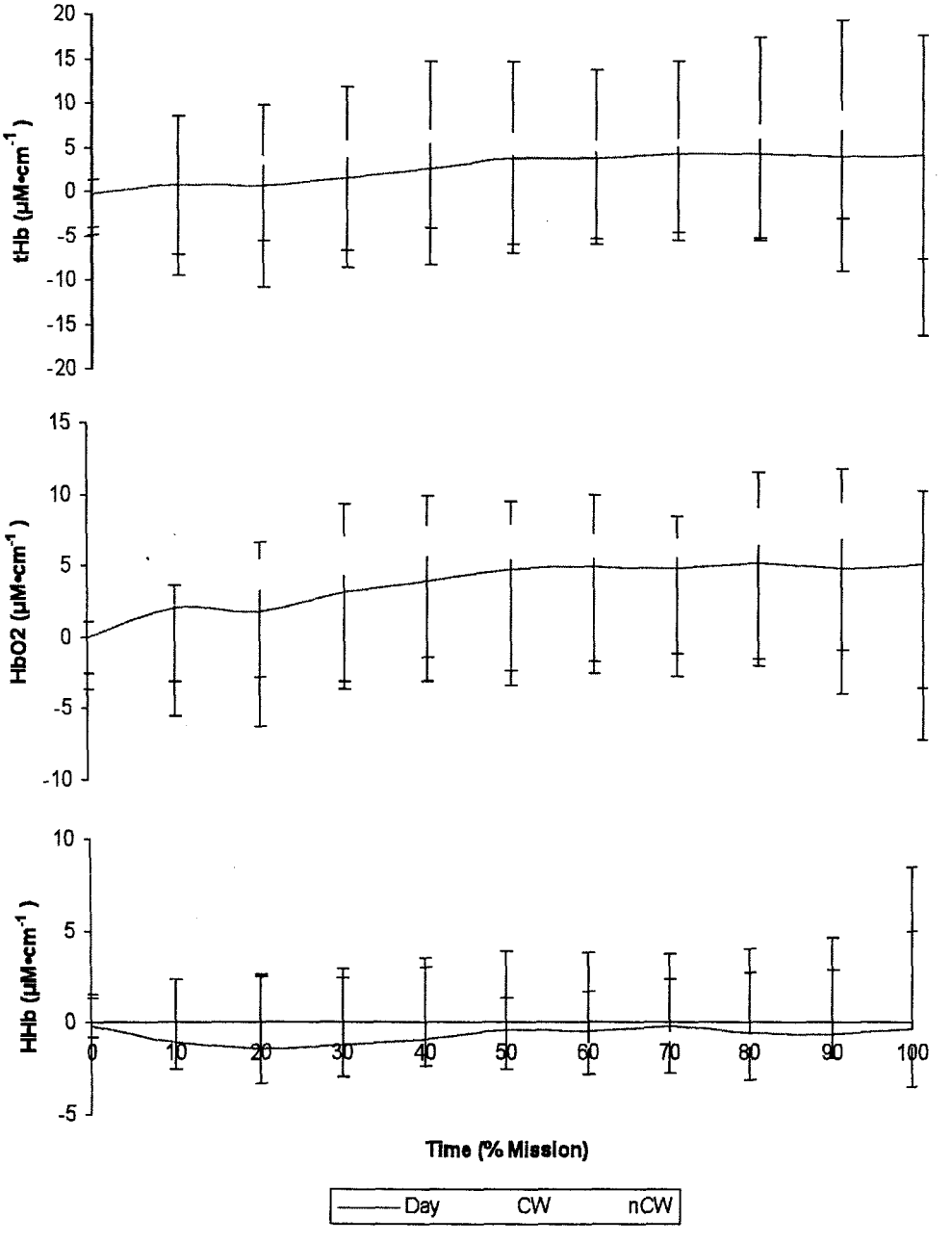


Figure B.2: Tissue oxygenation index (TOI), total haemoglobin (tHb), oxygenated (HbO_2) and deoxygenated Haemoglobin (HHb) values for trapezius muscles of CH-146 pilots using counterweight epuipment (CW; N=25) and not using counterweight equipment (nCW; N=6) during simulated Night missions compared to Day Missions.

DISCUSSION

This study's results were contrary to the original hypothesis. Based on the limited available literature in this area of research, it was hypothesised (due to the demands for dynamic neck postures and head positions during flying tasks) that the presence of additional mass in the form of CW would increase the metabolic demand and stress experienced by CF helicopter pilots using NVG. This hypothesis was reached after considering the effects of the additional mass on the moment experienced by the neck musculature when the pilots were required to operate in a posture that was deviated from neutral. Specfically, the author believed that the need to rotate the head to the side and to tilt the head forward and aft to look at items above and below the horizon would negate the balancing effect of the CW. This study's results, however, indicate that CW does actually act as a balance that minimizes the metabolic stress of the trapezius musculature during NVG missions.

This is the first study to investigate the effectiveness of CW equipment in mitigating the effects of helmet mounted devices such as NVGs on the oxygenation and hemodynamics of the trapezius muscle. It has been proposed that NVG use is a determining factor responsible for inducing neck strain in flight crewmembers (Wickes et al., 2005; Thuresson, 2005; Thuresson et al., 2005; Thomae et al., 1998; Adam, 2004; Weirstra, 2001; Harrison et al., 2007a) and anecdotal reports from aircrew members indicate that the use of CW equipment simultaneously with NVG may minimize muscle strain in some individuals because of the correction in center of gravity. CW equipment is

issued to CF flight crewmembers at their option and operational restrictions allow them to be modifiable after the fact by the individual. As per CF regulations, issued CW equipment is bound by an upper limit of 650 g but can be made lighter to suit personal preference. The methods of determining which crewmembers use and do not use CW is governed with similar laxness as determining which crewmembers use Non-Steroidal Anti-inflammatory Drugs (NSAIDs) to provide relief to NVG-induced neck strain. This is the first study to provide empirical evidence to support the use of CW (in a flight simulator) as well as providing arguments to support continuing to provide flight crewmembers with a choice.

Although there was an imbalance in the sample population (CW=25; nCW=6), the results demonstrated that the NIRS could detect differences in muscle oxygenation between these self-selected groups. The reader is reminded that the objective of this research was to collect the data in a manner that was as true to the operational setting of the CF CH-146 Griffon crewmembers as possible. As such, these results indicate that the metabolic cost of the trapezii muscles is increased when a pilot opts not to use CW equipment during NVG missions. Hemodynamic responses in terms of tHb, HHb, and HbO_2 were lower in day simulations and CW night simulations as compared to nCW night simulations. While CW may not be the best method of alleviating physiological strain in the neck extensors for certain postures and positions, these results indicate that the cumulative effective of this equipment is beneficial over the course of simulated missions that included numerous realistic scenarios that were very similar to what pilots encounter during real missions.

Despite the small effect sizes, trends toward significantly greater right tHb and HbO_2 responses were observed in the hemodynamic responses of the neck musculature between the CW and nCW subjects. Side differences have been noted in previously reported literature related to helicopter pilots and have been attributed to the configuration of the cockpit controls (Harrison et al., 2007a; Lopez-Lopez et al., 2001; Pelham et al., 2005). The CH-146 Griffon and many other helicopter models are flown in a right-handed manner. The cyclic control is positioned between the legs and is manipulated by the pilot's right hand while the collective control is located to the left of the pilot's seat and is manipulated by the left hand. The pilot maintains almost constant contact with both of these controls and the resulting adopted twisted posture has been described in the literature (Lopez-Lopez et al., 2001; Pelham et al., 2005). The above stated studies investigated the physiological strain of the muscles of the lumbar spine in helicopter pilots and no link has yet been made between the stresses experienced by the lumbar and cervical spine structures during flight operations in helicopter crewmembers. The metabolic findings measured bilaterally in the trapezius muscles indicate that there is a link between the segments of the spinal column but cannot comment with certainty on the causative relationship. Does the lumbar posture and fatigue influence the cervical results or vice versa? However, considering the deviated posture in the lower spinal column, it is possible that the thoracic and the cervical spine are not in a neutral posture for a great deal of the time spent in-flight. The change in posture likely places more stress on the muscles of a particular side of the spinal column as compared to the contralateral muscle group. It is also possible that these non-neutral postures influence the

effectiveness of the CW equipment in re-establishing a balanced center of gravity in the presence of NVG, resulting in the side differences noted in this study.

The present study possessed strengths and limitations that do need to be discussed. In terms of strengths, this sample population represented all active CF squadrons and represented approximately 15% of the CF's operational Griffon pilots, obviously covering a range of experience levels and ages. However, significant differences in subject characteristics between the CW and nCW populations were not found with respect to age, height, weight, and fitness values. Three subjects in the CW population reported that they were tobacco smokers. No information was available with respect to prescription medications but it should be noted that all subjects were on active flight duty. This fact alone limits the possibility that they were taking a prescription medication that would have had physiological side effects.

Second, the data in this study was collected during simulated flight conditions that lasted an extended period of time. Previous studies with helicopter pilots have investigated the effects of helmet weight on neck musculature during discrete activities in a laboratory setting (Thuresson, 2005; Thuresson et al., 2005; Butler, 1992). These results show that the baseline values of the hemodynamic variables were not significantly different, and that significant differences did not appear until at least 20% of the mission time had elapsed (approximately 20 minutes, depending on the subject). Therefore, it is possible that these previous studies underestimated or underreported the true extent of the physiological stresses caused by head supported masses.

In terms of limitations, and factors motivating further research into this area in the future, the use of only male pilots was included. This was not an intentional study design.

Female pilots do exist in the CF helicopter community but none were scheduled for training sessions in the simulator during the data collection (6 month) period. Only two female pilots cycled through the simulator recurrency training during this time period but this research was not able to collect data during their simulated missions. Additionally, such a small sample size would have been impossible to analyze statistically. Barazanji & Alem (2000) have found the tolerances to helmet mounted weights of female helicopter pilots in the US military to be similar to the tolerances of their male colleagues. However, further investigation into the hemodynamic responses of the cervical muscles of female pilots would still be beneficial if only to confirm these previous findings.

This study was designed with the intent of being as convenient and non-disruptive to the CF's regular simulator operations. As such, this study was not able to make use of a randomized design that would have seen half of the sampled pilots perform the NVG mission before the Day mission. Additionally, it is important to note that no recovery data was recorded at this time due to time constraints with the scheduling of the crews in the flight simulator unit. During the course of a normal day, at least three crews rotated through two flights per day with the average mission length being two hours (i.e. 12-14 hours of simulator operation). However, the author is confident in the legitimacy of the study's results and that adequate recovery time was provided between the simulated missions. As illustrated in Figure B.2, the baseline starting value for the hemodynamic variables was nearly identical for all both the CW and nCW groups during both Day and NVG settings.

Finally, 10 subjects used the HUD during their NVG mission. This resulted in an additional 200g of mass being added to the helmet. Of these subjects, 3 did not use CW.

However, the removal of 3 subjects from the nCW group would have rendered the sample useless for statistical analyses. The removal of the 7 CW subjects who used the HUD did not have a statistically significant effect on the data set. After considering these two factors, it was decided to leave the data set as it was with 25 CW subjects and 6 nCW subjects. However, it would be prudent for future research to specifically investigate the effects of the HUD as its usage is going to become more prevalent among helicopter aircrew.

Given the noted differences in the NIRS responses between the aircrew who used CW equipment and those that that did not and given the small sample size of nCW. subjects, numerous questions are raised that warrant further investigation. For example, to what extent are individual pilots choosing to use or not to use CW based on various characteristics such as fitness, history of injury, age, flight experience, and lifestyle choices such as tobacco use that can cause altered vasomotor response? And can training programs specifically designed to augment the fitness of the musculature of the neck counteract the effects of these characteristics? Scientific methods of determining appropriate limits on mission durations, frequencies, or helmet mounted loads are few in number and only anecdotal evidence exists with respect to the timeline associated with the onset of NVG-induced neck strain. Longitudinal studies of helicopter crewmembers accounting for experience, medical history, and levels of physical fitness would also reveal a great deal of information about the pathology of NVG-induced neck strain.

CONCLUSION

In assessing tHb, HHb, and HbO_2 responses in those individuals who regularly use CW and those who do not use CW during NVG missions, significantly different responses were observed. Therefore, it can be concluded that the use of CW equipment during NVG missions in military helicopter pilots does minimize the metabolic and haemodynamic responses of the trapezii muscles. This study provides important "groundwork" in understanding the musculoskeletal demands of using CW equipment with NVG. These differences in the metabolic responses of the neck musculature suggest that a change in muscular demand is occurring during the course of the NVG flights. It is these changes that need to be better understood in order to fully address the effects of cumulative exposure and its relation to musculoskeletal injury.

ACKNOWLEDGEMENTS

The authors wish to thank all members of 1 Wing for their assistance with this project. Funding was provided by Canadian Forces Quality of Life Grant 1725981.

Appendix C

Cytochrome Oxidase Changes in Trapezius Muscles with Night Vision Goggle Usage

Michael F Harrison[1], J Patrick Neary[1], Wayne J Albert[2], Neil P McKenzie[3], Daniel W Veillette[4], and James C Croll[2]

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INSERT: This paper, in press with the International Journal of Industrial Ergonomics, represents the initial investigation of the response of cytochrome oxidase in the trapezius muscles of helicopter pilots using NVG during simulated missions. This work complements previous aspects of this thesis in which increased blood volume was observed to occur during the NVG missions.

INTRODUCTION

Chronic neck and shoulder pain that is muscular in nature is becoming an increasing concern in the workforce of Western society with one reference indicating that the one-year prevalence of "neck myalgia" or "tension neck" can be as high as 48% (Eriksen, 2004). One of the proposed mechanisms of injury, for the trapezius muscles in

particular, is called the ""Cinderella hypothesis"" and a summary of the relevant literature is described elsewhere (Thorn, 2005; Visser & van Dieën, 2006). Simply, the "Cinderella hypothesis" states that uninterrupted periods of low level work over an extended period of time leads to muscular fatigue and muscular damage that can result in reports of chronic pain. This hypothesis appears to be especially relevant to situations that involve long-time static postures (Hagg, 2000).

Furthermore, tasks that demand mental acuity, focused concentration and head forward posture are linked to neck myalgia and possibly the "Cinderella hypothesis" (Sjogaard et al., 2000; Eriksen, 2004). The thought is that the individual assumes a more tense posture that results in static muscular contractions that are not necessarily relaxed regularly, if at all, during the task and that this posture involves leaning forward as if to get a better vantage point. This makes the "Cinderella hypothesis" of interest to those studying helicopter aircrew using night vision goggles (NVG) and the relation to reported neck strain. As such, it should not be surprising that military helicopter aircrew have been reporting increased levels of discomfort and disability in the cervical region that is most commonly attributed to the use of NVG but for which the primary causative factors are not well understood (Wickes et al. 2005; Adam, 2004; Thuresson, 2005, Harrison et al. 2007a, 2007b, 2007c).

In the Canadian Forces (CF), this issue is becoming a serious concern. Of the pilots who have in excess of 150 hrs of NVG flight time logged, 90% report neck pain (Adam, 2004). It is reported that the flight helmet mass is well below the values for maximal voluntary contractions (MVC) of the cervical muscles in flexion, extension or lateral flexion directions as reported by Taylor et al. (2006) for military recruits and by

previous research from our laboratory for helicopter aircrew (Harrison et al., 2009b). However, as the missions progress in length, the mass of an NVG-equipped helmet has been shown to cause increased metabolic stress (Harrison et al., 2007a, 2007b, 2007c).

Additionally, the "Cinderella hypothesis" states that uninterrupted low-level muscular contractions can lead to localized ischemia as a result of occlusion of the blood vessels that perfuse the affected muscle (Sjogaard et al., 2000; Eriksen, 2004). Ischemia has been found to be especially prevalent during prolonged periods of static head-down posture, commonly associated with computer work or other tasks that require focused attention (Eriksen, 2004). This ischemia will cause a shift towards anaerobic metabolism that will result in increased concentrations of lactate and free radicals within the muscle. These accumulations have been shown to negatively affect the redox state of mitochondrial cytochrome oxidase (CtOx) (Eriksen, 2004). CtOx is the final electron acceptor in the electron transport chain and is therefore vital to aerobic metabolism in the myocyte (Cooper & Springett, 1997). The changes in concentration of this protein occur slowly (i.e. days) but the changes in the redox state occur rapidly and are quantifiable with near infrared spectroscopy (NIRS) (Cooper & Springett, 1997; Cooper et al., 1998).

The use of CtOx monitoring with NIRS to detect changes in muscular activity may be beneficial in quantifying neck myalgia. Testing the "Cinderella hypothesis" using EMG analysis has proved difficult (Knardahl, 2002; Eriksen, 2004). Given the low level of the muscular activity coupled with the small portion of the muscle that is responsible for producing that force, it is difficult to develop a surface EMG protocol that is sensitive and robust enough to detect these changes (Knardahl, 2002; Eriksen, 2004). However, the efficacy of NIRS has not been determined.

The mass of the helmet, the duration of missions, and the operating environment specific to military helicopter aircrew lead us to hypothesize that the use of NVG equipment will result in concentration changes in hemodynamic and metabolic variables that will indicate that the muscle is experiencing stress as a result of uninterrupted low-level muscular contraction that will be in keeping with the "Cinderella" hypothesis. Specifically, we hypothesized that a decrease in concentrations of CtOx in a redox state that will allow complete aerobic metabolism in the trapezius muscles will be observed in the later stages of a mission with NVG.

METHODS

Subjects. Ethical approval was obtained from a university Ethics Review Board, and subjects were provided verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Twenty male military helicopter pilots representing all CF helicopter squadrons that use the CH-146 volunteered to participate in the study and signed informed consent forms. The pilots also agreed to provide information about their level of experience (hrs/yrs flying), their history of exposure to NVG flights (hrs), and permitted access to their most recent annual fitness test results. All subjects were on active flight duty and none reported neck pain that was greater than "occasional" and "very mild".

Protocol. The pilots were monitored during two simulated flights – one with NVG equipment mounted on the helmet and one without NVG equipment. Monitored flights

occurred at approximately the same time of day on two successive days, with the day (DAY) flight performed on Day 1, and the Night (NVG) flight performed on Day 2. Timing and structure of missions was beyond the control of the research team due to the CF mandated schedule for recurrency training curriculum. As such, randomization of the trial order was not possible. All missions were performed in the full-motion CH-146 Griffon Flight Simulator (CAE, Montreal, Canada) at Base Gagetown, New Brunswick. Mission design was determined by CF recurrency training guidelines and was not interrupted or altered in any way by this research design. Therefore, the missions were similar in task and duration (DAY= 99.1 ± 15.6 min; NVG= 99.6 ± 16.5 min) between conditions and subjects. Pilots wore their own helmets in their normal configuration with counterweight. A counterweight is a 600 g lead mass that is affixed to the inferior posterior aspect of the helmet to alleviate some of the metabolic and neuromuscular stress of NVG flight (Thuresson, 2005; Harrison et al., 2007b). All pilots sat in the same cockpit seat for their monitored flights (12 in right seat; 8 in left seat). Seat side has been shown to have no impact on the magnitude of the metabolic stress in trapezius muscles during NVG simulated flights (Harrison et al., 2007c).

Near Infrared Spectroscopy Monitoring. The trapezius muscle monitoring was performed bilaterally using the NIRO-300 Spatially Resolved Spectroscopic oximeter (Hamamatsu Photonics KK, Hamamatsu City, Japan). NIRS is an optical tool used to measure quantitative changes in muscle oxygenation, deoxygenation, and blood volume during periods of physical activity and rest. The technical description and operation of the NIRO-300 has been described in detail elsewhere (Suzuki et al., 1999). Briefly, the NIRO-300 uses different laser diodes with 4 wavelengths (776, 826, 845, 905 nm) as the

light source. Utilization of four lasers increases the sensitivity of instrument measurement as more chromophores can be detected in the localized region of light interrogation. The muscles are penetrated with near infrared light (700-1000nm) and the attenuation of this light is measured. The absorbency is calculated using a modified Beer-Lambert law (Delpy & Cope, 1997).

The NIRS probes were placed bilaterally at the C7 level of the upper trapezius with the skin marked to ensure the same location of placement on the subsequent night mission. Placement location was selected as a result of pilot work in the simulator, discussions with CF medical personnel, and previous work in our laboratory with whiplash sufferers (Leavins et al., 2005a; Leavins et al., 2005b; Harrison et al., 2007 a, 2007b, 2007c).

The NIRS optodes (source and receiver) were set at a separation distance of 5.0 cm for all subjects, allowing for a penetration depth of approximately 2.5 cm (Ferrari et al., 2004). Published values for differential path length factor (DPF) are not available for the trapezius muscle. Thus, a DPF was not used in this study, but instead the concentration changes from baseline are reported as a delta value in μmol/cm for total hemoglobin (tHb), oxyhemoglobin (HbO_2), deoxyhemoglobin (HHb), CtOx, as well, the total oxygenation index (TOI, %) was monitored to reflect changes in the percent of Hb saturation. Variables are reported with subscripts to identify the trapezius side ($_R$ for right; $_L$ for left) and mission setting ($_{nvg}$ for NVG; $_{day}$ for DAY); for example, $CtOx_{Rnvg}$ would refer to the cytochrome oxidase response of the right trapezius during the NVG mission. The sampling speed of the NIRO-300 was set at 6 Hz to account for rapid changes in the dependent variables during the separate missions.

A standardized segment of baseline data was collected during a two minute interval at the start of each mission with the pilots wearing their helmet and seated in the cockpit of the simulator. During simulated day missions, the helmet did not include any helmet mounted devices (i.e., NVG) or additional weight. During all simulated night missions, the pilots wore their helmets with the appropriate helmet mounted NVG devices.

Statistical Analyses. Data was compiled as the values at every tenth percent (i.e. 0, 10, 20...100% mission time) and analyzed using SPSS V16 (SPSS, Chicago, USA). Visual inspection was performed to identify values that were greater than three standard deviations from the mean. No data sets were found to contain outliers that warranted removal from the final analysis matrix. Results were also analyzed to assess normality. A split-plot repeated-measures analysis of variance (ANOVA) was conducted to determine left and right side differences, between conditions, and over the course of each mission. Bonferroni post-hoc tests were employed to identify the time period(s) at which significant differences occurred by trapezius (left or right) within mission settings (DAY vs NVG). Pearson-r correlations were performed to investigate the relationships between hemodynamic variable responses. Significance was set at $p \leq 0.05$.

RESULTS

Subject characteristics are described in Table 1. The left and right trapezius' hemodynamic responses to the DAY and NVG flight are illustrated in Figure 1. All

variables measured with NIRS were found to have 95% confidence intervals for skewness that included zero and were therefore determined to be normally distributed. The CtOx response was found to differ significantly during NVG operations as compared to DAY operations ($p=0.007$). The CtOx values decreased for the left (~1600%) and for the right (~150%) as compared to baseline during the NVG missions while during the DAY missions the values increased in the left (~2100%) and right (~430%) trapezius. No significant differences were noted between left and right trapezius responses as a function of DAY ($p=0.34$) or NVG operations ($p=0.20$). Significant differences were found between the DAY and NVG operating conditions for tHb ($p=0.047$) HbO_2 ($p<0.01$) while none were observed for TOI ($p=0.31$) or HHb ($p=0.19$). CtOx results were found to be significantly correlated with HbO_2 ($r=-0.696$, $p<0.001$) and tHb ($r=-0.655$, $p<0.01$) responses (Figure C.1).

Characteristic	Mean (SD)
Age (yrs)	36.9 (7.7)
Height (m)	1.78 (0.04)
Weight (kg)	82.1 (10.1)
Resting HR (bpm)	66.5 (13.4)
Resting systolic BP (mmHg)	112.8 (12.0)
Resting diastolic BP (mmHg)	70.8 (8.9)
VO_2Max ($mL \cdot kg^{-1} \cdot min^{-1}$)	42.2 (5.3)
Grip strength – right (kg)	56.8 (6.5)
Grip strength – left (kg)	55.0 (7.5)
Push Ups (without stopping)	29.5 (14.5)
Sit Ups (in one minute)	31.0 (10.6)
Total Flight Experience (yrs)	13.1 (9.2)
Total Flight Experience (hrs)	2835.0 (2681.7)
Helicopter Experience (yrs)	7.0 (6.8)
Helicopter Experience (hrs)	1548.3 (1504.8)
NVG experience (hrs)	127.3 (135.8)
Average NVG Flight (hrs/mission)	2.1 (0.4)
Maximal NVG Flight (hrs/mission)	3.4 (1.3)
Simulated DAY (min)	99.1 (15.6)
Simulated NVG (min)	99.6 (16.5)

Table C.1. Demographical and physical characteristics of 20 male helicopter pilots.

CtOx & Pilot Fitness & Experience Characteristics. When Pearson-r correlation values were calculated $CtOx_{L\text{-}nvg}$ and $CtOx_{R\text{-}nvg}$ and the detailed personal experience and fitness characteristics (Table C.1), only $CtOx_{L\text{-}nvg}$ provided any statistical significant results. $CtOx_{L\text{-}nvg}$ was moderately correlated with Helicopter Experience ($r=0.451$, $p=0.046$), and Max NVG Flight ($r=0.483$, $p=0.036$), but highly correlated with Avg NVG Flight ($r=0.867$, $p<0.001$). In terms of other hemodynamic variables from the NVG trials and the fitness and experience characteristics of the subjects, few correlations were found. $tHb_{L\text{-}nvg}$ results were correlated with NVG experience ($r=0.489$, $p=0.029$) and Max NVG Flight ($r=0.479$, $p=0.038$). $HHb_{L\text{-}nvg}$ were also correlated with Helicopter Experience ($r=0.498$, $p=0.026$), NVG experience ($r=0.569$, $p=0.009$) and Max NVG Flight ($r=0.726$, $p<0.01$).The NVG responses for the other hemodynamic variables did not provide any significant correlations for any of the variables reported in Table C1.

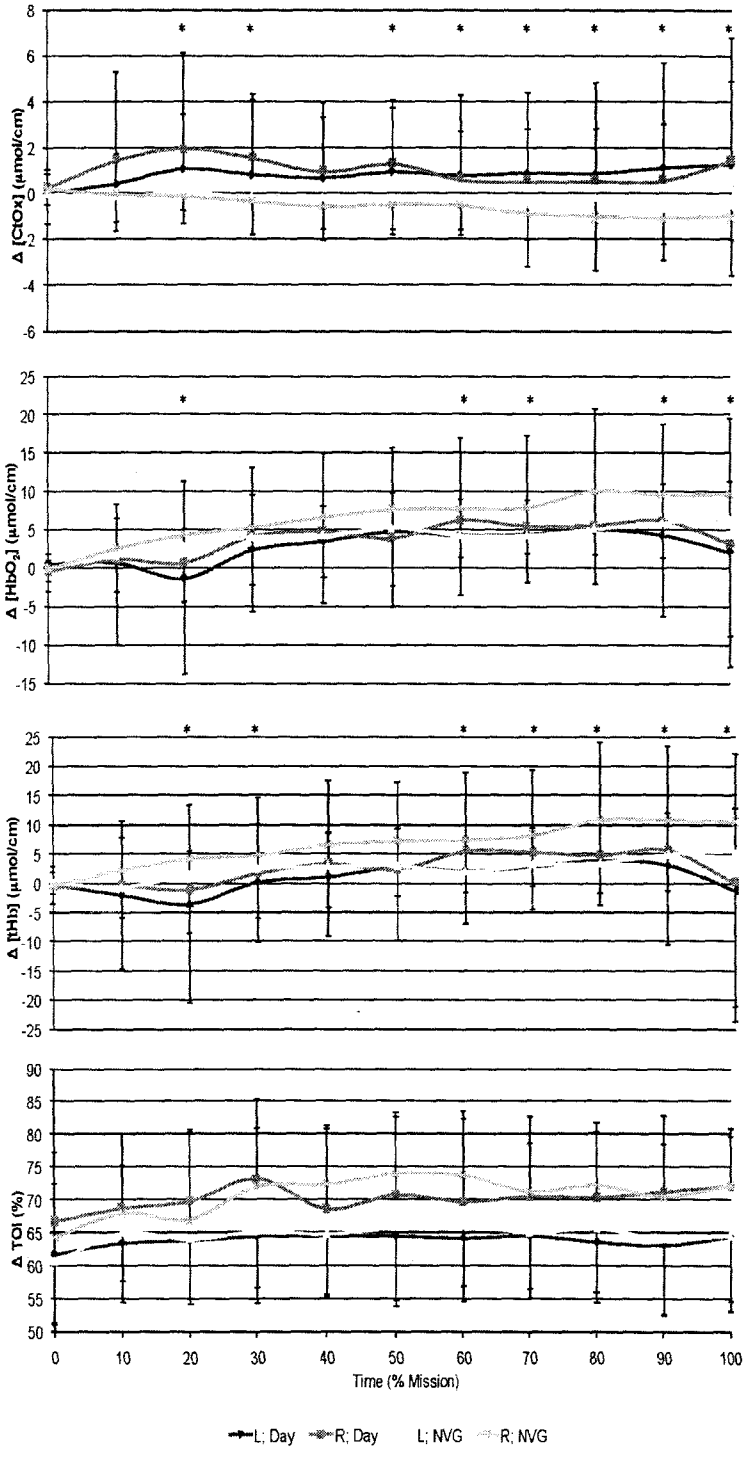


Figure C.1. Left and right trapezius muscle cytochrome oxidase (CtOx), oxyhemoglobin (HbO_2) and total hemoglobin (tHb) responses to DAY and NVG simulator flights in 20 male helicopter pilots.

DISCUSSION

The most significant finding of this study was that a quantitative change from baseline concentration in CtOx occurred during NVG conditions as compared to DAY conditions. This indicates shift towards a reduced (redox) state of CtOx of the trapezius muscles of our subjects, and thus potentially creating a negative impact on the enzyme's ability to accept electrons as they are transported along the electron transport chain. Thus, while the mass presented by a flight helmet with NVG may require a submaximal effort to support in upright seated posture, it can have a negative impact on the enzyme responsible for the final step in complete aerobic respiration at the cellular level. Such an impact could lead to the accumulation of lactate, H^+ ions and other end products of cellular aerobic respiration that can be potentially damaging to muscle cell. Occlusion does not occur in the trapezius muscles during simulated missions with NVG that are less than 2 hrs in duration. However, supplying the tissue with oxygen is of little value if the enzyme responsible for finishing aerobic respiration is impaired as a result of the additional stress from the NVG mass.

As mentioned in the description of the methodology, our subjects performed their monitored flights at the same time of day and they sat in the same cockpit seat for each of the monitored missions. The recurrency training sessions commenced at the same time of day and aircrew were expected to report for duty at 0700 each morning. As such, it can be inferred that the duration of wakefulness was consistent between days and no pilot reported changes in their sleep habits that were out of character with their normal behaviour. Furthermore, our selection of the split-plot ANOVA was done purposefully to

address variation within our research design for circumstances beyond the control of the research team (i.e., random assignment of mission order).

CtOx measurement using NIRS has been viewed by some with skepticism. However, available literature has demonstrated that CtOx measurement in muscle (and brain) tissue is possible with NIRS, in particular using the NIRO-300 and NIRO-500 (McGown et al., 2003; Nagdyman et al., 2003; Gagnon et al., 2005). Gagnon et al. (2005) used periods of circulatory arrest to validate CtOx signal in bovine tissue with the NIRO-300, the same oximeter used in our study, and found changes in the muscle that were comparable to our results in magnitude. Their work details the changes for each different muscle fiber type and indicates that decreases in CtOx redox state are magnified and occur at a faster rate for those individuals with a higher proportion of Type I fibers. Muscle histochemical analysis of the trapezius muscles of male foresters indicated a trend towards a decrease in Type I fiber expression in those workers with myalgia (60%) as compared to those without myalgia (65%) (Kadi et al., 1998). Additionally, the same workers reporting neck myalgia displayed an increased number of Type I fibers that were void of CtOx as compared to healthy controls or workers who were not reporting myalgia (Kadi et al., 1998). These findings are suggested by the authors to be related to the trapezius muscle's injury regeneration cycle which is likely to become increasingly exaggerated as the myalgia progresses.

In our work, the average mission length is ~2.0 hrs but careers can span many years. This may raise concerns with respect to the potential for a chronic mechanism of injury. We observed a decrease in CtOx over the course of the NVG mission in our aircrew population that was healthy. The results of Gagnon et al. (2005) indicate that an

individual with a greater number of Type I fibers in the trapezius muscle may not necessarily be at an advantage with respect to aerobic metabolism. The results of Kadi et al. (1998) suggest that once injured, the expression of Type I fiber becomes diminished and CtOx is suppressed in some of the remaining Type I fibers. As it relates to helicopter aircrew, this suggests that 90% of those with more than 150 hrs of NVG experience are potentially in the vicious cycle of myalgia and the associated changes at the muscular level.

NVG use has been linked to non-specific flight-induced neck pain through numerous studies, almost all of which have been questionnaires. Questionnaire results presented by Wickes & Greeves (2005, 2007) and by Adam (2004) indicate that aircrew perceive these items to be one of the primary causes of flight-induced neck pain and/or a contributing factor in making other items (such as vibration or instrument display location) a perceived cause of neck pain. Odds-ratio results have indicated that increasing the hours of experience with NVG results in a logarithmic increase in the risk of experiencing and/or reporting neck pain among helicopter aircrew (Behdinan & Fraser, 2007). Physiological results have demonstrated that the additional mass that the head and neck must support results in quantifiable differences in metabolism in the trapezius muscles during simulated missions (Harrison et al., 2007a, 2007b). Even under low to moderate +Gx exposure with NVG mass, the increase in muscle activation as assessed by EMG is quantifiable (Sovelius et al., 2008).

The results of the present study are promising. For NVG missions that were not statistically different from the reported average NVG mission length of our subjects, impairment of aerobic metabolism in the interrogated region of the trapezius muscles was

observed as determined by CtOx values. Occlusion of the blood supply to those same muscles in that region of interrogation was also not observed (based on tHb values). From a metabolic perspective and for the muscle as a whole, the stress presented by the additional mass of the helmet with NVG is not in compensable. On the other hand, while our results indicate that aerobic metabolism is not impaired in the area of the trapezius muscle interrogated during simulated missions, this is not to say that individual muscle fibers are not experiencing impaired aerobic metabolism. Previous studies have noted the difficulty in measuring the variables associated with the "Cinderella" hypothesis using surface EMG due to the work at the muscular level being low and isolated to one or a small number of individual muscle fibers (Knardahl, 2002; Eriksen, 2004). We have no way of knowing through non-invasive means exactly how many fibers and exactly how large an area was interrogated by NIRS.

Given our findings of significantly different CtOx responses during NVG (i.e., reduced CtOx) and DAY missions and an increased oxyhemoglobin presence during NVG missions as compared to the DAY missions, it is possible that aerobic metabolism is impaired on a small but significant scale for localized regions of the trapezius muscle. Further study with a larger sample size or newer NIRS systems that allow for "mapping" of the individual localized regions of the muscle may be able to address this issue (Miura et al 2003). Given our promising results, we present the following suggestions for further means of investigation.

The CtOx response for the left trapezius during the NVG mission was found to correlate significantly with helicopter flight experience and both average and maximum lengths of personal NVG flights. Our previous work with helicopter aircrew has

suggested that the right trapezius muscle experiences greater metabolic stress as compared to the left trapezius during NVG missions (Harrison et al., 2007a) regardless of cockpit seat side (Harrison et al., 2007c). Therefore, evidence is mounting that pilots experience a chronically induced state of fatigue in the right trapezius muscle. Given the impact of myalgia on Type I fiber expression as reported by Kadi et al. (1998), we hypothesize that this chronically stressed state in the right side musculature may be the reason why we are not able to report a significant correlation between of CtOx response in the right trapezius and NVG experience variables. While our subject pool is reasonably healthy, the CtOx responses are potentially impacted by the modest and transient neck pain reports of our subjects that would be expected from some of the more experienced aircrew members. While we feel overall that our heterogeneous population sample is a strength of our study, it has limited our ability to perform further statistical analyses on only the most experienced pilots. That is, we do not have enough numbers to perform a meaningful statistical analysis with those pilots with more than 150 hrs of NVG experience, the threshold for reports of neck pain in tactical helicopter aircrew (as identified by Adam, 2004), or on just those novice pilots who are under that threshold.

CONCLUSION

We found the CtOx response in the trapezius muscles of helicopter pilots to be negatively influenced by the additional mass of NVG. While our hemodynamic (tHb) responses indicate that occlusion of the blood supply does not occur during NVG flights in military helicopter pilots, the redox state of CtOx is altered as compared to DAY

flights. Further work along this area with more specific and invasive procedures is needed to support or discredit the "Cinderella" hypothesis as a source of neck myalgia in military helicopter aircrew.

ACKNOWLEDGEMENTS

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APPENDIX D

DIFFERENTIATION OF PHYSIOLOGICAL MEASURES OF NECK MYALGIA BY PRINCIPAL COMPONENT ANALYSIS

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INSERT: This paper, submitted to Journal of Electromyography & Kinesiology, represents the second of the end goals of this thesis. Using principal components analysis with the submaximal isometric ramping protocol described in the methods section and collected during the data collection for Chapter 4, the EMG signals of the symptomatic aircrew was found to differ significantly from the asymptomatic aircrew. This finding may have applications in the development of the appropriate treatment and prevention protocols with respect to night vision goggle induced neck strain.

INTRODUCTION

As research methods and techniques become more and more complex, the number of variables that can be collected simultaneously and with ease increases. To determine which of the variables are the most significant to the research question(s) from a statistical perspective, advanced statistical techniques can be very helpful in the data analysis. Principal components analysis (PCA) is a data reduction technique that allows "a large number of independent variables [to be] systematically reduced to a smaller, conceptually more coherent set of variables" (Dunteman, 1989). This aim is met through the identification of the fewest number of orthogonal factors or vertices that best describe the correlations and minimize variance among those original variables (Dunteman, 1989; Jackson, 1991; Jolliffe, 2002). PCA, for n variables, returns n components that explain a successively smaller proportion of the variation within the data set. Obviously n components are not retained but the successive ordering by magnitude of variance within the data set allows the researcher to retain fewer variables (k) that are indicative of the vast majority of the vital information within the data. From these k components that are deemed retainable, principal component scores that combine results from multiple variables into one number that is relative to the principal component scores of all other individual subjects within the sample population can be calculated. These values can then be used for subsequent analysis to test for statistically significant differences or relationships.

PCA, in kinesiological studies, has been used with success in investigations ranging from climbing performance to hockey player skating stride fatigue, to gait analysis and biomechanical lifting fatigue (Harrison, 2005; Wrigley et al., 2005 & 2006; Chester & Wrigley, 2008). Wrigley et al. (2005; 2006) have been successful in applying PCA to biomechanical waveforms related to lifting techniques. Through their analyses, PCA was shown to be sensitive enough to detect important differences within the variables of lifting mechanics where traditional methods failed while remaining insensitive to confounding factors. Further, Wrigley et al. (2005) were the first to discriminate lifting technique patterns between workers who later developed lower back pain from those who did not. Deluzio and Astephen (*In Press*) used PCA to identify the discriminating factors in knee kinematics related to osteoarthritis of that joint while Chester and Wrigley (2008) have further applied PCA to identify differences in gait mechanics across four age groups to describe the maturational process of bipedal walking. Given how effectively PCA has identified subtle but important differences in various waveform results related to human performance, the present study is proposing an expansion of PCA's application with the present study. Other researchers have successfully used PCA to identify differences in recruitment patterns during trunk stability testing for low-back pain subjects as compared to healthy controls (Hubley-Kozey & Vezina, 2002).

Initial work with military helicopter aircrew focused on using near infrared spectroscopy (NIRS) response of the trapezius muscles of aircrew employing night vision goggles (NVG) during training missions in a full-motion flight simulator (Harrison et al., 2007a,b). Follow-up work has involved the collection of NIRS data gathered in

conjunction with simultaneous electromyography (EMG) and isometric force with the same population (Harrison et al., *In Press* – a, c). From this research it has been demonstrated that the additional mass of the NVG does increase the metabolic activity of the trapezius muscles during simulated flight in helicopter pilots, regardless of left or right seat position (Harrison et al., 2007 a, b). This recorded data using NIRS was consistent with previously reported responses to submaximal activities summarized by Maikala and Bhambhani (2004). Coupled with complementary maximal voluntary contraction (MVC) results (Harrison et al., *In Press* – a, c), and the summary of forces and loads experienced by helicopter aircrew in the Canadian Forces (CF) (Weirstra, 2001), it appears that NVG-induced neck strain in Canadian aircrew is the result of chronic exposure to low-level forces and the associated submaximal activity related to flight duties.

However, while a logistic regression equation to identify and predict those aircrew that suffer neck pain as a result of NVG use has been presented (Harrison et al, *In Press* – b), this equation did not incorporate any of the physiological variables in the larger data set (Harrison et al., *In Press* – a). Previous publications have included a great number of physiological results but these are only made available through the analysis of submaximal endurance testing that is somewhat lengthy in nature and has been reported to be uncomfortable for the subject in the day(s) that followed testing. This study now proposes that a single maximal voluntary contraction and a single submaximal isometric ramping protocol in which the collected EMG and NIRS signals are collected at relative workloads and analyzed with PCA would allow both a quicker testing session as well as

less post-testing subject discomfort could be used to differentiate healthy from injured aircrew.

Therefore, it was hypothesized that the responses as measured with EMG and NIRS reconstructed waveforms related to submaximal isometric force production in the sagittal plane (extension) would differ significantly between aircrew reporting to suffer or experiencing NVG-induced neck strain and aircrew reporting to be asymptomatic.

METHODS

Subjects. Ethical approval was obtained from the University of Regina's Review of Ethics Board and subjects were provided verbal and written summaries of the goal of the project by both CF personnel and the members of the research team. Twenty-nine CF helicopter aircrew (25M, 4F) representing all Canadian tactical helicopter squadrons and levels of experience agreed to participate in the study and signed informed consent.

Electromyography Monitoring. Electrode placement was determined using predetermined measurements based on anatomical landmarks (Seniam, Enschede, Netherlands). Six EMG channels were collected with a commercially available 8-channel system (Bortec Biomedical Ltd., Calgary, Alberta, Canada). Silver-silver-chloride Red Dot (3M, St. Paul, Minnesota, USA) electrodes were placed in a bipolar arrangement over the right and left splenius capitis (SpC_R & SpC_L), right and left sternocleidomastoid (SCM_R & SCM_L), and right and left upper trapezius (T_R & T_L) muscles with an electrode spacing of 20mm. The placement sites were cleaned with 70% alcohol swab and lightly abraded with fine sandpaper to insure a clean signal. A reference electrode was affixed

over the bony protuberance of C7 and signal quality was visually assessed with custom oscilloscope software (US Army Aeromedical Research Laboratory [USAARL], Ft Rucker, Alabama) through the subject's performance of a series of test movements such as neck flexion and extension and shoulder shrugs.

The collected EMG signal was converted to a digital signal through a custom A/D board (USAARL, Ft Rucker, Alabama) using a 16-channel open-ended data acquisition card (Techmatron Instruments Inc., Montreal, Quebec, Canada). Collection frequency was 1024Hz. The digital signal was compiled and stored with custom data acquisition software designed for the testing sequences (USAARL, Ft Rucker, Alabama). Analysis of the EMG data was performed with custom MATLAB (The MathWorks, Natick, Massachusetts, USA) program software. From this signal, root mean squared (RMS) values for each muscle was determined at each of the submaximal work levels of the isometric exercise testing discussed in a subsequent subsection of the methods.

Near Infrared Spectroscopy Monitoring. The trapezii monitoring was performed bilaterally using the NIRO-300 Spatially Resolved Spectroscopic (SRS) oximeter (Hamamatsu Photonics KK, Hamamatsu City, Japan). NIRS is an optical tool used to measure quantitative changes in muscle oxygenation (HbO_2), deoxygenation (HHb) and total haemoglobin (tHb) during periods of physical activity and rest that has proved useful in evaluating responses in erector spinae in vivo (Albert et al., 2004). The technical description and operation of the NIRO-300 has been described in detail elsewhere (Suzuki et al., 1999) and previous work has demonstrated that NIRS is reliable between testing sessions (Bhambhani et al., 1998). Briefly, the NIRO-300 uses different laser

diodes with 4 wavelengths (776, 826, 845, 905nm) as the light source. Utilization of four lasers increases the sensitivity of instrument measurement as more chromophores can be detected in the localized region of light interrogation. The muscles are penetrated with near infrared light (700-1000nm) and the attenuation of this light is measured. The absorbency is calculated using the modified Beer-Lambert Law (Delpy & Cope, 1997).

The NIRS probes were placed bilaterally at the C7 level of the upper trapezius with the skin marked to ensure the same location of placement on the subsequent night mission. Placement location was selected as a result of pilot work in the simulator (Harrison et al., 2007a,b), discussions with CF medical personnel, and previous work in the laboratory with whiplash sufferers (Leavins et al., 2005). A standardized one minute of baseline data was collected at the start of each testing session with the subject seated in the seat with their head in neutral position.

Isometric Testing. Isometric testing was performed with subjects seated in a standard CH-146 helicopter cockpit seat with the appropriate 4-point flight-seat safety harness tightened and secured to minimize trunk movements during testing. As per Figure 6.1, a 2-inch webbing strap was secured around the subject's head and attached to a SSM-AJ-100 force transducer (Interface, Scottsdale, Arizona, USA) that was attached at head level to a 1-inch square steel pole. The pole was positioned 16-inches directly behind the subject and secured in place (Harrison et al., 2007a).



Figure D.1. Apparatus for testing isometric cervical flexion

Maximal Voluntary Contractions (MVC). Subjects were instructed to cross their arms on their chest to prevent them from generating additional leverage by grabbing the arm rests of the cockpit chair during the isometric contractions. During the MVC testing sequence, subjects were provided verbal instructions to "gradually ramp" their force up to maximal force production (approximately 2 sec) to avoid an injury of the neck muscles through a "jerking" movement and the large rate of force development and application related to those types of movements. Each subject performed three 5s isometric extension contractions with a two-minute rest period between contractions. The MVC trial of the greatest magnitude was saved as the true MVC score and it was this value that was used to calculate the subsequent 70% MVC submaximal target forces for the submaximal ramping protocol.

Submaximal Ramping Protocol. Subjects performed one submaximal ramping protocol trial in which they were instructed to increase force production by 7% of their MVC value every two seconds for ten equal intervals until they reached a value equivalent to 70% of their MVC value (i.e. rest, 7%, 14%, 21%...70%). At this time, subjects were instructed to decrease force production by the same magnitude at the same rate until they were no longer producing isometric force (i.e. 70%, 63%, 56%...rest). Visual feedback was provided with numeric indicators and color coded graphs that corresponded to actual force production and target force production in real time. Verbal feedback related to force production and relaxation was also provided in real time by the tester. The same tester was present for all subjects for all data collection sessions.

Data Analysis. Statistical analysis was performed with SPSS V16.0 (SPSS Inc., Chicago, USA). The EMG variable results included the RMS values that corresponded to each target force value (i.e. 7%, 14%, 21%...70% of MVC) of the submaximal ramping protocol for the splenius capitis, sternocleidomastoid, and upper trapezius muscles, bilaterally. The NIRS variable results included the values that corresponded to the same target force values for oxyhemoglobin, de-oxyhemoglobin, total hemoglobin, cytochrome oxidase and tissue oxygenation index, bilaterally. These results collected during the isometric ramping protocol were loaded into matrices for PCA. This resulted in a total of 16 variables being used in the PCA. While the EMG and NIRS data collected was continuous, each waveform was normalized to 21 data points. While the duration of the test was the same for all subjects, the exact timing at which subjects achieved the appropriate percentage of their MVC differed. In order to match the RMS value with the appropriate force value and to make certain that another source of variability was not introduced to the dataset, the curves were then reconstructed using the normalized 21 points. Each waveform was stored in a matrix with dimensions 29 x 21 (# of subjects x # of time points). The reconstructed waveform data was transformed into principal components through an eigenvector analysis of the covariance matrix using custom written MATLAB program software (Wrigley et al., 2005). A SCREE plot was generated and component retention for statistical comparison was determined using parallel analysis (Jackson, 1991; Wrigley et al., 2005). An example of this method is shown in Figure 6.2. This method retains only the principal components that captured an amount of variability that was greater than what would be expected by chance. The retained principal component scores were then stratified based upon aircrew position, pilots (n=15) or FE

(n=14) or pain reporting, symptomatic (n=16) or asymptomatic (n=13). A one-way analysis of variance (ANOVA) was performed to identify significant differences between the component scores across the stratified populations. The data was also stratified by position and status of neck pain symptoms for the appropriate analysis of variance (ANOVA). Significance was set at $p<0.05$.

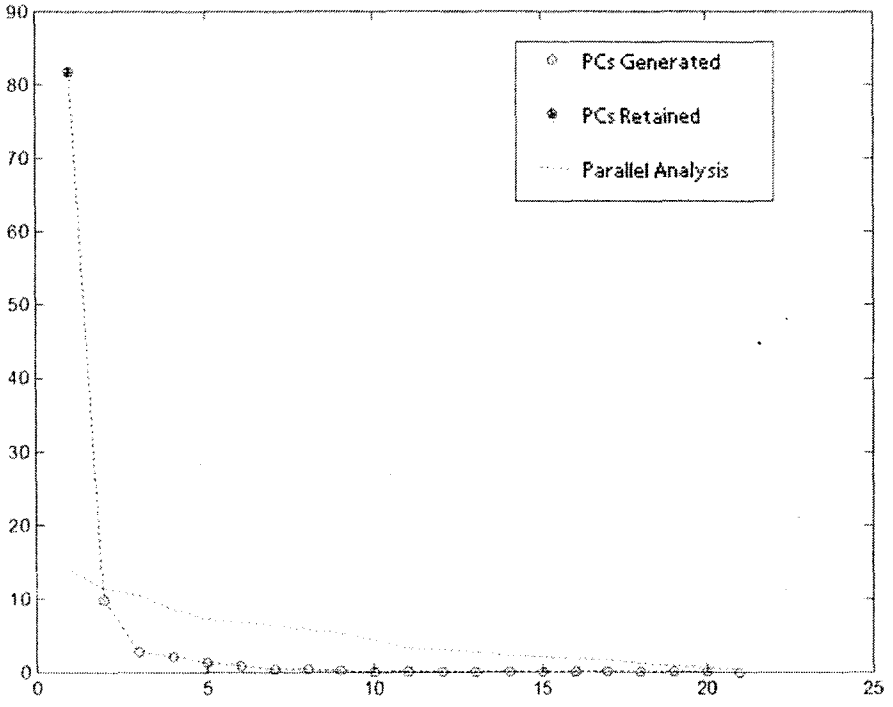


Figure D.2. Parallel analysis applied to the scaled eigenvalues of the 21 principal components (PCs) for root mean square (RMS) results of the left splenius capitis during submaximal ramping protocol with isometric extension. Only PCs for which the associated eigenvalue fell above the line generated by parallel analysis were retained for further analysis.

RESULTS

Subjects. Fifteen pilots (12M, 3F; Age = 32.9(5.6)yrs; Ht= 1.77(6.68)cm; Wt= 78.0(10.4)kg ; Flight Experience = 1531.9(1485.5)hrs) and fourteen Flight Engineers (13M, 1F; Age= 37.5(5.6)yrs; Ht= 1.77(7.86)cm; Wt= 85.1(11.8)kg; Flight Experience = 1128.3 (707.3)hrs) s volunteered to participate in this study. This created the following groups: asymptomatic pilots (n=8), asymptomatic FE (n=5), symptomatic pilots (n=7), and symptomatic FE (n=9). However, no significant differences were noted as a function of aircrew position in one-way ANOVA or as a function of aircrew position and pain in two-way ANOVA. Therefore, only one-way ANOVA of significant PCA scores was performed as a function of reported neck pain.

MVC Values. Each subject performed 3 MVC trials in each direction and a correlation matrix for each direction (i.e. extension, flexion, left flexion & right flexion) were generated. The summary of the correlation analyses are provided below.

Trial	Trial 1	2	3
1	1.00		
2	0.98 (0.01)	1.00	
3	0.97 (0.01)	0.95 (0.01)	1.00

100

Left Right

Table D.1. Correlation matrix for the 3 MVC trials for the extension protocol. Table values are R^2 (p-value).

Trial	Trial 1	2	3
1	1.00		
2	0.98 (0.01)	1.00	
3	0.92 (0.01)	0.96 (0.01)	1.00

Table D.2. Correlation matrix for the 3 MVC trials for the flexion protocol. Table values are R^2 (p-value).

Trial	Trial 1	2	3
1	1.00		
2	0.94 (0.01)	1.00	
3	0.94 (0.01)	0.94 (0.01)	1.00

Table D.3. Correlation matrix for the 3 MVC trials for the right flexion protocol. Table values are R^2 (p-value).

Trial	Trial 1	2	3
1	1.00		
2	0.91 (0.01)	1.00	
3	0.93 (0.01)	0.94 (0.01)	1.00

Table D.4. Correlation matrix for the 3 MVC trials for the left flexion protocol. Table values are R^2 (p-value).

Extension. The overall mean MVC result for the 29 subjects was 23.5(5.4)kg. No differences were found between groups as a function of position, pain or position and pain. No significant differences were found for the results of any of the EMG RMS variables or any of the NIRS variables using one-way ANOVA (symptomatic vs. asymptomatic). A summary of the PC scores is provided in Table 6.1, with means and standard deviations for the symptomatic and asymptomatic groups presented. The same summary of PC score results is provided for the NIRS variables in Table 6.2. Only the ANOVA of the PC scores for the RMS results of the right upper trapezius returned more than one statistically significant PC. However, of the 16 EMG and NIRS variables used in the PCA, only the PC scores for the left splenius capitis (SpC_L; p=0.021) and the right splenius capitis (SpC_R;p=0.016) RMS results were found to be statistically significant with one-way ANOVA as a function of pain. The reconstructed curves for the RMS of the submaximal ramp protocol and the variance explained by the PC (81.62%) with the data for SpC_L are included in Figure 6.3. Figure 6.4 provides the same information for the SpC_R PC, which captured 84.95% of the variance within the data set. In both cases, the symptomatic aircrew demonstrated suppressed RMS values as compared to their healthy counterparts.

Variable	Number of PCs	Captured Variance (%)	PC Scores Asymptomatic	Symptomatic
Splenius Capitis (left)	1	81.62	56.28 (108.26)*	-22.34 (63.41)*
Splenius Capitis (right)	1	84.95	66.66 (122.68)*	-26.20 (69.92)*
Sternocleidomastoid (left)	1	94.41	11.28 (35.54)	9.00 (57.15)
Sternocleidomastoid (right)	1	96.09	9.91 (32.81)	7.36 (54.19)
Upper Trapezius (left)	1	95.32	8.74 (34.52)	6.47 (56.19)
Upper Trapezius (right)	1	75.20	-4.24 (39.49)	1.55 (50.02)
Upper Trapezius (right)	2	18.21	-4.71 (14.42)	4.51 (21.34)

Table D.5. Summary of PCA results for EMG RMS variables. Only the scores for the significant PCs are presented as means and standard deviations. * denotes a significant difference between the asymptomatic and symptomatic populations ($p<0.05$).

Variable	Number of PCs	Captured Variance (%)	PC Scores Asymptomatic	Symptomatic
Oxyhemoglobin (left)	1	88.19	3.09 (22.45)	-0.70 (13.56)
Oxyhemoglobin (right)	1	92.64	-7.14 (34.15)	6.27 (15.06)
De-oxyhemoglobin (left)	1	88.78	-1.76 (11.71)	0.12 (8.01)
De-oxyhemoglobin (right)	1	95.10	-7.64 (21.35)	3.73 (12.44)
Total Hemoglobin (left)	1	88.78	1.45 (32.74)	0.77 (15.26)
Total Hemoglobin (right)	1	93.88	-14.78 (54.25)	10.01 (21.82)
Cytochrome Oxidase (left)	1	89.98	0.25 (7.62)	1.01 (3.07)
Cytochrome Oxidase (right)	1	94.70	4.05 (11.00)	-0.81 (4.03)
Tissue Oxygenation Index (left)	1	91.60	-11.10 (120.24)	23.49 (92.60)
Tissue Oxygenation Index (right)	1	99.23	-18.84 (145.68)	37.87 (105.37)

Table D.6. Summary of PCA results for NIRS variables. Only the scores for the significant PCs are presented as means and standard deviations. No significant difference between the asymptomatic and symptomatic populations were observed ($p<0.05$).

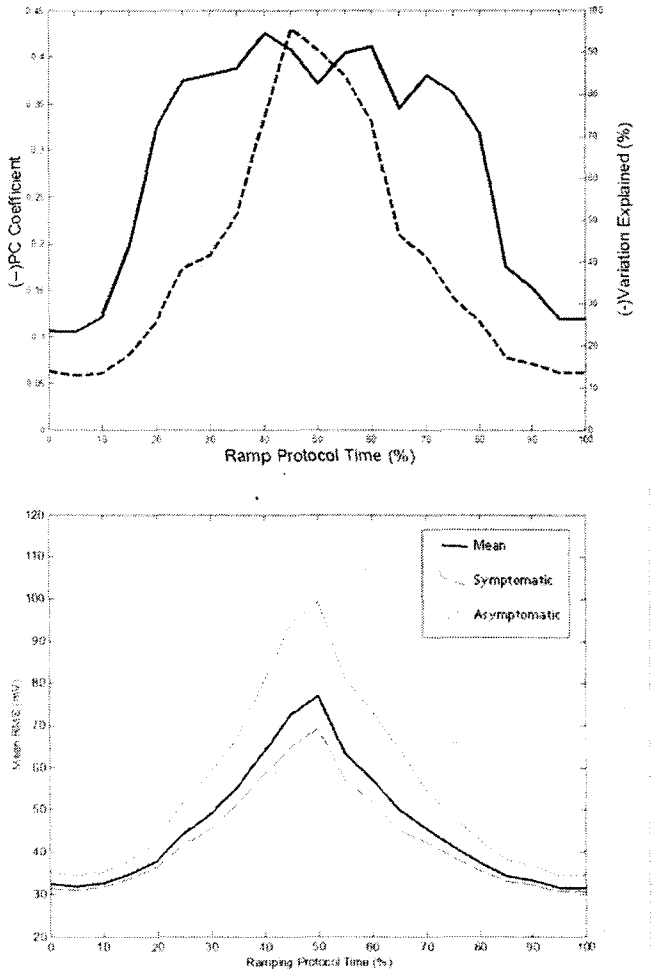


Figure D.3. The PC coefficient and proportion of variance explained by first PC of the left splenius capitis are presented above (top). The difference ($p=0.021$) between the reconstructed curves for symptomatic and asymptomatic subjects is also presented above (bottom). The submaximal ramping protocol was standardized to 21 points and presented with time as a percentage of the trial: Ramping Protocol Time <50% corresponds to gradual increases in force (i.e. 7%, 14%, 21%...63%MVC); Ramping Protocol Time =50% corresponds to 70%MVC; Ramping Protocol Time >50% corresponds to the gradual decreases in force (i.e. 63%, 56%, 49%...7%MVC) .

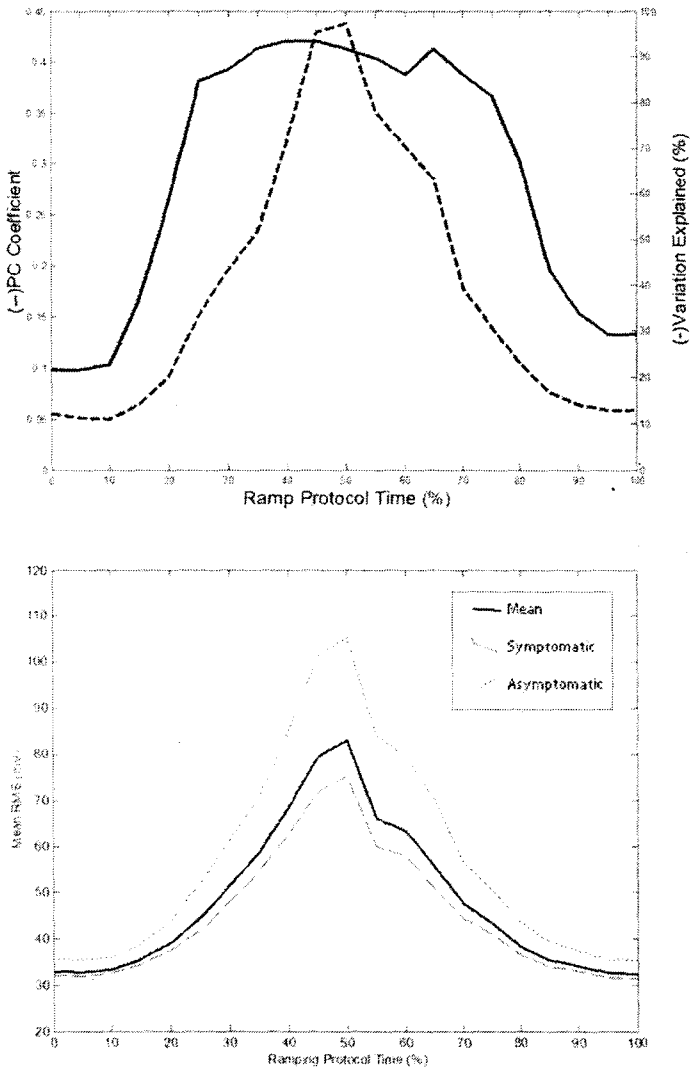


Figure D.4. The PC coefficient and proportion of variance explained by first PC of the right splenius capitis are presented above (top). The difference (p=0.016) between the reconstructed curves for symptomatic and asymptomatic subjects is also presented above (bottom). The submaximal ramping protocol was standardized to 21 points and presented with time as a percentage of the trial: Ramping Protocol Time <50% corresponds to gradual increases in force (i.e. 7%, 14%, 21%...63%MVC); Ramping Protocol Time =50% corresponds to 70%MVC; Ramping Protocol Time >50% corresponds to the gradual decreases in force (i.e. 63%, 56%, 49%...7%MVC).

DISCUSSION

This study has used PCA and the resultant PC scores to identify differences between helicopter aircrew with and without reported neck pain. To the best of the author's knowledge, this is the first time that PCA has been applied to data curves reconstructed from both EMG RMS and NIRS data as they relate to standardized force production of the cervical spine. Previous publications using PCA with waveforms have had success in analyzing gait to identify differences as a result of maturation that had been too subtle for detection with more conventional statistics (Chester & Wrigley, 2008), and to identify lifting mechanics related to back pain (Wrigley et al., 2005). Hubley-Kozey and Vezina (2002) noted a lack of uniformity as assessed with EMG within their low-back pain population with respect to recruitment co-activation across seven muscle groups. However, this line of research had previously been unable to differentiate aircrew who reported neck pain from their reportedly healthy colleagues using parameter-based analyses with various physiological measures including MVC, EMG (RMS and median frequency) and NIRS results (Harrison et al., *In Press* – a; c). In this paper, success has been achieved with the application of PCA to curves generated by only one isometric submaximal ramping protocol in differentiating neck-pain reporters from non-reporters.

In the previous papers presented in this dissertation, lengthy tests have been used that involved both maximal force production and maximal endurance of a submaximal force in four isometric directions (Harrison et al., *In Press* - a). The logistic regression

analysis paper (Chapter 5) indicated that these test protocols were more involved than was needed to accurately predict neck pain (Harrison et al., *In Press* – b). Data was collected from subjects performing a series of isometric tests of varying intensities in four different directions with the corresponding EMG and NIRS channels monitored. The resulting logistic regression equation, much to our surprise, did not provide any physiological insight. That is to say, of all the information collected, only the most basic of information was required to accurately identify aircrew as symptomatic or asymptomatic based upon their self-reported neck pain status. Once again, this chapter demonstrates that perhaps our original methodology was beyond what was needed to identify and monitor neck pain status among this aircrew population. While EMG and NIRS channels were monitored during submaximal trials that saw force increased and decreased in a successive manner that was proportional to each subject's maximal capacity for force production in each of the four directions, only the splenius capitis EMG RMS results for extension trial was required to differentiate the asymptomatic aircrew from their symptomatic colleagues. The result is a testing methodology that is simple, quick, and detailed in the physiological sense.

Electromyography

We selected to only use the extension isometric contraction to keep with the intent of PCA to achieve data reduction and parsimony. While this line of research had collected and could have analyzed results from the same 4 isometric contractions as have been previously reported (Harrison et al., In Press – a; c), the intent was to present as much meaningful data with as few tests as possible. The extension protocol seemed the

most logical test to include for a variety of reasons. First, research has indicated that the mass of the NVG is often forward and above the center of gravity of the head and the extensors are the muscles that are primarily responsible for maintaining a heads-up posture (Thuresson et al., 2003; Netto & Burnett, 2006). Second, the author wanted to make certain the research collected data from the musculature on the left and right side of the cervical spine. Previously presented results have consistently noted a left and right difference that is independent of cockpit seat side (Harrison et al., 2007a, c; Harrison et al., *In Press* – a; c) and complement the work of others in the assessment of lumbar spinal muscular activity in the same population (De Oliveira & Nadal, 2004). The use of the extension trial allowed the author to monitor the key neck muscles bilaterally with one test.

The PCA of the RMS of the primary agonist during isometric extension has provided a novel finding that may provide insight into some mechanisms of neck pain. The symptomatic aircrew demonstrated a decreased RMS response, bilaterally, in the splenius capitis as compared to the asymptomatic aircrew. A number of hypotheses could explain this phenomenon. The relative workloads (7%, 14%, 21%...70% of MVC) were not statistically different between the two groups (i.e., asymptomatic vs. symptomatic). Therefore, the level of muscular recruitment is decreased in the prime agonist for the symptomatic aircrew as compared to their healthy peers. This did not correspond with any increases in neuromuscular or metabolic function in other muscles monitored by this protocol. The first hypothesis is that this suggests that the symptomatic aircrew members might be relying on fewer muscle fibres to do the same amount of work. This postulation bears similarities to the already presented "Cinderella Hypothesis" (Thorn, 2005; Visser

& van Dieën, 2006; Sjogaard et al., 2000; Eriksen, 2004) that have been discussed as being a possibility in this population in their occupational settings (Harrison et al., *In Press* - c). Alternatively, the decrease in RMS could be indicative of an impairment of muscular recruitment coordination. The symptomatic aircrew population could be utilizing a comparable number of muscle fibers and associated motor units as the asymptomatic subjects but the firing sequences may be dysfunctional.

Another hypothesis exists that may further explain the apparent link between decrease in muscular activity in the splenius capitis during submaximal extension and neck pain. Panjabi (2006) presents a hypothesis that ligament subfailure injuries may lead to pain in the cervical and lumbar spine. These injuries are purported to be the result of microtrauma from a source that is either acute or cumulative in nature. The ligaments, the facet capsules, or the discs themselves are damaged and muscle control dysfunction results. The author suggests that "muscle coordination and individual muscle force characteristics, i.e. onset, magnitude, and shut-off, are disrupted." Further statements in the article suggest that this increases the stress and strain on a number of structures, including the slow-healing spinal ligaments and the imbedded mechanoreceptors. This line of thought complements the abnormal muscle recruitment patterns recorded with EMG in low-back pain subjects by Hubley-Kozey and Vezina (2002) as compared to healthy controls.

The injury model involving cumulative microtrauma and resulting impact on muscular response to load in the cervical spine described by Panjabi (2006) seems to have been arrived as a result of research in military flight involving NVG. Other studies have noted that the mass of the NVG-equipped helmet is submaximal in nature as

compared to the strength capacity of the neck in all directions (Ang et al., 2005 & 2006; Harrison et al., *In Press* – a). But the cumulative effects have been attributed to neck strain in numerous studies using helicopter aircrew (Wickes & Greeves, 2005; Adam, 2004; Harrison et al., *In Press* – b; Fraser et al., 2006). Thresholds of total flight hours logged with NVG have been identified (Adam, 2004; Fraser et al., 2006) and thresholds for maximum duration of single mission exposure have been determined (Harrison et al., *In Press* – b). With the present paper, evidence is provided of decreased muscle activity in conjunction with reported neck pain among that aircrew population.

It is possible that structures other than those monitored with EMG and hypothesized by Panjabi (2006) may be assisting in force production in the extension movement in ways that the research tools were not able to measure in this study. It is possible, in the sense that these results do not allow the author to refute the suggestion, that non-superficial muscles such as the longissimus capitis or semispinalis capitis are more active in the symptomatic subjects as compared to their asymptomatic counterparts during extension. This research can, though, suggest that this is unlikely to be true. Research has suggested that training programs should aim to recruit the deep flexors and extensors to address neck myalgia (Jull et al., 1999). It is therefore unlikely that the subjects who were reporting neck pain were recruiting their non-superficial muscles to a greater degree than their healthy colleagues and still reporting neck pain.

Near infrared spectroscopy

The author was surprised by the NIRS results. To date, earlier work has found NIRS to be an extremely variable measure that requires advanced statistical approaches to identify significant differences within a population (Harrison et al., 2007a). PCA by nature addresses variability and provides a powerful statistical tool to address such. However, these NIRS variables did not return more than one PC for each variable that was significant as determined by parallel analysis nor did they return PC scores that were significantly different when compared as a function of pain reports. Once again, this time with PC scores, it was observed that the variation with the NIRS results was quite large as compared to the mean values. The lack of significance suggests a number of potential explanations. First this research collected the NIRS results from the trapezius muscles. While this was sufficient in preliminary work in the applied setting for trials that were very lengthy in nature (Harrison et al., 2007a), the trapezius may not be the most appropriate muscle to monitor during shorter duration, i.e. 40 seconds, tests for cervical muscle function. However, repositioning the NIRS probes and effectively monitoring muscles such as the splenius capitis would prove difficult and would render EMG collection impossible.

In light of the available literature, a second explanation should also be considered with respect to the lack of difference in NIRS response between the two groups. Unpublished data from an associated laboratory (Leavins et al., 2005) found differences in the haemodynamics and metabolism in the upper trapezius of whiplash associated

disorder (WAD) patients as compared to healthy norms. In the case of our results, this may therefore be an indication of an effect on the avascular tissues such as the cervical spinal ligaments in the symptomatic group. If this is in fact the case, this would lend further support to the hypothesis put forth by Panjabi (2006); i.e., pain is the result of chronic microtrauma in structures such as ligaments and tendons.

Follow-up research should endeavour to evaluate the causes of these results. The author currently cannot state with definitive authority whether we are observing a decreased RMS in splenius capitis muscles as determined with PC scores in the symptomatic group as a *cause* of injury or as a *result* of injury. The author also cannot say with certainty that this decrease in activity is resulting in increased stress on the structures postulated by Panjabi (2006) or whether the deficit in muscular activity is being addressed by other deeper muscles that are not easily measured with EMG.

CONCLUSION

This research has successfully used PCA to differentiate between the neuromuscular activity curves created by the RMS of a progressive submaximal isometric cervical extension task by aircrew who did and who did not report suffering from neck pain. This finding provides dual benefit: it provides evidence to support the further use of this statistical approach for physiological data analyses, as well as provides support to a pre-existing and compelling hypothesized mechanism of cervical pain. Using sensitive statistical tools designed for analyzing variance within a data set, differences between symptomatic and asymptomatic aircrew can be identified at the muscular level.

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APPENDIX E

University of Regina – Faculty of Kinesiology & Health Studies
Night Vision Goggle Study Data Sheet

Please complete the information below honestly and accurately. Your responses will be held by the research team in the strictest of confidence and your individual responses will not be provided to CF medical personnel or your superior officers for any reason.

Name & Rank: _____ Age: _____

Height: _____ Weight: _____

Position: ☐ Pilot ☐ Flight Engineer ☐ Other

Dominant Hand: ☐ Right ☐ Left
Counterweight: ☐ Use Counterweight ☐ Do NOT Use Counterweight

Flight experience

Total years of flight experience: _____

Total hours of flight experience: _____

Years of rotary wing flight experience: _____

Hours of rotary wing flight time: _____

Hours of Night Vision Goggle (NVG) experience: _____

Average length of a typical NVG flight (hours): _____

Maximum length of personal NVG flight (hours): _____

Neck Pain History

Do you experience neck pain? (please check all that apply)

☐ Yes, constant & severe
☐ Occasionally
☐ During the summer months
☐ During repeated exposure to NVG
☐ During simulator exercises
☐ Yes, constant but moderate
☐ During the winter months
☐ During extended exposure to NVG
☐ During actual flight time
☐ Never

Do you smoke tobacco? ☐ Yes ☐ No

Physical Activity History

How often, per week, do you engage in aerobic exercise for a period of 20 minutes or more?

☐ 0 ☐ 1-2 ☐3-4 ☐5+

How often, per week, do you engage in strength training exercise that includes at least 3 different exercises?

☐ 0 ☐ 1-2 ☐3-4 ☐5+

My physical fitness regimen includes the following: (please check all that apply)

☐ Aerobic training
☐ Stretching and flexibility
☐ Organized team sports
☐ Strength training
☐ Yoga
☐ Other

Neck Pain Treatment History

I have sought treatment from the following professionals for neck pain: (please check all that apply)

☐ Flight Surgeon
☐ Massage therapist
☐ Personal trainer
☐ Physiotherapist
☐ Acupuncture specialist
☐ Other

If "Other", please specify: _____

_____

_____

_____

Thank you for your time and input

VITA

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Saba University School of Medicine
(2008-)
Doctor of Medicine, Masters of Science (Hyperbaric Oxygen Therapy), Bachelor of Science (Health Science)

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